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ModuFlex ERP: Customizable Management Solution

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Bachelor of Science in Software Engineering (2021-2025)

The candidate confirms that the work submitted is their own and appropriate credit has been given where reference has been made to the work of others.



COMSATS University, Islamabad Pakistan

ModuFlex ERP: Customizable Management Solution

**A project presented to
COMSATS University, Islamabad**

**In partial fulfillment
Of the requirement for the degree of**

Bachelor of Science in Software Engineering (2021-2025)

By

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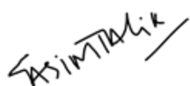


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CERTIFICATE OF APPROVAL

It is to certify that the final year project of BS (SE) “ModuFlex ERP: Customizable Management Solution” was developed by **Muhammad Aqsam Qurashi (CIIT/FA21-BSE-051)** and **Syed Sirdan Ali (CIIT/FA21-BSE-084)** under the supervision of “**Mr. Qasim Malik**” and that in his opinion; it is fully adequate, in scope and quality for the degree of Bachelors of Science in Software Engineering.



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Executive Summary

Businesses today often struggle with fragmented processes, rigid systems, and inefficient resource management, which impede productivity and decision-making. ModuFlex ERP addresses these challenges by providing a customizable, scalable, and comprehensive Enterprise Resource Planning solution designed to integrate and streamline business operations effectively.

ModuFlex ERP is modular, allowing organizations to customize functionalities, including User and Role Management, HR Management, Task and Notification Management, Billing and Financing, Order and Inventory Management, Development and Integration tools, Dashboard and Reporting, and an intelligent Chatbot module. This flexibility ensures tailored user experiences, enhanced operational efficiency, real-time data visibility, and informed decision-making capabilities.

Built upon robust modern technologies, ModuFlex ERP minimizes operational complexities, reduces overhead costs, and promotes seamless integration across departments. With user-friendly interfaces and comprehensive reporting capabilities, businesses can optimize performance, reduce manual workloads, and foster strategic growth, positioning them competitively within their industries.

Acknowledgement

All praise is to Almighty Allah who bestowed upon us a minute portion of His boundless knowledge by virtue of which we were able to accomplish this challenging task.

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And we are also thankful to our parents and family who have been a constant source of encouragement for us and brought us with the values of honesty & hard work.

Muhammad Aqsam Qurashi



Syed Sirdan Ali



Abbreviations

SRS	Software Requirement Specification
SDS	Software Design Specification Document
FS	File System
ERP	Enterprise Resource Planning
HRM	Human Resource Management
KPI	Key Performance Indicator
SKU	Stock Keeping Unit
POS	Point of Sale
CRUD	Create, Read, Update, Delete

Table of Contents

1	Introduction.....	1
1.1	Vision Statement.....	1
1.2	Related System Analysis/Literature Review.....	1
1.3	System Deliverables.....	2
1.3.1	Application Modules The application will include the following five modules:	2
1.4	System Limitations/Constraints	2
1.5	Tools and Technologies	3
1.6	Relevance to Course Modules.....	3
2	Problem Definition.....	4
2.1	Problem Solution	4
2.2	Objectives of the Proposed System.....	4
2.3	Scope.....	5
2.4	Modules.....	5
2.4.1	<i>Module 1: User Management.....</i>	5
2.4.2	<i>Module 2: Role and Permission Management</i>	5
2.4.3	<i>Module 3: Development and Integration Module</i>	6
2.4.4	<i>Module 4: Human Resource Management (HRM).....</i>	6
2.4.5	<i>Module 5: Dashboard and Reporting</i>	6
2.4.6	<i>Module 6: Billing and Financing.....</i>	6
2.4.7	<i>Module 7: Task and Notification Management</i>	7
2.4.8	<i>Module 8: Order and Inventory Management</i>	7
2.4.9	<i>Module 9: Chatbot</i>	7
2.5	Project Contribution.....	7
3	Requirement Analysis.....	9
3.1	User classes and characteristics	9
3.2	Requirement Identifying Technique	9
3.2.1	Use Case Diagram.....	9
3.2.2	Detailed Use cases.....	13
3.2.2.1	Module 1: User Management.....	13
3.2.2.2	Module 2: Role and Permission Management.....	17
3.2.2.3	Module 3: Integration and Development	21
3.2.2.4	Module 4: Human Resource Management (HRM):	23
3.2.2.5	Module 5: Dashboard and Reporting.....	28
3.2.2.6	Module 6: Billing and Financing.....	31
3.2.2.7	Module 7: Task and Notification Management	35
3.2.2.8	Module 6: AI chatbot Assistance	37
3.3	Functional Requirements	39
3.3.1	Functional Requirements for UC-01: Login	39
3.3.1.1	FR-1.1: User Authentication.....	39
3.3.1.2	FR-1.2: Password Recovery Process.....	40

3.3.1.3	FR-1.3: Post-Login Redirection	40
3.3.2	Functional Requirements for UC-02: Edit User Profile	40
3.3.2.1	FR-2.1: Access Profile Editing	41
3.3.2.2	FR-2.2: Validate User Input	41
3.3.2.3	FR-2.3: Update User Profile	41
3.3.3	Functional Requirements for UC-03: Change Password	42
3.3.3.1	FR-3.1: Access Change Password Page	42
3.3.3.2	FR-3.2: Verify Current Password	42
3.3.3.3	FR-3.3: Validate New Password	43
3.3.3.4	FR-3.4: Update Password in System	43
3.3.4	Functional Requirements for UC-04: Delete User Account.....	43
3.3.4.1	FR-4.1: Select User Account	44
3.3.4.2	FR-4.2: Confirm Deletion Action	44
3.3.4.3	FR-4.3: Execute Account Deletion.....	44
3.3.5	Functional Requirements for UC-05: Logout	45
3.3.5.1	FR-5.1: Display Logout Prompt	45
3.3.5.2	FR-5.2: Confirm Logout	45
3.3.5.3	FR-5.3: Redirect to Login Page	46
3.3.5.4	FR-5.4: Cancel Logout	46
3.3.6	Functional Requirements for UC-06: Customize User Interfaces.....	46
3.3.6.1	FR-6.1: Access Customization Interface	47
3.3.6.2	FR-6.2: Select Role for Customization.....	47
3.3.6.3	FR-6.3: Configure Dashboard Items	47
3.3.6.4	FR-6.4: Manage Custom Menus.....	48
3.3.7	Functional Requirements for UC-07: Create New User Roles	48
3.3.7.1	FR-7.1: Add New Role	48
3.3.7.2	FR-7.2: Validate Role Uniqueness	49
3.3.7.3	FR-7.3: Save Role Information	49
3.3.8	Functional Requirements for UC-08: Assign User to Dashboard.....	49
3.3.8.1	FR-8.1: Select User for Dashboard Assignment	50
3.3.8.2	FR-8.2: Display Available Dashboards	50
3.3.8.3	FR-8.3: Assign Dashboard to User.....	50
3.3.8.4	FR-8.4: Confirm Dashboard Assignment	51
3.3.9	Functional Requirements for UC-09: Assign User Role.....	51
3.3.9.1	FR-9.1: Select User for Role Assignment.....	51
3.3.9.2	FR-9.2: Display Available Roles	52

3.3.9.3	FR-9.3: Assign Role to User	52
3.3.10	Functional Requirements for UC-10: Define Database Schema	52
3.3.10.1	FR-10.1: Define Schema Fields	53
3.3.10.2	FR-10.2: Save Database Schema	53
3.3.11	Functional Requirements for UC-11: Import Data Structure from Excel	53
3.3.11.1	FR-11.1: Upload Excel Sheet	53
3.3.11.2	FR-11.2: Validate Excel Format	54
3.3.11.3	FR-11.3: Read Data from Excel	54
3.3.11.4	FR-11.4: Create Database Tables	55
3.3.12	Functional Requirements for UC-12: Generate Code	55
3.3.12.1	FR-12.1: Initiate Code Generation	55
3.3.12.2	FR-12.2: Integrate Generated Code	56
3.3.13	Functional Requirements for UC-13: Create New User Profile	56
3.3.13.1	FR-13.1: Enter User Information	56
3.3.13.2	FR-13.2: Validate User Information	56
3.3.13.3	FR-13.3: Save User Profile	57
3.3.13.4	FR-13.4: Display Confirmation	57
3.3.14	Functional Requirements for UC-14: Edit User Profile	58
3.3.14.1	FR-14.1: Select User to Edit	58
3.3.14.2	FR-14.2: Update User Information	58
3.3.14.3	FR-14.3: Validate Updated Information	59
3.3.14.4	FR-14.4: Confirm and Save Changes	59
3.3.15	Functional Requirements for UC-15: Submit and Track Expense Reports	59
3.3.15.1	FR-15.1: Create New Expense Report	60
3.3.15.2	FR-15.2: Validate Expense Report Details	60
3.3.15.3	FR-15.3: Submit Expense Report	60
3.3.15.4	FR-15.4: Track Expense Report Status	61
3.3.15.5	FR-15.5: Display Confirmation Message	61
3.3.16	Functional Requirements for UC-16: Apply for and Track Leave Requests	62
3.3.16.1	FR-16.1: Access Leave Request Section	62
3.3.16.2	FR-16.2: Input Leave Details	62
3.3.16.3	FR-16.3: Validate Leave Request	62
3.3.16.4	FR-16.4: Submit and Notify	63
3.3.16.5	FR-16.5: Track Leave Status	63
3.3.17	Functional Requirements for UC-17: Manage Timesheets	64
3.3.17.1	FR-17.1: Access Timesheet Section	64

3.3.17.2	FR-17.2: Enter and Update Work Hours	64
3.3.17.3	FR-17.3: Validate and Save Timesheet.....	65
3.3.17.4	FR-17.4: Display Confirmation Message	65
3.3.17.5	FR-17.5: Calculate Working Hours	65
3.3.18	Functional Requirements for UC-18: Review and Approve Requests.....	66
3.3.18.1	FR-18.1: Review Pending Requests.....	66
3.3.18.2	FR-18.2: Approve or Reject Requests	66
3.3.18.3	FR-18.3: Notify Users	67
3.3.19	Functional Requirements for UC-19: Generate HR Reports.....	67
3.3.19.1	FR-19.1: Specify Report Parameters	67
3.3.19.2	FR-19.2: Generate Report	68
3.3.19.3	FR-19.3: Validate Report Parameters	68
3.3.19.4	FR-19.4: Save and Display Report	69
3.3.20	Functional Requirements for UC-20: Generate Custom Reports.....	69
3.3.20.1	FR-20.1: Specify Report Criteria	69
3.3.20.2	FR-20.3: Display Generated Report.....	70
3.3.21	Functional Requirements for UC-21: Export Reports.....	70
3.3.21.1	FR-22.1: Choose Export Format	70
3.3.21.2	FR-22.2: Process and Export Report.....	70
3.3.22	Functional Requirements for UC-22: Customize Dashboards by Role.....	71
3.3.22.1	FR-22.1: Customize Dashboard Elements	71
3.3.22.2	FR-22.2: Select User Role for Dashboard Assignment	71
3.3.22.3	FR-22.3: Apply Customized Settings to Role	72
3.3.22.4	FR-22.4: Confirm and Log Customization.....	72
3.3.23	Functional Requirements for UC-23: Interactive Data Visualizations.....	73
3.3.23.1	FR-23.1: Interact with Visualization Elements	73
3.3.23.2	FR-23.2: Display Detailed Data.....	73
3.3.23.3	FR-23.3: Customize Visualization Placement	74
3.3.24	Functional Requirements for UC-24: Customize and Update Dashboards.....	74
3.3.24.1	FR-24.1: Select Dashboard Elements	74
3.3.24.2	FR-24.2: Arrange Dashboard Layout	75
3.3.24.3	FR-24.3: Save Customizations and Refresh Dashboard	75
3.3.25	Functional Requirements for UC-25: Create Invoices	76
3.3.25.1	FR-25.1: Enter Customer and Product Details	76
3.3.25.2	FR-25.2: Calculate Total Cost	76
3.3.25.3	FR-25.3: Review and Finalize Invoice	77

3.3.26	Functional Requirements for UC-26: Manage Receipts and Payments	77
3.3.26.1	FR-26.1: Select Transaction Option.....	77
3.3.26.2	FR-26.2: Enter Payment Details	77
3.3.26.3	FR-26.3: Process Payment.....	78
3.3.26.4	FR-26.4: Generate and Download Receipt.....	78
3.3.27	Functional Requirements for UC-27: Monitor Invoice Status	79
3.3.27.1	FR-27.1: Display Invoice Status	79
3.3.27.2	FR-27.3: Alert for Overdue Invoices	79
3.3.28	Functional Requirements for UC-28: Create Income and Expense Category	80
3.3.28.1	FR-28.1: Access Income and Expense Category Section	80
3.3.28.2	FR-28.2: Add New Category	80
3.3.28.3	FR-28.3: Validate Category Details.....	81
3.3.28.4	FR-28.4: Submit and Confirm Category Creation.....	81
3.3.29	Functional Requirements for UC-29: Add Account Incomes and Expense	81
3.3.29.1	FR-29.1: Access Account Income and Expense Section	81
3.3.29.2	FR-29.2: Input Financial Details.....	82
3.3.29.3	FR-29.3: Validate Financial Entry	82
3.3.29.4	FR-29.4: Submit and Update Financial Records	83
3.3.30	UC-30: Manage To-Do List.....	83
3.3.30.1	FR-30.1: Access To-Do List Section.....	83
3.3.30.2	FR-30.2: Add and Specify Task Details	84
3.3.30.3	FR-30.3: Edit Task Details	84
3.3.30.4	FR-30.4: Delete Task.....	84
3.3.31	Functional Requirements for UC-31: Schedule Events in Calendar	85
3.3.31.1	FR-31.1: Add New Event	85
3.3.31.2	FR-31.2: Modify and Delete Events.....	85
3.3.32	Functional Requirements for UC-32: Set Reminders and Receive Notifications	86
3.3.32.1	FR-32.1: Access Reminder Settings	86
3.3.32.2	FR-32.2: Specify Reminder Details	86
3.3.33	Functional Requirements for UC-33: Chatbot Assistance for Schema Creation	87
3.3.33.1	FR-33.1: Access Chatbot Interface	87
3.3.33.2	FR-33.2: Process Schema Queries	87
3.3.34	Functional Requirements for UC-34: Format Guidance for Excel Schema	88
3.3.34.1	FR-34.1: Provide Format Guidance	88
3.3.35	Functional Requirements for UC-35: Collect Feedback for Performance Improvement.....	88
3.3.35.1	FR-35.1: Collect Feedback	88
3.3.35.2	FR-35.2: Analyze Feedback	89

3.4	Non-Functional Requirements	89
3.4.1	Reliability	89
3.4.2	Usability	89
3.4.3	Performance	90
3.4.4	Security	90
3.5	External Interface Requirements.....	90
3.5.1	User Interfaces Requirements	90
3.5.2	Software interfaces.....	91
3.5.3	Hardware interfaces	91
3.5.4	Communications interfaces.....	91
4	Design and Architecture	92
4.1	Architectural Design	92
4.2	Design Models	92
4.2.1	Activity Diagram.....	92
4.2.1.1	Login.....	93
4.2.1.2	Edit Profile.....	94
4.2.1.3	Delete User Account	95
4.2.1.4	Role Creation and Assignment.....	96
4.2.1.5	CRUD Generator.....	97
4.2.1.6	Create New User Profile.....	98
4.2.1.7	Submit and Track Expense Report	99
4.2.1.8	Leave Request	100
4.2.1.9	Timesheets	101
4.2.1.10	Generate Report	102
4.2.1.11	Create and Customize Dashboard.....	103
4.2.1.12	Invoices Payment and Generate Receipts	104
4.2.1.13	Financial Inputs	105
4.2.1.14	Task and Progress Tracking.....	106
4.2.1.15	Order and Inventory Management.....	107
4.2.2	Data Flow Diagram.....	108
4.2.2.1	Context Diagram	108
4.2.2.2	Level 1 DFD.....	109
4.2.2.3	Level 2 DFD.....	111
4.2.3	State Transition Diagram	116
4.2.3.1	User Authentication and Session Management	116
4.2.3.2	Database Schema Generation.....	117
4.2.3.3	Submit and Track Expense Reports.....	118
4.2.3.4	Managing Timesheets and Reports	119

4.2.3.5	Manage Tasks.....	120
4.2.3.6	Events Scheduling	121
4.2.3.7	Predictive Analytics	122
4.2.3.8	Inventory Management	123
4.3	Data Design.....	123
4.3.1	User Management Schema.....	123
4.3.2	Role and Permission Schema	124
4.3.3	Human Resource Management Schema.....	124
4.3.4	Dashboard and Reporting Schema	125
4.3.5	Billing and Financing Schema	125
4.3.6	Task and Notification Schema	126
4.3.7	Data Dictionary	126
4.4	Human Interface Design	129
4.4.1	Login Screen	129
4.4.2	CRUD Generator Screen.....	129
4.4.3	Dashboard Screen.....	130
4.4.4	Stripe Payment Screen	130
4.4.5	Role Screen	131
4.4.6	Expense Report Screen.....	131
5	Implementation.	132
5.1	Algorithm.....	132
5.1.1	Calculating WorkHours.....	132
5.1.2	Report Generation	132
5.1.3	Leave Balance Calculation.....	133
5.2	External APIs/SDKs	134
6	Testing and Evaluation	135
6.1	User management module.....	135
6.1.1	Unit Testing.....	135
6.1.2	Functional Testing.....	135
6.1.3	Business Rule Testing.....	136
6.1.4	Integration Testing	136
6.2	Role and Permission Management.....	137
6.2.1	Unit Testing.....	137
6.2.2	Functional Testing.....	138
6.2.3	Business Rule Testing.....	138
6.2.4	Integration Testing	138
6.3	Human Resource Management	139
6.3.1	Unit Testing.....	139
6.3.2	Functional Testing.....	139
6.3.3	Business Rule Testing.....	140
6.3.4	Integration Testing	140
6.4	Task Management	141
6.4.1	Unit Testing.....	141
6.4.2	Functional Testing.....	142
6.4.3	Integration Testing	142
6.5	Dashboard and Reporting.....	143
6.5.1	Unit Testing.....	143
6.5.2	Functional Testing.....	143

6.5.3	Integration Testing	144
6.6	Billing and Financing.....	144
6.6.1	Unit Testing.....	144
6.6.2	Functional Testing.....	144
6.6.3	Integration Testing	145
6.7	Chatbot.....	145
6.7.1	Unit Testing.....	145
6.7.2	Functional Testing.....	146
6.7.3	Integration Testing	146
6.8	Order and Inventory	147
6.8.1	Unit Testing.....	147
6.8.2	Functional Testing.....	147
6.8.3	Integration Testing	148
7	Conclusion and Future Work.....	149
7.1	Conclusion	149
7.2	Future Work	149
8.	References	151
9.	Turnitin Report.....	152

List of Figures

Figure 3-1 Use case Diagram for Admin.....	10
Figure 3-2 Use case Diagram for Admin.....	11
Figure 3-3 Use case diagram for Admin and User	12
Figure 4-1 MVC Diagram of ModuFlex ERP System.....	92
Figure 4-2 Login Activity Diagram	93
Figure 4-3 Edit profile Activity diagram	94
Figure 4-4 Delete User Account Activity Diagram	95
Figure 4-5 Role and Permission Activity Diagram.....	96
Figure 4-6 Integration and development Activity Diagram.....	97
Figure 4-7 Create new user profile Activity Diagram	98
Figure 4-8 Submit and track expense report Activity Diagram.....	99
Figure 4-9 Leave Request Activity Diagram	100
Figure 4-10 Timesheets Activity Diagram	101
Figure 4-11 Generate Reports Activity Diagram.....	102
Figure 4-12 Dashboard creation Activity Diagram	103
Figure 4-13 Invoices and payment Activity Diagram.....	104
Figure 4-14 Financial Inputs Activity Diagram.....	105
Figure 4-15 Task management Activity Diagram	106
Figure 4-16 Order Management Activity Diagram	107
Figure 4-17 Context Diagram	108
Figure 4-18 User and Role Management Level 1 DFD.....	109
Figure 4-19 Development and HR management Level 1 DFD	110
Figure 4-20 Login Level 2 DFD	111
Figure 4-21 Integration and Development Level 2 DFD.....	112
Figure 4-22 HR management Level 2 DFD.....	113
Figure 4-23 Dashboard Level 2 DFD	114
Figure 4-24 Task and billing Management Level 2 DFD.....	115
Figure 4-25 STATE TRANSITION DIAGRAM user authentication.....	116
Figure 4-26 STATE TRANSITION DIAGRAM database schema generation.....	117
Figure 4-27 STATE TRANSITION DIAGRAM submit and track expense report	118
Figure 4-28 STATE TRANSITION DIAGRAM timesheets and reports	119
Figure 4-29 STATE TRANSITION DIAGRAM managing tasks	120
Figure 4-30 STATE TRANSITION DIAGRAM Event Scheduling.....	121
Figure 4-31 STATE TRANSITION DIAGRAM Predictive Analytics.....	122
Figure 4-32 STATE TRANSITION DIAGRAM Inventory management	123
Figure 4-33 Login Mockup Screen	129
Figure 4-34 CRUD Generator Mockup Screen	129
Figure 4-35 Dashboard Mockup Screen	130
Figure 4-36 Stripe Mockup Screen	130
Figure 4-37 Role Mockup Screen	131
Figure 4-38 Expense Report Mockup Screen	131

1 Introduction

In the current business technology environment, organization often faces slowdowns due to the repetitive coding of similar functions across inflexible and expensive management systems. The ModuFlex ERP system is to provide an efficient and customizable enterprise resource planning solution specifically designed for retailers and e-commerce businesses that aim to simplify the complexities of traditional management software. By incorporating key features such as user management, financial tracking, integrated product cataloging, inventory control, order management, code generation, and customizable dashboards, ModuFlex ERP seeks to streamline core retail and online sales operations, enhance productivity, and minimize development costs. The system's primary goal is to enable businesses both with physical stores and online presence to easily configure and manage their operations without any complexity. Ultimately, ModuFlex ERP aims to foster efficiency, flexibility, and cost-effectiveness for retailers and e-commerce merchants operating in a competitive marketplace.

1.1 Vision Statement

*For businesses **who** want to streamline their management processes and reduce the overhead associated with custom software development, **the** ModuFlex ERP is web based application **that** offers extensive customization capabilities, scalable architecture, and user-friendly interfaces. **Unlike** traditional ERP systems that are rigid and expensive to adapt, **our product** (ModuFlex ERP) provides a flexible, cost-effective solution that allows users to adapt the system to their specific needs without extensive coding knowledge, significantly reducing development time and costs, while enhancing operational efficiency.*

1.2 Related System Analysis/Literature Review

Table 1.1 shows the Related System Analysis of ModuFlex ERP

Table 1.1 Related System Analysis with proposed project solution

Application Name	Weakness	Proposed Project Solution
Quick Admin Panel [1]	Provides pre-defined and already created module which user can add into his system	Our system provide the integration and development module which is used to create user defined module
Odoo ERP [2]	Limited flexibility in customizing dashboards and reporting features to meet unique business requirements	ModuFlex ERP allows users to fully customize dashboards and real-time reporting according to specific user roles and needs.
SAP Business One [3]	Complex implementation process, requiring extensive	ModuFlex ERP simplifies implementation through intuitive modules and automated

	technical knowledge and high initial setup costs	integration tools, significantly reducing complexity and costs
--	--	--

1.3 System Deliverables

- **Software Requirements Specification (SRS):** A comprehensive document detailing the functional and non-functional requirements of the ModuFlex ERP system, including user stories, use cases, and business objectives.
- **Software Design Specification (SDS):** A technical document outlining the system architecture, database design, API integrations, and module-specific designs.

1.3.1 Application Modules

The application will include the following five modules:

- **User Management Module:**
 1. User registration, login, and authentication.
 2. User profile management (create, update, delete).
- **Role and Permission Module:**
 1. Role-based access control (RBAC).
 2. Assigning and managing permissions for different user roles.
- **Dashboard and Reporting Module:**
 1. Interactive dashboard with key metrics and visualizations.
 2. Customizable reports and export functionality.
- **Task Management Module:**
 1. Create, assign, and track tasks.
 2. Task prioritization, deadlines, and status updates.
- **Human Resource Management Module:**
 1. Employee data management.
 2. Leave management, attendance tracking, and performance evaluation.

1.4 System Limitations/Constraints

LI-1: Due to the modular nature of the system, the integration complexity between different modules might vary, requiring additional customization or configuration efforts to ensure seamless functionality.

LI-2: The ModuFlex ERP is designed primarily for small to medium-sized enterprises; thus, the system might not scale adequately or provide specific functionalities required by large corporations without significant customization.

LI-3: While the system will support a broad range of industries, highly specialized sectors might require development of custom modules which could extend the implementation timeline and increase costs.

LI-4: The system's performance, particularly in terms of real-time data processing, may be limited by the underlying hardware and network capabilities of the user environment.

LI-5: The Generate Code module, while powerful, requires users to have a basic understanding of system architecture and design principles to fully utilize its capabilities, which might be a barrier for some non-technical users.

1.5 Tools and Technologies

Table 1.2 shows the Tool and Technology of ModuFlex ERP

Table 1.2: Tools and Technologies for Proposed Project

Tools And Technologies	Tools	Version	Rationale
	MS Visual Studio	2022	IDE
	MS SQL Server	2022	DBMS
	Canva	Latest	Design Work
	Technology	Version	Rationale
	React JS	18	Front-end Development
	Express.JS & Node JS	3 & 22	Back-end Development
	Python	3.11	Chatbot Development

1.6 Relevance to Course Modules

The **ModuFlex ERP System** project is highly relevant to the courses studied during the BSE program, particularly **Web Application Development using the MERN stack**, **Database Management with SQL**, and **Artificial Intelligence (AI)**. The project is being developed using the **MERN stack** (Express.js, React.js, Node.js), which aligns with the skills acquired in the web application course. The frontend, built with **React.js**, provides a dynamic and responsive user interface for modules like customizable dashboards and reporting, while **Node.js** and **Express.js** handle the backend logic, enabling seamless communication between the frontend and backend through RESTful APIs. Although MongoDB is part of the MERN stack, the project also incorporates **MySQL**.

2 Problem Definition

In today's fast-paced and highly competitive business environment, organizations face significant challenges in managing their operations efficiently. Many companies rely on disparate systems for various functions such as finance, human resources, supply chain, inventory, and customer relationship management. This approach often leads to inefficiencies, data inconsistencies, poor decision-making, and increased operational costs. To address these issues, there is a growing need for an integrated Enterprise Resource Planning (ERP) system that can streamline processes, enhance data visibility, and improve overall organizational performance. The primary problem is the lack of a unified platform that can consolidate all business processes into a single, cohesive system. Existing legacy systems are often outdated, inflexible, and unable to scale with the organization's growth. Additionally, manual data entry and redundant processes result in errors, delays, and inefficiencies. The absence of real-time data access further hampers the ability of decision-makers to respond quickly to market changes or internal issues.

2.1 Problem Solution

The ModuFlex ERP system addresses the inefficiencies of fragmented business processes by integrating all core functions like task, HR, inventory, and user management into a single, unified platform. This eliminates data problems, ensuring consistency and accuracy across the organization. By automating routine tasks and reducing manual data entry, the ModuFlex ERP systems minimize errors, save time, and improve operational efficiency. Real-time data access enables decision-makers to respond quickly to market changes, enhancing agility and competitiveness. Additionally, The ModuFlex ERP systems provide advanced dashboard and reporting modules, offering actionable insights for better resource allocation and strategic planning. They streamline workflows, reduce operational costs, and improve collaboration across departments. With scalable and customizable features, the ModuFlex ERP systems adapt to the organization's growth and evolving needs. By centralizing and optimizing operations, the ModuFlex ERP solutions empower businesses to enhance productivity, customer satisfaction, and overall performance, driving long-term success.

2.2 Objectives of the Proposed System

BO-1: Reduce development time for management systems by streamlining the process of coding and deployment through the use of modular components.

BO-2: Facilitate the transition for businesses from manual record-keeping to digital solutions, enhancing operational efficiency and accuracy while minimizing the administrative burden associated with manual processes.

BO-3: Reduce the costs associated with custom software development by providing a scalable and easily configurable modular system.

BO-4: Increase in productivity by integrating real-time data processing and updates, allowing users to make informed decisions swiftly.

2.3 Scope

The scope of the ModuFlex ERP project focuses on delivering a modular enterprise resource planning system tailored for the retail and e-commerce sectors. Key modules include Billing and Financing, allowing businesses to manage income and expenses while maintaining accurate financial records; Dashboard and Reporting, enabling users to customize their dashboards and generate insightful reports on sales performance, inventory status, and customer behavior. Role and Permission capabilities ensure that each user has access only to the tools and data they need, while the Integration and Development module allows for rapid creation of industry-specific features such as product and catalog management. Through the Human Resource Management (HRM) module, businesses can oversee employee profiles, track leave requests, manage expenses, and maintain productivity metrics. Task and Notification Management ensures that key deadlines, promotions, or inventory alerts never go unnoticed, while the system's cloning capability facilitates quick deployment of similar configurations across multiple store locations or online ventures. By centralizing all essential functions into a cohesive, adaptable platform, ModuFlex ERP delivers a scalable and responsive solution designed to enhance operational efficiency, support data-driven decisions, and ultimately boost customer satisfaction and profitability in the retail and e-commerce landscape.

2.4 Modules

2.4.1 Module 1: User Management

FE-1: Users can securely log in to the system using their credentials (username/email and password).

FE-2: Users can log out from the system, terminating their active session.

FE-3: Users can edit and update their profile information, including personal details, contact information, and profile picture.

FE-4: Users can change their password using the current password and a secure password update form.

FE-5: Users can reset their password via email if they forget their login credentials.

FE-6: Admins can deactivate or delete user accounts when needed.

2.4.2 Module 2: Role and Permission Management

FE-1: Control and assign different levels of access and permissions for system users based on roles.

FE-2: Admin can create new user roles and assign them to users, defining their access to specific modules and data.

FE-3: Admin can decide the role of each user and assign them a customized menu and dashboard tailored to their responsibilities.

FE-4: Admin can create different dashboards and menus for each user role, ensuring they see only relevant information and tools.

2.4.3 Module 3: Development and Integration Module

FE-1: Users can create new modules by defining a database schema. This involves filling out forms and selecting choices that specify fields, data types, and other attributes.

FE-2: The system integrates the generated code into the existing system, automatically providing the necessary database schema and configurations.

FE-3: The module ensures that all generated modules are consistent with the provided schema, maintaining data integrity and system coherence.

2.4.4 Module 4: Human Resource Management (HRM)

FE-1: Admin can create and manage employee profiles, including personal information, job roles, and departments

FE-2: Employees can submit and track expense reports with detailed categorizations and receipt attachments.

FE-3: Employees can apply for and track leave requests

FE-4: Concerned authority can review, approve, or reject expense reports, leave requests.

FE-5: Users can receive automated alerts and notifications for pending approvals, leave status, and timesheet deadlines.

2.4.5 Module 5: Dashboard and Reporting

FE-1: Users can generate real-time reports based on specific criteria (e.g., financial performance, sales trends, and employee productivity).

FE-2: Admins can create and assign different dashboards for different user roles, ensuring each user sees tailored data.

FE-3: Dashboards support interactive visualizations (e.g., charts, graphs, tables) that allow users to drill down into specific data points.

FE-4: Dashboard allows users to customize the layout and content according to their needs, and visualizations are kept updated with real-time data to reflect the most current performance metrics.

2.4.6 Module 6: Billing and Financing

FE-1: Users can create invoices for their products or services, ensuring accurate billing records.

FE-2: Users can either download the receipt or make payments through a payment gateway, tracking the status of each transaction.

FE-3: Users can monitor the status of open invoices, receiving notifications and alerts for overdue payments.

FE-4: Users can manage the records of expenses and income, categorizing them to enhance financial tracking and analysis

2.4.7 Module 7: Task and Notification Management

FE-1: Users can create and manage a to-do list with tasks, setting priorities and deadlines.

FE-2: Users can schedule events, tasks, or appointments in a calendar view, with the ability to add recurring events.

FE-3: Users can set reminders for tasks or events, and the system will send notifications via email or in-app alerts.

2.4.8 Module 8: Order and Inventory Management

FE-1: Centralize and manage all product information (names, images, SKUs, variants, categories) in a single repository.

FE-2: Monitor stock levels in real time across multiple warehouses and storage locations.

FE-3: Configure automated reorder points and safety stock thresholds, receiving alerts when inventory runs low.

FE-4: Capture orders from various channels (e.g., online storefronts, POS systems) for a unified view of all sales orders.

FE-5: Manage order fulfillment and payment workflows

FE-6: Handle order status tracking and exchanges seamlessly, maintaining accurate records and ensuring customer satisfaction.

2.4.9 Module 9: Chatbot

FE-1: The chatbot dynamically retrieves real-time data such as attendance summaries, inventory status, sales performance, and task progress directly from the database, facilitating timely business decisions.

FE-2: Users can query data through simple, natural-language prompts (e.g., "attendance summary between dates" or "best-selling products this month"), making data access intuitive without technical complexity.

2.5 Project Contribution

ModuFlex ERP introduces significant improvements over conventional ERP solutions by offering a modular, customizable, and AI-enhanced approach to enterprise management. Below are the core innovations:

1. Modular & No-Code ERP Customization

- **Dynamic Module Creation:** Users can build new ERP modules (e.g., inventory, HR, finance) without coding by defining database schemas via forms or Excel uploads.

- **Instant System Integration:** New modules automatically integrate with existing workflows, APIs, and dashboards, eliminating manual development.

2. Unified Multi-Channel Business Management

- **Centralized POS & E-Commerce:** Combines in-store (POS), online sales, and inventory in one system for real-time synchronization.
- **Omnichannel Order Processing:** Captures orders from all sales channels (web, mobile, in-person) into a single dashboard.

3. AI-Driven Automation & Insights

- **Smart Chatbot for Analytics:** Translates sales forecasts, inventory trends, and error metrics into plain-language business insights.

6. Real-Time Inventory & Warehouse Control

- **Multi-Warehouse Tracking:** Monitors stock levels across locations with low-stock alerts.

3 Requirement Analysis

3.1 User classes and characteristics

Table 3.1 shows user classes and characteristics for ModuFlex ERP

Table 3.1 User classes and characteristic for ModuFlex ERP

User classes	Description
Admin	Administrators have comprehensive control within the ERP system. Their responsibilities include managing user profiles, roles, and permissions; approving and managing financial transactions like invoices and receipts; generating and exporting reports for various business needs; defining and importing schemas, as well as generating code for system functionality; customizing user interfaces and dashboards to enhance user experience; and handling system-wide authentication and login processes.
User	Regular users engage with the ERP system to execute daily operational tasks. This includes logging into the system; updating personal profiles and changing passwords; submitting requests such as expense reports, leave applications, and timesheets; recording financial transactions including incomes and expenses; managing their personal calendar with events and reminders; providing feedback to improve system functionalities; and receiving confirmations and notifications regarding their transactions and requests.
Bank	The Bank acts as an external entity that interfaces with the ERP system primarily for financial transaction processing. It handles receiving transaction requests from the ERP system, providing confirmations of payments, and sending back transaction details and responses to ensure accurate and secure financial operations within the ERP system.

3.2 Requirement Identifying Technique

3.2.1 Use Case Diagram

Use Case diagrams for all actors are provided below:

Figure 3.1 shows the use cases performed by User in ModuFlex ERP

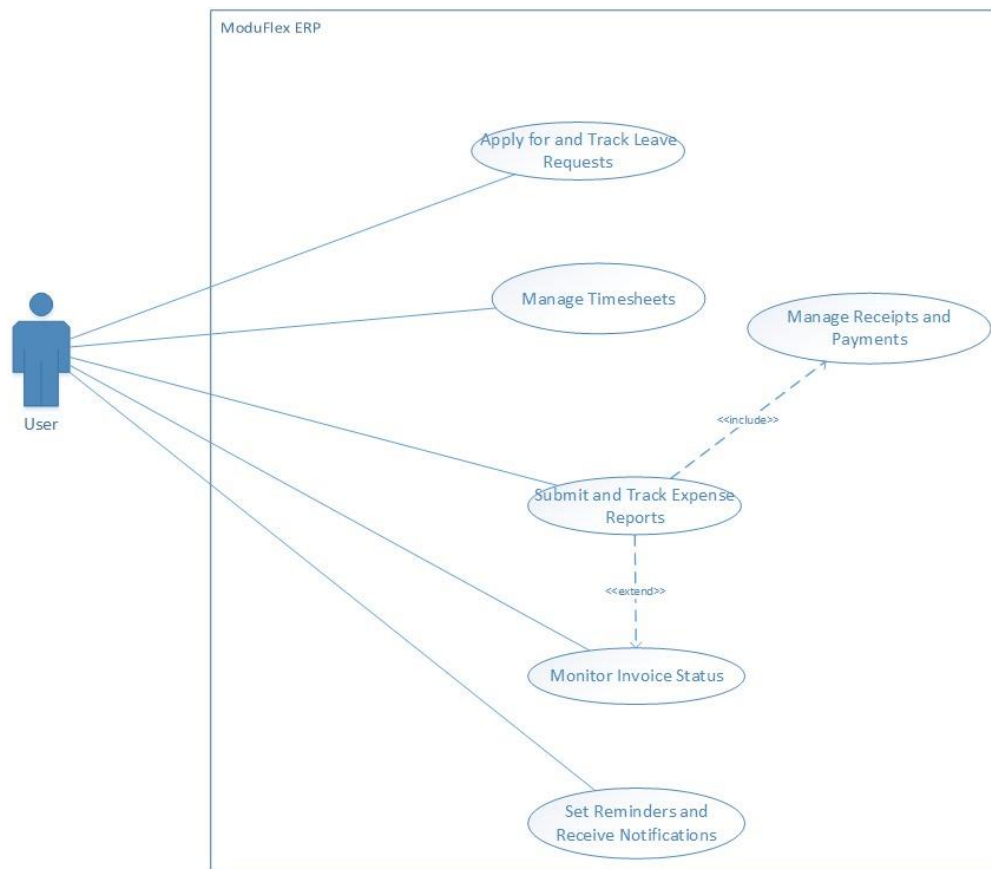


Figure 3-1 Use case Diagram for Admin

Figure 3.2 shows the use cases performed by admin in ModuFlex ERP

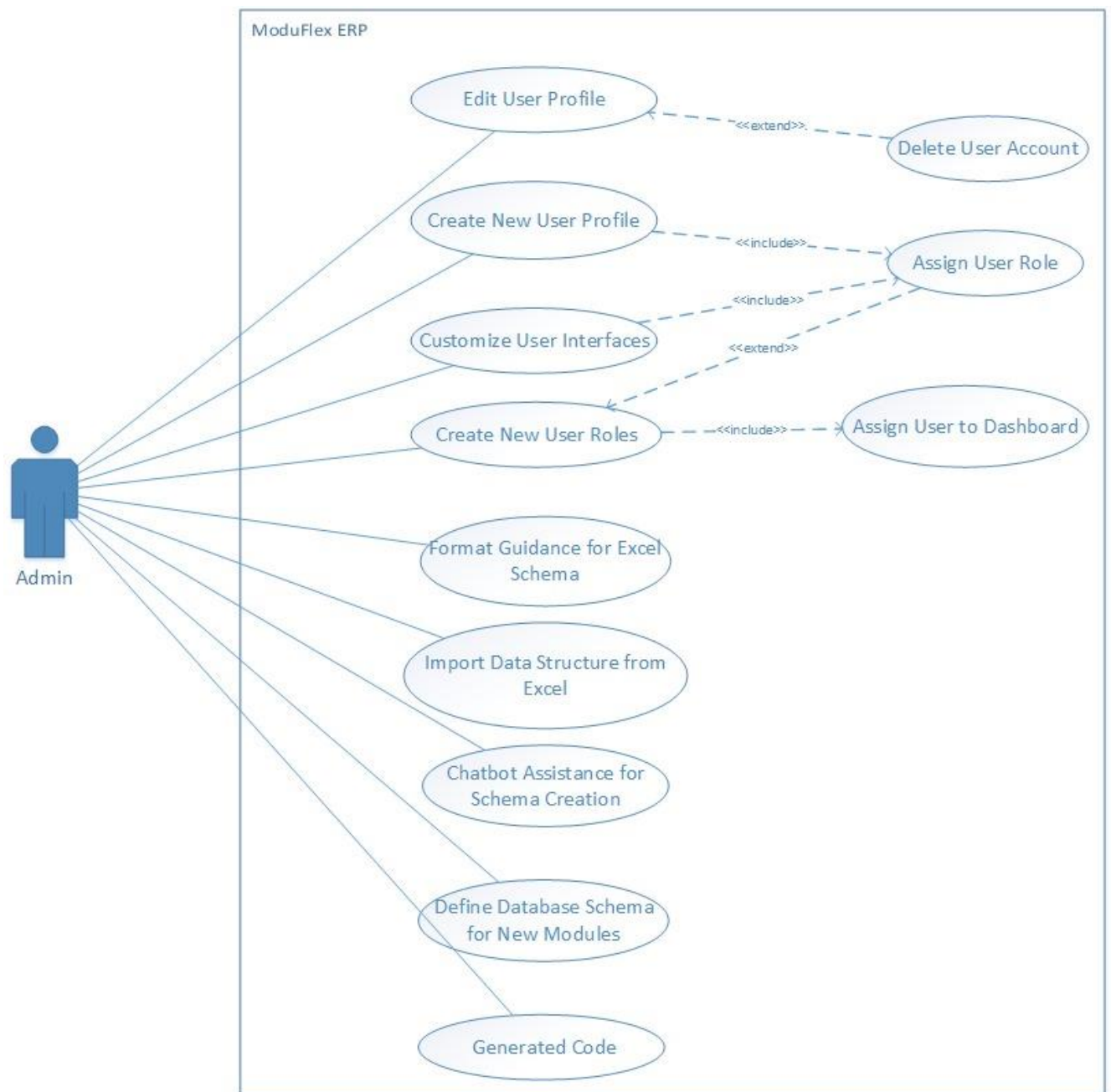


Figure 3-2 Use case Diagram for Admin

Figure 3.3 shows the diagram of combine use cases performed by admin and user in ModuFlex ERP

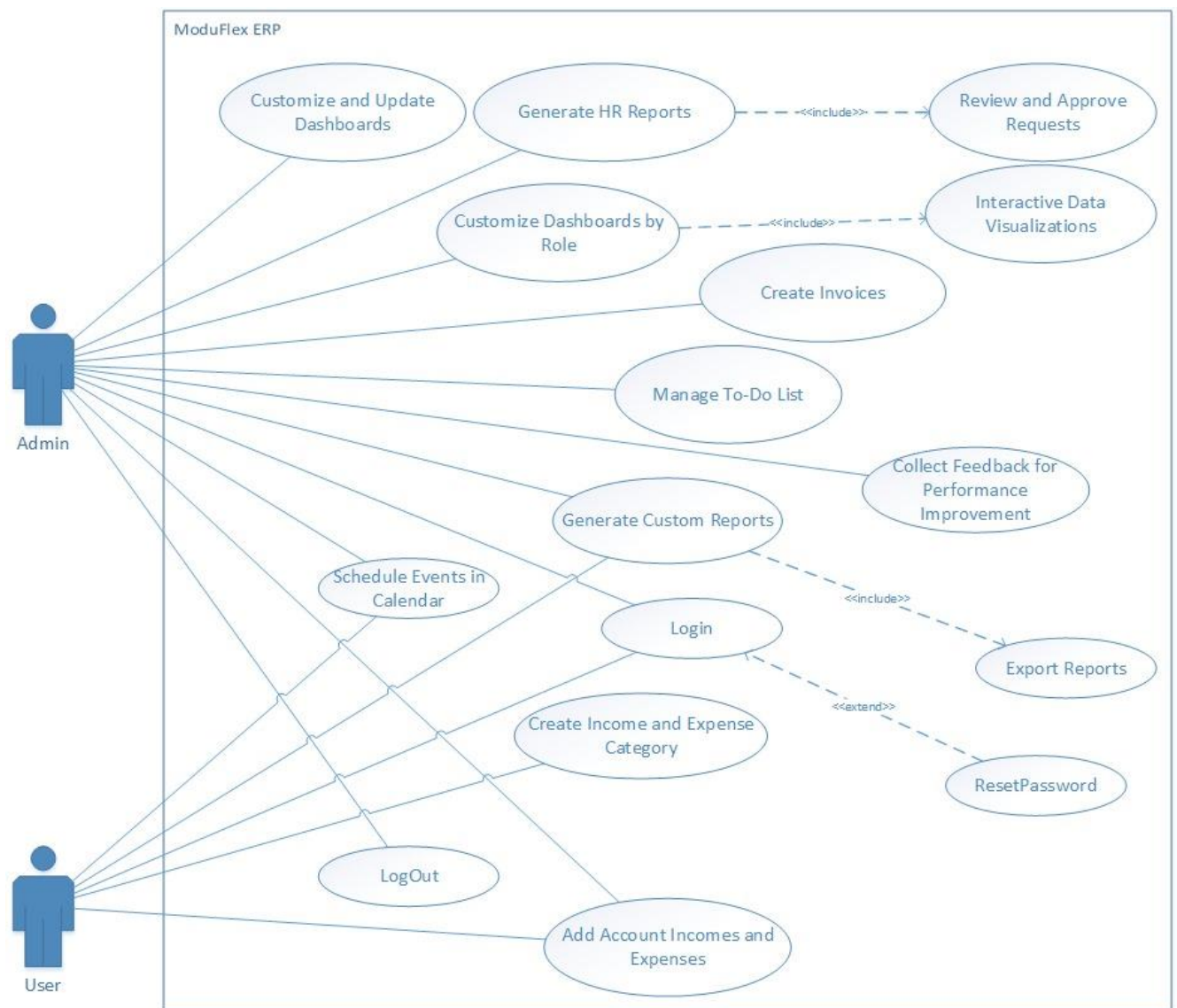


Figure 3-3 Use case diagram for Admin and User

3.2.2 Detailed Use cases

3.2.2.1 Module 1: User Management

3.2.2.1.1 UC-01: Login

The user wants to access the ERP system by providing valid credentials.

Actors	Primary Actor: User, Admin
Trigger	The user indicates the intention to log in by navigating to the login page.
Preconditions	PRE-1: The user account must already exist in the system. PRE-2: The user must have valid login credentials (username/email and password).
Postconditions	POST-1: The user is successfully authenticated and logged into the system. POST-2: The user is redirected to the dashboard or home page after successful login.
Normal Flow	1. User navigates to the login page. 2. User enters their username/email and password. 3. The system verifies the credentials. 4. The user is granted access to the system. 5. The system redirects the user to the dashboard/home page.
Alternative Flows	2b: The user forgets their password and chooses to recover it. 1. The user clicks on the "Forgot Password" link. 2. The system prompts the user to enter their registered email address. 3. The system sends a password recovery link or code to the user's email. 4. The user follows the instructions to reset their password. 5. The flow returns to step 2 of the normal flow once the password is reset.
Exceptions	2a: The user enters an incorrect username/email or password. 1. The system displays an error message indicating that the credentials are incorrect. 2. The user is prompted to re-enter their credentials or recover their password. 3. The flow returns to step 2 of the normal flow.
Business Rules	BR-1: Passwords must be at least 8 characters long.
Assumptions	It is assumed that users are familiar with the login process and understand the error messages provided. It is assumed that users have access to the email account associated with their user profile for password recovery.

3.2.2.1.2 UC-02: Edit User Profile

The user wants to update personal or professional information in their profile.

Actor	Primary Actor: User, Admin
Trigger	The user initiates the profile edit by navigating to the profile settings page.
Preconditions	PRE-1: The user must be logged into the system. PRE-2: User clicked on edit profile button.
Postconditions	POST-1: The user's profile is successfully updated with the new information.
Normal Flow	1. User navigates to the user management profile page. 2. User edits the required fields. 3. System validates the provided information. 4. System saves the changes to the user's profile.
Alternative Flows	N/A
Exceptions	3a: The user provides invalid value in the field. 1. The system displays an error message specifying the issue with the provided information. 2. The user is prompted to correct the information and resubmit. 3. Return to step 2 of the normal flow.
Business Rules	BR-1: The system should validate all fields according to the defined input criteria before saving the profile. BR-2: The system should ensure that any uploaded data meet the required format and size constraints.
Assumptions	1: It is assumed that the user has the necessary permissions to edit their profile. 2: It is assumed that the system provides feedback to the user in case of any validation errors during profile editing.

3.2.2.1.3 UC-03: Change Password

The user wants to change its current password and add a new password.

Actor	Primary Actor :User, Admin
Trigger	The user initiates the process by navigating to the change password page.
Preconditions	PRE-1: The user must be logged into the system. PRE-2: The user must know their current password.
Postconditions	POST-1: The user's password is successfully changed and updated in the system.

Normal Flow	1. User navigates to the change password page. 2. User enters the current password and the new password. 3. System verifies the current password for accuracy. 4. System validates the new password (e.g., length, complexity). 5. System updates the password in the system.
Alternative Flows	N/A
Exceptions	3a: The user enters an incorrect current password. 1. The system displays an error message indicating that the current password is incorrect. 2. The user is prompted to re-enter the current password. 3. Return to step 2 of the normal flow. 3b: The user enters a new password that does not meet the required criteria. 1. The system displays an error message specifying the issue with the new password. 2. The user is prompted to enter a valid new password. 3. Return to step 2 of the normal flow.
Business Rules	BR-1: Passwords must be at least 8 characters long.
Assumptions	1: It is assumed that the user has a valid current password. 2: It is assumed that the system will provide immediate feedback on password validity.

3.2.2.1.4 UC-04: Delete User Account

The admin can delete the user account.

Primary Actor	Admin
Trigger	The Admin decides to delete a user account.
Preconditions	PRE-1: Admin must be logged in and have the necessary permissions to deactivate or delete user accounts. PRE-2: Admin must be in User Management page and have list of user and their role. PRE-3: User must be included in the user list.
Postconditions	POST-1: The user account is either deactivated or deleted from the system, depending on the action taken by the Admin.
Normal Flow	1. Admin navigates to the user management page. 2. Admin selects a user account to delete. 3. Admin confirms the action.

	4. System deletes the user account. 5. System provides a confirmation message to the Admin about the successful deactivation or deletion of the account.
Alternative Flows	3a. if user did not confirm and press “NO” button 1. System will move to step 2 of normal flow
Exceptions	4a. he selected user account is not found in the system. 1. The system displays an error message indicating that the account is not found. 2. Admin is prompted to reselect a valid user account. 3. Return to step 2 of the normal flow. 4b: Admin chooses to deactivate the user account instead of deleting it. 1. System deactivates the account, disabling the user's ability to log in but retaining the account in the system. 2. Admin is informed of the successful deactivation
Business Rules	BR-1: Deactivating an account should retain the user's data for audit purposes, while deletion should remove all personal data in accordance with data protection regulations.
Assumptions	1: It is assumed that the Admin has verified the need to deactivate or delete the account before proceeding. 2: It is assumed that the system correctly identifies and processes the account selected by the Admin.

3.2.2.1.5 UC-05: Logout

The user wants to end their session and logout of the system

Actor	Primary Actor: Admin, User
Trigger	The user indicates the intention to log out by clicking the "Logout" button.
Preconditions	PRE-1: The user must be logged into the system.
Postconditions	POST-1: The user is redirected to the login page.
Normal Flow	1. User clicks on the "Logout" button. 2. System displays a prompt message “are you sure to logout “. 3. User clicks “Yes”
Alternative Flows	3a. In step 3 of the normal flow, if the user clicks “No”, 3. The user will be re-directed to the dashboard
Exceptions	N/A
Business Rules	N/A

Assumptions	1: It is assumed that users will log out after completing their tasks for security reasons.
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3.2.2.2 Module 2: Role and Permission Management

3.2.2.2.1 UC-06: Customize User Interfaces

Admin wants to create customize menus and dashboard for users

Section	Details
Actors	Primary Actor: Admin
Trigger	The admin needs to create a custom dashboard template and menus for specific roles.
Preconditions	PRE-1: The admin is logged into the ERP system. PRE-2: The user for whom dashboard and menus is created is registered into the system
Postconditions	POST-1: A custom dashboard template is created and assigned to a role or user group.
Normal Flow	1. Admin navigates to role and permission . 2. Admin selects the option to assign to users 3. Admin will select the role registered 4. Admin will select dashboard 4. Admin will select the dashboard item that he wants to show to that selected user 5. Admin clicks the submit button
Alternative Flows	4a. if user selects menus option: 1. System will display him the available menu 2. Admin will select the menus he wants to show to user. 3. Admin clicks the save button
Exceptions	3a. if user selects role that is already assigned 1. System will prompt user that it is already assigned.
Business Rules	BR-1: Only admins can create and assign dashboards and menu. BR-2: The custom dashboard and menu must only be assigned to registered users.
Assumptions	1. The ERP system has sufficient capacity to create and manage multiple dashboard templates and menu. 2. Roles and user groups are already established in the system.

3.2.2.2.2 UC-07: Create New User Roles

Admin can create new user roles in the system and assign it to user for role defining purpose.

Section	Details
Actors	Primary Actor: Admin
Trigger	The admin needs to create a new user role for managing access levels and permissions.
Preconditions	PRE-1: The admin is logged into the ERP system. PRE-2: Admin has permissions to manage roles and user access.
Postconditions	POST-1: A new user role is created. POST-2: The role is available for user assignment.
Normal Flow	1. Admin navigates to user profile. 2. Admin selects User Role. 3. Admin will select the add new role. 4. Admin will select the name of the role. 5. Admin will put dashboard URL to show to that specific user 6. Admin will click the submit button to save the data
Alternative Flows	4a: The role name entered by the admin already exists in the system: 1. The system prompts the admin that the role name is already in use. 2. Admin changes the role name and returns to step 4. 6a: Admin selects "Save and Add More" to add multiple roles in one session: 1. The system saves the current role and allows the admin to return to step 3 to add another role.
Exceptions	1.0. E1: System encounters an error during the role creation process, and the role is not saved. 1. The system logs the error and notifies the admin, allowing them to retry the process or contact support.
Business Rules	BR-1: Only admins can create and assign new roles. BR-2: Role names must be unique across the system.
Assumptions	Admin has sufficient knowledge of system permissions.

3.2.2.2.3 UC-08: Assign User to Dashboard

Admin can assign a specific user to a pre-configured dashboard

Section	Details
Actors	Primary Actor: Admin

Trigger	The admin wants to assign a specific user to a pre-configured dashboard.
Preconditions	PRE-1: The admin is logged into the ERP system. PRE-2: The user is registered into the system PRE-3: The Dashboard is already created into the system
Postconditions	POST-1: The user is assigned to the specified dashboard. POST-2: The user's dashboard is updated.
Normal Flow	1. Admin navigates to user Profile 2. Admin selects a user. 3. Admin selects "Assign Dashboard". 4. System shows available dashboards. 5. Admin assigns the dashboard. 6. Admin will click the save button.
Alternative Flows	4a: No dashboards are available or the list of dashboards is empty: 1.The system notifies the admin that no dashboards are available for assignment. 2.Admin must create or configure a dashboard first, then return to step 3. 5a: Admin attempts to assign a dashboard to a user who already has one assigned: 1. The system prompts the admin to confirm if the existing dashboard should be replaced. 2.Admin can confirm the replacement or cancel the operation, returning to step 4.
Exceptions	1.0. E1: The system fails to retrieve the list of available dashboards due to a database error: 1. The system displays an error message and prevents further action until the issue is resolved.
Business Rule	BR-1: Only admins can assign users to dashboards. BR-2: A user must be registered and have a valid profile before assigning dashboard BR-3: The system should ensure that a user can only have one active dashboard at a time.
Assumptions	All dashboards are pre-configured and ready for assignment.

3.2.2.2.4 UC-09: Assign User Role

Admin can assign already added role in the system to user for its permission and restriction.

Section	Details
Actors	Primary Actor: Admin
Trigger	The admin wants to assign a specific role to a user within the ERP module.
Preconditions	PRE-1: The admin is logged into the ERP system. PRE-2: The user is registered in the system. PRE-3: The role is already created and available within the system.
Postconditions	POST-1: The user is assigned the specified role within the ERP module. POST-2: The user's access and permissions are updated according to the role.
Normal Flow	1. Admin navigates to the "User Management" section. 2. Admin selects the user from the list. 3. Admin selects the "Assign Role" option. 4. System displays available roles. 5. Admin selects the appropriate role to assign. 6. Admin clicks the "Save" button. 7. System updates the user's access and permissions.
Alternative Flows	4a: No roles are available or the list of roles is empty: 1. The system notifies the admin that no roles are available for assignment. 2. Admin must create or configure a role first, then return to step 3. 5a: Admin attempts to assign a role that the user already holds: 1. The system notifies the admin that the user already has this role assigned. 2. Admin can either cancel the operation or assign additional roles by returning to step 4.
Exceptions	1.0. E1: The system encounters an error while assigning the role, and the assignment fails. 1. The system logs the error and notifies the admin. The admin can retry the operation or contact support. 1.0. E2: The system fails to retrieve the list of available roles due to a database error. 1. The system displays an error message and prevents further action until the issue is resolved.
Business Rules	BR-1: Only admins can assign roles to users.

Assumptions	The ERP system supports role-based access control (RBAC) and handles permission updates in real-time.
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3.2.2.3 Module 3: Integration and Development

3.2.2.3.1 UC-10: Define Database Schema for New Modules

Admin can create a database schema for its new module.

Actor	Primary Actor : Admin
Trigger	The admin decides to create a new module by defining its database schema.
Preconditions	PRE-1: Admin must be logged into the system. PRE-2: Admin in on Integration and Development page
Postconditions	POST-1: The database schema for the new module is successfully defined and stored.
Normal Flow	1. Admin navigates to the Integration and Development page. 2. Admin fills out the form to specify fields, data types, and relationships. 3. Admin saves the database schema. 4. System validates the schema and confirms creation.
Alternative Flows	2b: The Admin want to edit created schema. 1. Admin edits the specific schema table or column 2. System moves to step 2 of normal flow
Exceptions	2b: The user enters invalid details schema . 1. The system displays an error message. 2. The user is prompted to correct the invalid entries.
Business Rules	BR-1: Fields must have defined data types, and relationships must comply with database normalization rules.
Assumptions	1: It is assumed that the user has sufficient knowledge of database design principles.

3.2.2.3.2 UC-11: Import Data Structure from Excel

Admin can export excel file to import multiple table at a time for module creation

Actor	Primary Actor :Admin
Trigger	The user decides to import data from an Excel sheet.
Preconditions	PRE-1: Admin must be logged into the system. PRE-2: Admin must have a valid Excel sheet ready for upload.

Postconditions	POST-1: The database tables are created based on the Excel sheet's structure and data.
Normal Flow	1. Admin navigates to the Integration and Development page. 2. Admin uploads the Excel sheet. 3. System reads the structure and data from the sheet. 4. Admin is notified of successful import.
Alternative Flows	N/A
Exceptions	3a: The uploaded Excel sheet does not match the required format. 1. The system displays an error message. 2. The user is prompted to upload a valid Excel sheet. 3. The control goes to normal flow 2.
Business Rules	BR-1: The Excel sheet must follow a predefined structure to ensure successful table creation.
Assumptions	1: It is assumed that the user is aware of the correct Excel format for data import.

3.2.2.3.3 UC-12: Generated Code

The admin can create a module from schema and integrate it into the system.

Actor	Primary Actor : Admin
Trigger	The system finishes generating new modules or importing data structures.
Preconditions	PRE-1: Admin must be logged into the system. PRE-1: New modules must have been successfully defined or imported.
Postconditions	POST-1: The generated code is integrated into the existing system, and the necessary database schema is configured.
Normal Flow	1. The admin clicked on generate code button. 2. The system detects that new modules have been created or imported. 3. The system integrates the generated code into the software. 4. The system configures the necessary database schema. 5. The system notified of successful integration.
Alternative Flows	N/A
Exceptions	2a: The system encounters an issue during the integration process. 1. The system displays an error message. 2. The integration process is rolled back, and users are notified of the issue. 3. The controlled backed to normal flow 2.
Business Rules	N/A

Assumptions	1: It is assumed that the system has access to the necessary resources for integration.
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3.2.2.4 Module 4: Human Resource Management (HRM):

3.2.2.4.1 UC-13: Create New User Profile

Admin creates new User profiles, entering personal information, job roles, and department assignments.

Attribute	Details
Actors	Primary: Admin
Trigger	Admin needs to add a new User to the system.
Preconditions	PRE-1: Admin is logged into the system. PRE-2: Admin has the necessary permissions to create new User profiles.
Postconditions	POST-1: New User profile is created in the system. POST-2: System logs the creation action.
Normal Flow	1. Admin opens the HR management section. 2. Admin navigates to the User management section. 3. Admin selects the option to create a new profile. 4. Admin enters personal information, job roles, and department assignments for the new User. 5. Admin submits the information. 6. System validates and saves the new profile. 7. System displays a confirmation message.
Alternative Flows	N/A
Exceptions	4a. In step 4, if invalid data is entered or required fields are left blank: 1. System highlights the fields in red. 2. System prompts the admin to correct the information. 3. Use case resumes at step 4.
Business Rules	BR-1: Only authorized admins can create new User profiles.
Assumptions	Assume all input data is validated real-time against existing database constraints.

3.2.2.4.2 UC-14: Edit User Profile

Admin edits existing User profiles, updating personal information, job roles, and department assignments.

Attribute	Details
Actors	Primary: Admin
Trigger	Admin needs to update existing User information.
Preconditions	PRE-1: Admin is logged into the system. PRE-2: Admin has the necessary permissions to edit User profiles.
Postconditions	POST-1: User profile is updated in the system. POST-2: System logs the update action.
Normal Flow	1. Admin opens the HR management section. 2. Admin navigates to the User management section. 3. Admin selects an User to edit. 4. Admin enters or updates personal information, job roles, and department assignments. 5. Admin submits the changes. 6. System validates and saves the updates. 7. System displays a confirmation message.
Alternative Flows	N/A
Exceptions	4a. In step 4, if invalid data is entered or required fields are left blank: 1. System highlights the fields in red. 2. System prompts the admin to correct the information. 3. Use case resumes at step 4.
Business Rules	BR-1: Only authorized admins can update job roles and departments.
Assumptions	Assume all input data is validated real-time against existing database constraints.

3.2.2.4.3 UC-15: Submit and Track Expense Reports

Users submit expense reports with categorizations and attachments, and track the approval status.

Attribute	Details
Actors	Primary: User
Trigger	User incurs an expense that needs to be reported.
Preconditions	PRE-1: User is logged into the system. PRE-2: User has receipts or proof of expense.
Postconditions	POST-1: Expense report is submitted and is pending approval. POST-2: Concerned person receives the expense report for review.
Normal Flow	1. User logs into the HR management section. 2. User navigates to the expense reporting section. 3. User selects the option to create a new expense report. 4. User enters expense details, categorizes the expense, and attaches

	required receipts. 5. User reviews the details and submits the report. 6. System sends the report to the concerned department. 7. System displays a confirmation message to the User. 8. User can track the status of the report through the system.
Alternative Flows	5a. In step 5, if the User realizes an error in the report: 1. User opts to edit the report. 2. Use case resumes at step 4.
Exceptions	4a. In step 4, if required fields are left blank or attachments are missing: 1. System highlights the missing fields or indicates missing attachments. 2. System does not proceed with submission. 3. User is prompted to complete the fields or attach documents. 4. Use case resumes at step 4.
Business Rules	BR-1: All expenses must be categorized according to company policy.
Assumptions	Assume Users are trained on using the expense report module.

3.2.2.4.4 UC-16: Apply for and Track Leave Requests

Users apply for leave and monitor the status of their leave requests through the system.

Attribute	Details
Actors	Primary: User
Trigger	User needs to request leave.
Preconditions	PRE-1: User is logged into the system.
Postconditions	POST-1: Leave request is recorded in the system. POST-2: Concerned person is notified of the leave request.
Normal Flow	1. User logs into the HR management system. 2. User navigates to the leave request section. 3. User selects the type of leave and inputs the dates. 4. User submits the leave request. 6. System updates the leave status and notifies concerned person 7. User receives confirmation and can track the leave status.
Alternative Flows	N/A
Exceptions	3a. if user input number of days more than specified days available for leave 1. System notifies that only that specific number of days' leave is allowed 2. Return to step 2 of normal flow
Business Rules	BR-1: Leave requests must be submitted at least 5 days in advance.

Assumptions	Assume that Users are aware of their leave balances and company leave policies.
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3.2.2.4.5 UC-17: Manage Timesheets

Users create and update timesheets on a daily, weekly, or monthly basis, recording regular and overtime hours worked.

Attribute	Details
Actors	Primary: User
Trigger	User needs to record work hours.
Preconditions	PRE-1: User is logged into the system.
Postconditions	POST-1: Timesheet is updated in the system. POST-2: System calculates working hours for User.
Normal Flow	1. User logs into the system. 2. User navigates to the timesheet section. 3. User enters or updates hours worked for the selected period. 4. User reviews and submits the timesheet. 5. System validates and saves the timesheet. 6. System displays confirmation to the User.
Alternative Flows	5a. In step 5, if User clicks save and add more: 1. System moves to step 3 of normal flow.
Exceptions	5a. In step 5, if there are errors in the timesheet: 1. System highlights the errors. 2. User corrects the timesheet. 3. Use case resumes at step 4.
Business Rules	BR-1: Timesheets must be completed by the end of each pay period.
Assumptions	Assume that Users have been trained on how to use the timesheet module.

3.2.2.4.6 UC-18: Review and Approve Requests

Admin can review, approve, or reject submitted expense reports and leave requests.

Attribute	Details
Actors	Primary: Admin
Trigger	An expense report or leave request is submitted.
Preconditions	PRE-1: Admin is logged into the system. PRE-2: There are pending requests to review.

Postconditions	POST-1: Requests are approved or rejected. POST-2: Users are notified of the decision.
Normal Flow	1. Admin logs into the HR management section. 2. Admin navigates to the approval section. 3. Admin reviews pending requests. 4. Admin approves or rejects requests. 5. System updates the request status and notifies Users.
Alternative Flows	4a. if Admin rejects the request 1. System open the section to write reason for rejection 2. Reason is then forwarded to user 3. System return step 5 of normal flow
Exceptions	4a. In step 4, if further information is needed: 1. User requests additional details from the User. 2. Use case is paused until information is provided. 3. Use case resumes at step 3.
Business Rules	BR-1: Only authorized personnel can approve or reject requests.
Assumptions	Assume Users are familiar with the approval process and policies.

3.2.2.4.7 UC-19: Generate HR Reports

Admin can generate detailed reports and analytics on User attendance, expenses, and overall workforce productivity.

Attribute	Details
Actors	Primary: Admin
Trigger	Need to review the performance of the user.
Preconditions	PRE-1: Admin is logged into the HR system.
Postconditions	POST-1: Reports are generated and available in the system
Normal Flow	1. Admin logs into the HR management section. 2. HR Analyst selects the reports section. 3. HR Analyst specifies the type of report and parameters. 4. System generates the report based on the specified parameters. 6. Report is saved into the system
Alternative Flows	N/A
Exceptions	4a. In step 4, if the report parameters are incorrect: 1. System alerts the HR Analyst. 2. HR Analyst adjusts the parameters. 3. Use case resumes at step 4.
Business Rules	N/A

Assumptions	Assume that HR Analysts are trained on the reporting tools and understand the company's data policies.
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3.2.2.5 Module 5: Dashboard and Reporting

3.2.2.5.1 UC-20: Generate Custom Reports

Users generate real-time reports based on specific criteria such as financial performance, sales trends, and User productivity.

Attribute	Details
Actors	Primary: Admin, User
Trigger	Need to analyze specific aspects of business performance.
Preconditions	PRE-1: User is logged into the Dashboard and Reporting system.
Postconditions	POST-1: Custom report is generated and available for review.
Normal Flow	1. User logs into the system. 2. User navigates to the reporting section. 3. User selects or defines the criteria for the report. 4. System processes the data based on the selected criteria. 5. System displays the generated report to the user.
Alternative Flows	N/A
Exceptions	4a. If the system cannot process the report due to data errors: 1. System alerts the user. 2. User modifies the criteria or corrects the data errors. 3. Use case resumes at step 3.
Business Rules	N/A
Assumptions	Assume that the system is capable of handling complex data processing in real-time.

3.2.2.5.2 UC-21: Export Reports

Users export reports in various formats such as PDF, Excel, or CSV for further analysis or sharing.

Attribute	Details
Actors	Primary: Admin, User
Trigger	User needs to share or analyze report data outside the system.
Preconditions	PRE-1: Report is generated and available for export. PRE-2: User is logged into the Dashboard and Reporting system.

Postconditions	POST-1: Report is exported in the selected format.
Normal Flow	1. User navigates to Dashboard and Reporting section. 2. User opens the report within the system. 3. User selects the export option. 4. User chooses the format for export. 5. System processes and exports the report. 6. System provides a download link or sends the report via email.
Alternative Flows	4a. if user selects the PDF format for report: 1. System checks if report is available in selected format. 2. return to step 5 of normal flow
Exceptions	4a. If the export fails due to a system error: 1. System alerts the user. 2. User attempts to export again. 3. Use case resumes at step 2. 4b. if selected report is not available in selected format 1. System shows the available format. 2. Return to step 3 of normal flow
Business Rules	N/A
Assumptions	Assume that the system supports multiple data export formats and is integrated with email services for direct sharing.

3.2.2.5.3 UC-22: Customize Dashboards by Role

Admins create and assign different dashboards for different user roles, ensuring each user sees tailored data.

Attribute	Details
Actors	Primary: Admin
Trigger	Need to tailor dashboard data visibility based on user roles.
Preconditions	PRE-1: User is logged into the system. PRE-2: User has administrative privileges.
Postconditions	POST-1: Dashboards are customized and assigned based on user roles.
Normal Flow	1. Admin navigates to dashboard and reporting. 2. Admin customizes the dashboard elements and data views suitable for the role. 3. Admin selects a user role. 4. System applies the customized dashboard settings to the selected role.
Alternative Flows	N/A

Exceptions	5a. If the system fails to save the customized settings: 1. System alerts the administrator. 2. Administrator attempts to save again. 3. Use case resumes at step 4.
Business Rules	N/A
Assumptions	Assume that the system provides flexible tools for dashboard customization.

3.2.2.5.4 UC-23: Interactive Data Visualizations

Dashboards support interactive visualizations such as charts, graphs, tables that allow users to drill down into specific data points.

Attribute	Details
Actors	Primary: Admin
Trigger	Admin needs to visualize data to check performance
Preconditions	PRE-1: Admin is logged into the system.
Postconditions	POST-1: Admin interacts with visual data elements to gain deeper insights.
Normal Flow	1. Admin accesses the dashboard. 2. Admin interacts with a visualization element 3. System displays detailed data related to the selected visualization element. 4. Admin selects the position of visualization on dashboard. 5. Admin clicks submit and visualization is created.
Alternative Flows	3a if user selects elements which show comparison of data: 1. System asks user to select multiple table. 2. Return to step 4 of normal flow
Exceptions	3a. If user selects data of string type: 1. System prompts user to select numerical data for visualization 2. Return to step 2 of normal flow 3b. If the system does not respond to interaction: 1. System alerts the user of a technical issue. 2. User retries or reports the issue. 3. Use case resumes at step 2.
Business Rules	BR-1: Interactive elements must be accessible and provide accurate data.
Assumptions	Assume the system is equipped with advanced data visualization tools that support interactivity.

3.2.2.5.5 UC-24: Customize and Update Dashboards

Admin can customize the layout and content of their dashboards according to their needs, and visualizations are kept updated with real-time data.

Attribute	Details
Actors	Primary: admin
Trigger	Admin needs a personalized view of the dashboard that reflects current data.
Preconditions	PRE-1: Admin is logged into the Dashboard and Reporting system.
Postconditions	POST-1: Dashboard is customized and displays up-to-date data.
Normal Flow	1. Admin logs into the system. 2. Admin navigates to the dashboard customization section. 3. Admin selects elements to add, remove, or modify on the dashboard. 4. Admin arranges the layout to fit their preferences. 5. System saves the customized settings and refreshes the dashboard to show current data.
Alternative Flows	3a. If user selects add more item : 1. System moves to add visualization screen 2. After selecting the element and data system moves to step 4 of normal flow
Exceptions	5a. If the system fails to save the customization settings: 1. System alerts the user. 2. User attempts to customize again. 3. Use case resumes at step 3.
Business Rules	BR-1: Customizations must not interfere with the overall performance of the dashboard.
Assumptions	Assume users have the ability to understand and manipulate dashboard settings effectively.

3.2.2.6 Module 6: Billing and Financing

3.2.2.6.1 UC-25: Create Invoices

Admin can create invoices for their customer products or services, ensuring accurate billing records.

Attribute	Details
Actors	Primary: Admin
Trigger	Need to bill a customer for products or services.

Preconditions	PRE-1: Admin is logged into the Billing and Financing system. PRE-2: Admin has access to pricing and customer details.
Postconditions	POST-1: An invoice is created and stored in the system. POST-2: Customer receives the invoice.
Normal Flow	1. Admin logs into the system. 2. Admin navigates to the invoice creation section. 3. Admin enters or selects customer details and the products or services. 4. System calculates the total cost 5. Admin reviews the invoice details. 6. Admin finalizes and issues the invoice. 7. System sends the invoice to the customer and records the transaction.
Alternative Flows	N/A
Exceptions	3a. If customer details are incomplete: 1. System alerts the user. 2. User updates the customer information. 3. Use case resumes at step 3.
Business Rules	N/A
Assumptions	Assume users are trained on how to use the invoice creation tools.

3.2.2.6.2 UC-26: Manage Receipts and Payments

Users either download receipts or make payments through a payment gateway, tracking the status of each transaction.

Attribute	Details
Actors	Primary: User
Trigger	Receipt needs to be generated or payment needs to be processed.
Preconditions	PRE-1: Financial transaction has occurred. PRE-2: User is logged into the Billing and Financing system.
Postconditions	POST-1: Receipt is generated and available for download. POST-2: Payment is processed and recorded in the system.
Normal Flow	1. User logs into the system. 2. User navigates to the payment management section. 3. User selects payment and proceed option 4. User enter the detail of card. 5. System updates the transaction status and notifies relevant parties.
Alternative Flows	3a. if user selects the download the report: 1. system open the receipt for him and ask user to download receipt.

	2. Return to step 2 of normal flow
Exceptions	3a. If the payment gateway is not responding: 1. System notifies the user of the issue. 2. User retries the transaction when the system is back online. 3. Use case resumes at step 3.
Business Rules	BR-1: Payments must be processed securely using approved gateways.
Assumptions	Assume the system has real-time access to payment gateways and financial data.

3.2.2.6.3 UC-27: Monitor Invoice Status

Users monitor the status of open invoices, receiving notifications and alerts for overdue payments.

Attribute	Details
Actors	Primary: User
Trigger	An invoice is issued and its status needs monitoring.
Preconditions	PRE-1: User is logged into the system.
Postconditions	POST-1: Updated status of the invoice is available in the system.
Normal Flow	1. User logs into the system. 2. User navigates to the invoice monitoring section. 3. User reviews the status of various invoices. 4. System alerts the user to any invoices that are overdue. 5. User takes appropriate action on overdue invoices.
Alternative Flows	3a. if user ant to open any invoices: 1. System gets the detail and show details to user 2. User closes the detail and system return to step 3 of normal flow
Exceptions	4a. If the system fails to update invoice statuses automatically: 1. User manually checks the status. 2. IT support is notified to resolve the system issue. 3. Use case resumes at step 3.
Business Rules	BR-1: System must alert users to overdue invoices within one day of due date.
Assumptions	Assume that all invoices are tracked in real-time within the system.

3.2.2.6.4 UC-28: Create Income and expense Category

User can create a new income and expense category in the accounting module.

Section	Details
Actors	Primary Actor: Admin, User

Trigger	Admin wants to create a new income and expense category in the accounting module.
Preconditions	PRE-1: User is logged into the ERP system. PRE-2: User has defined income and expense category
Postconditions	POST-1: New income and expense category is added to the system. POST-2: The category is available for assigning incomes.
Normal Flow	1. User navigates to "Income and expense Categories". 2. Selects "Add New Category". 3. User selects vendor name from vendor list 4. User add category name 5. User adds description of category 6. User clicks submit button
Alternative Flows	4a: User edits existing category 1. System will move to category section for editing 2. Returns to step 3 of normal flow
Exceptions	3a: User add category without vendor name selected 1. System will show an error message to user 3b: Insufficient permissions to create categories. 1. System will show error message
Business Rules	BR-1: Income categories must be unique and properly defined.
Assumptions	N/A

3.2.2.6.5 UC-29: Add Account Incomes and Expense

User can record a new income entry in the system.

Section	Details
Actors	Primary Actor: Admin, User
Trigger	User wants to record a new income entry in the system.
Preconditions	PRE-1: User is logged into the ERP system. PRE-2: User has all the details of income category
Postconditions	POST-1: Income and expense entry is recorded and categorized. POST-2: Financial records are updated.
Normal Flow	1. User navigates to "Account Incomes and expense". 2. Selects "Add New Income and expense". 3. User select vendor name from list 4. User selects category name from list 5. User adds expense name and date 6. User adds total amount, paid amount and due amount

	7. User add payment mode and bank name 8. User clicks submit button
Alternative Flows	4a. User can add new category name 1. System moves to category page and user adds its category 2. Returns to step 5 of normal flow
Exceptions	1.0. E1: Insufficient permissions to add income. 1.1 E2: Insufficient storage to add income to database
Business Rules	BR-1: The system must validate the accuracy of all income data before saving
Assumptions	N/A

3.2.2.7 Module 7: Task and Notification Management

3.2.2.7.1 UC-30: Manage To-Do List

Admin can create a list of task and assign it to user

Actor	Primary Actor : Admin
Trigger	The user decides to manage their tasks using the to-do list.
Preconditions	PRE-1: Admin must be logged into the system. PRE-2: Admin must be in task and notification management page
Postconditions	POST-1: The Admin's to-do list is updated with new tasks, priorities, and deadlines.
Normal Flow	1. Admin navigates to the to-do list page. 2. Admin adds a new task, specifying details such as title, priority, and deadline. 3. Admin saves the task.
Alternative Flows	2a: Admin wants to modify an existing task. 1. Admin selects an existing task to edit. 2. Admin updates the task details. 3. System saves the changes. 2b: Admin wants to remove an existing task. 1. Admin selects an existing task to delete. 2. Admin confirms deletion. 3. System removes the task.
Exceptions	2a: Admin enters invalid details. 1. The system displays an error message. 2. Admin is prompted to re-enter valid task details.

Business Rules	BR-1: Tasks must have a title and can have optional deadlines and priority levels.
Assumptions	1: It is assumed that the user is familiar with the interface for managing the to-do list.

3.2.2.7.2 UC-31: Schedule Events in Calendar

User can add an event like meeting and any other event in the calendar for remainder purpose.

Actor	Primary Actor :User, Admin
Trigger	The user decides to schedule an event or appointment on a calendar.
Preconditions	PRE-1: User must be logged into the system. PRE-2: User must be in on the Task and Notification page
Postconditions	POST-1: The event is successfully added to the calendar.
Normal Flow	1. User navigates to the Task and Notification page. 2. User navigates to the calendar page. 3. User selects a date and time for the event. 4. User enters event details, including title, description, and recurrence settings. 5. User saves the event.
Alternative Flows	4a: The user wants to modify an existing event. 1. User selects an existing event on the calendar. 2. User updates the event details. 3. System saves the changes. 4b: The user wants to remove an existing event. 1. User selects an existing event on the calendar. 2. User confirms deletion. 3. System removes the event from the calendar.
Exceptions	E1: The system encounters an error while saving the event. 1. The system displays an error message. 2. The user is prompted to retry the operation.
Business Rules	BR-1: Events must have a title and can optionally include a description.
Assumptions	1: It is assumed that the user knows how to use the calendar interface effectively.

3.2.2.7.3 UC-32: Set Reminders and Receive Notifications

User get the notification about the event added in the calendar.

Actor	Primary Actor : User
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Trigger	The user decides to set a reminder for an upcoming task or event.
Preconditions	PRE-1: User must be logged into the system.
Postconditions	POST-1: The reminder is successfully set and scheduled for notification.
Normal Flow	1. User navigates to the reminder settings page. 2. User selects a task or event for which to set a reminder. 3. User specifies the reminder time 4. User saves the reminder.
Alternative Flows	N/A
Exceptions	2a: The user selects a reminder time that is invalid. 1. The system displays an error message. 2. The user is prompted to choose a valid reminder time. E1: The system fails to send the notification at the scheduled time. 1. The system logs the error. 2. The user may receive a message indicating a failure to deliver the notification.
Business Rules	BR-1: Reminders can be set for any task or event that has a defined date and time.
Assumptions	1: It is assumed that the user has access to notifications and has provided the necessary contact information.

3.2.2.8 Module 6: AI chatbot Assistance

3.2.2.8.1 UC-33: Chatbot Assistance for Schema Creation

The chatbot provides instant answers to common questions regarding schema creation for new modules, enhancing user understanding and efficiency.

Attribute	Details
Actors	Primary: Admin
Trigger	Admin wants to know about schema of specific module.
Preconditions	PRE-1: User is logged into the system
Postconditions	POST-1: User get the schema of that specific module for module creation.
Normal Flow	1. Admin clicks AI chatbot. 2. Admin asks questions related to schema creation of module with name. 3. Chatbot processes the query using its knowledge base. 4. Chatbot provides a response with guidance on schema structure. 5. Admin uses the information to assist in module development.
Alternative Flows	N/A.

Exceptions	3a. If the chatbot cannot find an answer: 1. Chatbot informs the developer it cannot provide a specific answer. 2. Chatbot offers to log the query for human support. 3. Developer decides whether to wait for human support.
Business Rules	N/A
Assumptions	Assume the chatbot is continuously updated with the latest schema creation standards and practices.

3.2.2.8.2 UC-34: Format Guidance for Excel Schema

The chatbot advises users on the appropriate format and structure required for schemas within an Excel sheet, ensuring data consistency and correctness.

Attribute	Details
Actors	Primary: Admin
Trigger	User needs assistance with formatting schemas in Excel for multiple table creation.
Preconditions	PRE-1: Admin is logged into the system
Postconditions	POST-1: Admin gets the dummy format for exporting excel file to create multiple tables for module creation
Normal Flow	1. Admin initiates a query about Excel schema formatting. 2. Chatbot provides dummy excel file for user to check formatting of Excel. 3. Admin applies the guidance to create or adjust schemas in Excel.
Alternative Flows	Admin paste the schema and ask to create Excel file: 1. System prompt user that cannot handle that request 2. Return step 1 of normal flow.
Exceptions	3a. If the guidance provided is insufficient or incorrect: 1. Data Analyst notifies the chatbot of the issue. 2. Chatbot logs the feedback for further improvement. 3. Data Analyst seeks alternative help or resources.
Business Rules	N/A
Assumptions	Assume the chatbot is equipped with the latest standards and practices for data schema in Excel.

3.2.2.8.3 UC-35: Collect Feedback for Performance Improvement

The chatbot collects user feedback and ratings to assess its performance and improve user experience.

Attribute	Details
Actors	Admin
Trigger	Completion of interaction with the chatbot.
Preconditions	PRE-1: Admin has completed an interaction with the AI chatbot.
Postconditions	POST-1: Feedback is collected and analyzed to improve chatbot performance.
Normal Flow	1. After an interaction, the chatbot prompts the user to provide feedback. 2. Admin submits feedback and ratings based on their experience. 3. Chatbot processes and stores the feedback. 4. System analyzes the feedback to identify areas for improvement. 5. Updates and enhancements are made to the chatbot based on the feedback.
Alternative Flows	N/A
Exceptions	2a. If the user chooses not to provide feedback: 1. Chatbot logs the lack of feedback. 2. Interaction concludes without further action.
Business Rules	N/A
Assumptions	Assume that users are willing to provide constructive feedback to help improve the system.

3.3 Functional Requirements

3.3.1 Functional Requirements for UC-01: Login

The user wants to access the ERP system by providing valid credentials

3.3.1.1 FR-1.1: User Authentication

Attribute	Details
Identifier	FR-1.1
Title	User Authentication
Requirement	The system shall authenticate users by verifying their username/email and password against stored credentials.
Source	User
Rationale	To ensure that only registered and authorized users gain access to the ERP system functionalities.

Business Rule	BR-1: Passwords must be at least 8 characters long.
Dependencies	None
Priority	High

3.3.1.2 FR-1.2: Password Recovery Process

Attribute	Details
Identifier	FR-1.2
Title	Password Recovery Process
Requirement	The system shall provide a mechanism for users to recover or reset their passwords via a registered email link or code.
Source	User
Rationale	To enable users who have forgotten their passwords to regain access to their accounts securely without administrator intervention.
Business Rule	None
Dependencies	None
Priority	High

3.3.1.3 FR-1.3: Post-Login Redirection

Attribute	Details
Identifier	FR-1.3
Title	Post-Login Redirection
Requirement	Upon successful authentication, the system shall redirect the user to their designated dashboard or home page.
Source	User
Rationale	To provide a seamless user experience by directing users immediately to a landing page where they can begin their session's activities.
Business Rule	None
Dependencies	FR-1.1
Priority	Medium

3.3.2 Functional Requirements for UC-02: Edit User Profile

The user wants to update personal or professional information in their profile.

3.3.2.1 FR-2.1: Access Profile Editing

Attribute	Details
Identifier	FR-2.1
Title	Access Profile Editing
Requirement	The system shall allow users to navigate to the user management profile page and initiate the editing process by selecting the edit profile option.
Source	User
Rationale	To enable users to update their personal or professional information easily.
Business Rule	None
Dependencies	None
Priority	High

3.3.2.2 FR-2.2: Validate User Input

Attribute	Details
Identifier	FR-2.2
Title	Validate User Input
Requirement	The system shall validate all user inputs against predefined criteria for correctness and completeness before saving any changes.
Source	System
Rationale	To ensure data integrity and prevent errors in user profile information.
Business Rule	BR-1: The system should validate all fields according to the defined input criteria before saving the profile. BR-2: Ensure that any uploaded data meet the required format and size constraints.
Dependencies	FR-2.3
Priority	High

3.3.2.3 FR-2.3: Update User Profile

Attribute	Details
Identifier	FR-2.3
Title	Update User Profile

Requirement	Upon successful validation, the system shall save the updated user profile information to the database.
Source	System
Rationale	To permanently store the changes made by the user to their profile.
Business Rule	None
Dependencies	None
Priority	High

3.3.3 Functional Requirements for UC-03: Change Password

The user wants to change its current password and add a new password.

3.3.3.1 FR-3.1: Access Change Password Page

Attribute	Details
Identifier	FR-3.1
Title	Access Change Password Page
Requirement	The system shall allow users to navigate to the change password page to initiate the password change process.
Source	User
Rationale	To provide users with a dedicated interface for updating their password, enhancing security and user experience.
Business Rule	None
Dependencies	None
Priority	High

3.3.3.2 FR-3.2: Verify Current Password

Attribute	Details
Identifier	FR-3.2
Title	Verify Current Password
Requirement	The system shall verify the accuracy of the current password entered by the user to authenticate their identity before allowing a password change.
Source	System
Rationale	To ensure that the password change request is initiated by the rightful account holder.

Business Rule	None
Dependencies	None
Priority	High

3.3.3.3 FR-3.3: Validate New Password

Attribute	Details
Identifier	FR-3.3
Title	Validate New Password
Requirement	The system shall validate the new password entered by the user to ensure it meets predefined complexity and security criteria.
Source	System
Rationale	To enhance account security by enforcing strong password criteria.
Business Rule	BR-1: Passwords must be at least 8 characters long.
Dependencies	3.4
Priority	High

3.3.3.4 FR-3.4: Update Password in System

Attribute	Details
Identifier	FR-3.4
Title	Update Password in System
Requirement	Upon successful validation, the system shall update the user's password with the new password in the system database.
Source	System
Rationale	To complete the process of changing the user's password, ensuring the new password is stored securely.
Business Rule	None
Dependencies	None
Priority	High

3.3.4 Functional Requirements for UC-04: Delete User Account

The admin can delete the user account.

3.3.4.1 FR-4.1: Select User Account

Attribute	Details
Identifier	FR-4.1
Title	Select User Account
Requirement	The system shall allow the Admin to select a user account from a list for deletion or deactivation.
Source	Admin
Rationale	To provide a method for specifying which user account should be affected by deletion or deactivation actions.
Business Rule	None
Dependencies	None
Priority	High

3.3.4.2 FR-4.2: Confirm Deletion Action

Attribute	Details
Identifier	FR-4.2
Title	Confirm Deletion Action
Requirement	The system shall require the Admin to confirm the deletion or deactivation of a selected user account to prevent accidental data loss.
Source	Admin
Rationale	To ensure that account deletion or deactivation is intentional and confirmed to avoid erroneous actions.
Business Rule	BR-1: Deactivating an account should retain the user's data for audit purposes, while deletion should remove all personal data in accordance with data protection regulations.
Dependencies	FR-4.1
Priority	High

3.3.4.3 FR-4.3: Execute Account Deletion

Attribute	Details
Identifier	FR-4.3
Title	Execute Account Deletion
Requirement	Upon confirmation, the system shall delete or deactivate the user account according to the Admin's choice.

Source	Admin
Rationale	To process the Admin's decision to remove or deactivate a user account, effectively updating the system's user database.
Business Rule	BR-1: Deactivating an account should retain the user's data for audit purposes, while deletion should remove all personal data in accordance with data protection regulations.
Dependencies	FR-4.2
Priority	High

3.3.5 Functional Requirements for UC-05: Logout

The user wants to end their session and logout of the system

3.3.5.1 FR-5.1: Display Logout Prompt

Attribute	Details
Identifier	FR-5.1
Title	Display Logout Prompt
Requirement	The system shall display a prompt message asking the user to confirm if they wish to log out.
Source	User
Rationale	To confirm the user's intent to log out and prevent accidental session termination.
Business Rule	None
Dependencies	None
Priority	Medium

3.3.5.2 FR-5.2: Confirm Logout

Attribute	Details
Identifier	FR-5.2
Title	Confirm Logout
Requirement	Upon user confirmation by clicking “Yes” on the logout prompt, the system shall proceed to log out the user.
Source	User
Rationale	To ensure the user intentionally ends their session for security purposes.

Business Rule	None
Dependencies	FR-5.1
Priority	High

3.3.5.3 FR-5.3: Redirect to Login Page

Attribute	Details
Identifier	FR-5.3
Title	Redirect to Login Page
Requirement	After successfully logging out, the system shall redirect the user to the login page.
Source	system
Rationale	To securely end the user's session and redirect them to the login interface, facilitating potential subsequent logins.
Business Rule	None
Dependencies	FR-5.2
Priority	High

3.3.5.4 FR-5.4: Cancel Logout

Attribute	Details
Identifier	FR-5.4
Title	Cancel Logout
Requirement	If the user selects "No" in response to the logout prompt, the system shall cancel the logout process and redirect the user back to the dashboard.
Source	User
Rationale	To allow users to continue their session without interruption if they change their mind about logging out.
Business Rule	None
Dependencies	FR-5.1
Priority	Low

3.3.6 Functional Requirements for UC-06: Customize User Interfaces

Admin wants to create customize menus and dashboard for users

3.3.6.1 FR-6.1: Access Customization Interface

Attribute	Details
Identifier	FR-6.1
Title	Access Customization Interface
Requirement	The system shall allow the Admin to access the interface for customizing dashboards and menus from the role and permission management section.
Source	Admin
Rationale	To provide the Admin with access to tools required for creating and managing customized user interfaces based on role or user group.
Business Rule	None
Dependencies	None
Priority	High

3.3.6.2 FR-6.2: Select Role for Customization

Attribute	Details
Identifier	FR-6.2
Title	Select Role for Customization
Requirement	The system shall provide the Admin with the capability to select a specific role or user group for which to customize the dashboard and menus.
Source	Admin
Rationale	To ensure that customizations are applicable and appropriately targeted to the correct user groups or roles within the ERP system.
Business Rule	BR-1: Only registered roles or user groups can be selected for customization.
Dependencies	FR-6.1
Priority	High

3.3.6.3 FR-6.3: Configure Dashboard Items

Attribute	Details
Identifier	FR-6.3
Title	Configure Dashboard Items
Requirement	The system shall allow the Admin to select and configure individual dashboard items to be displayed based on the selected role's needs.
Source	Admin

Rationale	To facilitate a role-specific dashboard setup that enhances usability and relevance for the user's daily tasks.
Business Rule	BR-1: Configurations must adhere to the security and access privileges associated with the role.
Dependencies	FR-6.2
Priority	High

3.3.6.4 FR-6.4: Manage Custom Menus

Attribute	Details
Identifier	FR-6.4
Title	Manage Custom Menus
Requirement	The system shall allow the Admin to create and assign custom menus to different roles, complementing the dashboard assignment.
Source	Admin
Rationale	To ensure that each role or user group has access to relevant functionalities through customized menus, thereby improving navigation and efficiency.
Business Rule	BR-1: Only admins have the authority to customize and assign menus.
Dependencies	FR-6.1
Priority	High

3.3.7 Functional Requirements for UC-07: Create New User Roles

Admin can create new user roles in the system and assign it to user for role defining purpose.

3.3.7.1 FR-7.1: Add New Role

Attribute	Details
Identifier	FR-7.1
Title	Add New Role
Requirement	The system shall provide the Admin with functionality to add a new role by specifying the role's name and associated permissions.
Source	Admin
Rationale	To facilitate the creation of new roles within the ERP system, enabling customized access control and functionality assignment.
Business Rule	BR-1: Role names must be unique across the system.

Dependencies	None
Priority	High

3.3.7.2 FR-7.2: Validate Role Uniqueness

Attribute	Details
Identifier	FR-7.2
Title	Validate Role Uniqueness
Requirement	Before saving a new role, the system shall check if the role name already exists and prompt the Admin to change it if necessary.
Source	System
Rationale	To prevent the duplication of roles within the system and ensure role management integrity.
Business Rule	BR-2: Role names must be unique.
Dependencies	FR-7.1
Priority	High

3.3.7.3 FR-7.3: Save Role Information

Attribute	Details
Identifier	FR-7.3
Title	Save Role Information
Requirement	Upon successful validation, the system shall save the new role information, including its name, permissions, and any other relevant settings.
Source	System
Rationale	To ensure that all new role data is correctly stored in the system's database for future reference and application.
Business Rule	BR-1: Only admins can create and assign new roles.
Dependencies	FR-7.2
Priority	High

3.3.8 Functional Requirements for UC-08: Assign User to Dashboard

Admin can assign a specific user to a pre-configured dashboard

3.3.8.1 FR-8.1: Select User for Dashboard Assignment

Attribute	Details
Identifier	FR-8.1
Title	Select User for Dashboard Assignment
Requirement	The system shall provide the Admin with the capability to select a registered user from a list for dashboard assignment.
Source	Admin
Rationale	To facilitate the assignment of dashboards to specific users, ensuring personalized access to functionalities.
Business Rule	BR-1: A user must be registered and have a valid profile before assigning a dashboard.
Dependencies	None
Priority	High

3.3.8.2 FR-8.2: Display Available Dashboards

Attribute	Details
Identifier	FR-8.2
Title	Display Available Dashboards
Requirement	The system shall display a list of available dashboards to the Admin after selecting a user.
Source	System
Rationale	To provide the Admin with options to assign a suitable dashboard based on the user's role or requirements.
Business Rule	None
Dependencies	None
Priority	High

3.3.8.3 FR-8.3: Assign Dashboard to User

Attribute	Details
Identifier	FR-8.3
Title	Assign Dashboard to User
Requirement	Upon Admin's selection, the system shall assign the chosen dashboard to the selected user and update the user's profile accordingly.

Source	Admin
Rationale	To ensure users have access to the necessary tools and interfaces for their roles within the ERP system.
Business Rule	BR-1: The system should ensure that a user can only have one active dashboard at a time.
Dependencies	FR-8.2
Priority	High

3.3.8.4 FR-8.4: Confirm Dashboard Assignment

Attribute	Details
Identifier	FR-8.4
Title	Confirm Dashboard Assignment
Requirement	The system shall prompt the Admin to confirm the dashboard assignment before finalizing changes, especially if replacing an existing dashboard.
Source	System
Rationale	To prevent unintentional changes and ensure Admins are aware of and agree to the modifications being made.
Business Rule	None
Dependencies	FR-8.3
Priority	Medium

3.3.9 Functional Requirements for UC-09: Assign User Role

Admin can assign already added role in the system to user for its permission and restriction.

3.3.9.1 FR-9.1: Select User for Role Assignment

Attribute	Details
Identifier	FR-9.1
Title	Select User for Role Assignment
Requirement	The system shall provide the Admin with the ability to select a registered user from a list for role assignment.
Source	Admin
Rationale	To enable the assignment of roles to specific users, ensuring appropriate access control and functional permissions are granted.

Business Rule	None
Dependencies	None
Priority	High

3.3.9.2 FR-9.2: Display Available Roles

Attribute	Details
Identifier	FR-9.2
Title	Display Available Roles
Requirement	The system shall display a list of available roles to the Admin after selecting a user.
Source	System
Rationale	To provide the Admin with options for assigning appropriate roles based on the user's needs or responsibilities.
Business Rule	None
Dependencies	None
Priority	High

3.3.9.3 FR-9.3: Assign Role to User

Attribute	Details
Identifier	FR-9.3
Title	Assign Role to User
Requirement	Upon Admin's selection, the system shall assign the chosen role to the selected user and update the user's profile accordingly.
Source	Admin
Rationale	To ensure that users are provided with the necessary permissions and access as per their assigned roles, enhancing system security and functionality.
Business Rule	BR-1: Only admins can assign roles to users.
Dependencies	FR-9.2
Priority	High

3.3.10 Functional Requirements for UC-10: Define Database Schema

Admin can create a database schema for its new module.

3.3.10.1 FR-10.1: Define Schema Fields

Attribute	Details
Identifier	FR-10.1
Title	Define Schema Fields
Requirement	The system shall allow the Admin to define fields, specifying their names, data types, and any relationships with other data entities within the form for creating a new module's database schema.
Source	Admin
Rationale	To enable structured input of data elements and relationships necessary for the new module, ensuring data integrity and system functionality.
Business Rule	BR-1: Fields must have defined data types, and relationships must comply with database normalization rules.
Dependencies	None
Priority	High

3.3.10.2 FR-10.2: Save Database Schema

Attribute	Details
Identifier	FR-10.2
Title	Save Database Schema
Requirement	Upon successful validation, the system shall save the new database schema, making it available for the module's implementation and further development.
Source	System
Rationale	To store the defined and validated database schema in the system's repository, enabling its use in module development and integration.
Business Rule	None
Dependencies	None
Priority	High

3.3.11 Functional Requirements for UC-11: Import Data Structure from Excel

Admin can export excel file to import multiple table at a time for module creation

3.3.11.1 FR-11.1: Upload Excel Sheet

Attribute	Details
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Identifier	FR-11.1
Title	Upload Excel Sheet
Requirement	The system shall provide an interface for the Admin to upload an Excel sheet that contains the data structure for multiple tables.
Source	Admin
Rationale	To enable the Admin to initiate the process of importing data structures for module creation directly from an Excel sheet.
Business Rule	BR-1: The Excel sheet must follow a predefined structure to ensure successful table creation.
Dependencies	None
Priority	High

3.3.11.2 FR-11.2: Validate Excel Format

Attribute	Details
Identifier	FR-11.2
Title	Validate Excel Format
Requirement	The system shall validate the format and structure of the uploaded Excel sheet to ensure it meets the required criteria for data import.
Source	System
Rationale	To ensure that the Excel sheet format is correct and can be used to successfully create database tables without errors.
Business Rule	BR-1: Excel sheets must conform to a specified structure including data types and relationships between tables.
Dependencies	FR-11.1
Priority	High

3.3.11.3 FR-11.3: Read Data from Excel

Attribute	Details
Identifier	FR-11.3
Title	Read Data from Excel
Requirement	Upon successful validation, the system shall read the structure and data from the Excel sheet to prepare for table creation.
Source	System
Rationale	To extract necessary information from the validated Excel sheet for the creation of database tables in the ERP system.

Business Rule	None
Dependencies	FR-11.2
Priority	High

3.3.11.4 FR-11.4: Create Database Tables

Attribute	Details
Identifier	FR-11.4
Title	Create Database Tables
Requirement	The system shall create database tables based on the structure and data extracted from the Excel sheet.
Source	System
Rationale	To actualize the data structure into usable database tables within the ERP system, enhancing module functionality.
Business Rule	BR-1: All created tables must follow the predefined database schemas and integrity constraints as per the system's standards.
Dependencies	FR-11.3
Priority	High

3.3.12 Functional Requirements for UC-12: Generate Code

The admin can create a module from schema and integrate it into the system.

3.3.12.1 FR-12.1: Initiate Code Generation

Attribute	Details
Identifier	FR-12.1
Title	Initiate Code Generation
Requirement	The system shall allow the Admin to initiate the code generation process for new modules by clicking the "Generate Code" button.
Source	Admin
Rationale	To enable the Admin to start the automated process of generating code based on new or imported module schemas.
Business Rule	None
Dependencies	None
Priority	High

3.3.12.2 FR-12.2: Integrate Generated Code

Attribute	Details
Identifier	FR-12.2
Title	Integrate Generated Code
Requirement	Upon successful code generation, the system shall integrate the generated code into the existing system architecture and update relevant components.
Source	System
Rationale	To seamlessly incorporate new functionalities into the ERP system, ensuring that new modules work in harmony with existing ones.
Business Rule	None
Dependencies	None
Priority	High

3.3.13 Functional Requirements for UC-13: Create New User Profile

Admin creates new User profiles, entering personal information, job roles, and department assignments.

3.3.13.1 FR-13.1: Enter User Information

Attribute	Details
Identifier	FR-13.1
Title	Enter User Information
Requirement	The system shall provide an interface for the Admin to enter personal information, job roles, and department assignments for new users.
Source	Admin
Rationale	To allow the Admin to input comprehensive details necessary for creating a user profile in the system.
Business Rule	BR-1: Only authorized admins can create new user profiles.
Dependencies	None
Priority	High

3.3.13.2 FR-13.2: Validate User Information

Attribute	Details
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Identifier	FR-13.2
Title	Validate User Information
Requirement	The system shall validate the personal information, job roles, and department assignments entered by the Admin to ensure they meet the system's standards and requirements.
Source	System
Rationale	To ensure data integrity and prevent errors in user profile creation.
Business Rule	BR-1: User information must comply with defined validation rules, including data format and completeness.
Dependencies	FR-13.1
Priority	High

3.3.13.3 FR-13.3: Save User Profile

Attribute	Details
Identifier	FR-13.3
Title	Save User Profile
Requirement	Upon successful validation, the system shall save the new user profile to the database.
Source	System
Rationale	To securely store the newly created user profile, making it available for system operations and access.
Business Rule	BR-1: User profiles must be securely stored and logged for audit and operational purposes.
Dependencies	FR-13.2
Priority	High

3.3.13.4 FR-13.4: Display Confirmation

Attribute	Details
Identifier	FR-13.4
Title	Display Confirmation
Requirement	After saving the new user profile, the system shall display a confirmation message to the Admin.
Source	System
Rationale	To confirm to the Admin that the user profile has been successfully created and stored in the system.

Business Rule	None
Dependencies	FR-13.3
Priority	Medium

3.3.14 Functional Requirements for UC-14: Edit User Profile

Admin edits existing User profiles, updating personal information, job roles, and department assignments.

3.3.14.1 FR-14.1: Select User to Edit

Attribute	Details
Identifier	FR-14.1
Title	Select User to Edit
Requirement	The system shall provide the Admin with the ability to select an existing user profile from a list for editing.
Source	Admin
Rationale	To enable the Admin to choose which user profile to update, ensuring the correct data is accessed and modified.
Business Rule	None
Dependencies	None
Priority	High

3.3.14.2 FR-14.2: Update User Information

Attribute	Details
Identifier	FR-14.2
Title	Update User Information
Requirement	The system shall allow the Admin to update personal information, job roles, and department assignments for the selected user profile.
Source	Admin
Rationale	To facilitate modifications to user profiles to reflect changes in personal details, job roles, or departmental alignments.
Business Rule	BR-1: Only authorized admins can update job roles and departments.
Dependencies	FR-14.1

Priority	High
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3.3.14.3 FR-14.3: Validate Updated Information

Attribute	Details
Identifier	FR-14.3
Title	Validate Updated Information
Requirement	The system shall validate the updated information entered by the Admin to ensure it meets all required data standards and constraints.
Source	System
Rationale	To ensure the integrity and accuracy of the updated user profile data, preventing errors and data inconsistencies.
Business Rule	BR-1: User information must comply with defined validation rules, including data format and completeness.
Dependencies	FR-14.2
Priority	High

3.3.14.4 FR-14.4: Confirm and Save Changes

Attribute	Details
Identifier	FR-14.4
Title	Confirm and Save Changes
Requirement	Upon successful validation, the system shall save the updated user profile data and provide a confirmation message to the Admin.
Source	System
Rationale	To finalize changes to the user profile, ensuring the updates are permanently reflected in the system and acknowledged to the Admin.
Business Rule	None
Dependencies	FR-14.3
Priority	High

3.3.15 Functional Requirements for UC-15: Submit and Track Expense Reports

Users submit expense reports with categorizations and attachments, and track the approval status.

3.3.15.1 FR-15.1: Create New Expense Report

Attribute	Details
Identifier	FR-15.1
Title	Create New Expense Report
Requirement	The system shall provide an interface for the User to create a new expense report by entering expense details, categorizing the expense, and attaching required receipts or proofs of expense.
Source	User
Rationale	To enable Users to report their expenses conveniently and ensure all necessary information is captured accurately for processing.
Business Rule	BR-1: All expenses must be categorized according to company policy.
Dependencies	None
Priority	High

3.3.15.2 FR-15.2: Validate Expense Report Details

Attribute	Details
Identifier	FR-15.2
Title	Validate Expense Report Details
Requirement	The system shall validate the completeness and correctness of the entered details and attached documents before allowing the expense report to be submitted.
Source	System
Rationale	To ensure that all submitted expense reports are complete and meet the requirements for approval and reimbursement.
Business Rule	None
Dependencies	FR-15.1
Priority	High

3.3.15.3 FR-15.3: Submit Expense Report

Attribute	Details
Identifier	FR-15.3
Title	Submit Expense Report
Requirement	Upon successful validation, the system shall submit the expense report to the concerned department for approval.

Source	System
Rationale	To facilitate the processing and approval of expense claims, ensuring timely review and action by the concerned department.
Business Rule	None
Dependencies	FR-15.2
Priority	High

3.3.15.4 FR-15.4: Track Expense Report Status

Attribute	Details
Identifier	FR-15.4
Title	Track Expense Report Status
Requirement	The system shall provide the User with the capability to track the approval status of their submitted expense reports.
Source	User
Rationale	To keep Users informed of the progress of their expense report processing and provide transparency in the approval process.
Business Rule	None
Dependencies	FR-15.3
Priority	Medium

3.3.15.5 FR-15.5: Display Confirmation Message

Attribute	Details
Identifier	FR-15.5
Title	Display Confirmation Message
Requirement	After submitting an expense report, the system shall display a confirmation message to the User, indicating successful submission.
Source	System
Rationale	To confirm to the User that their expense report has been successfully received by the system and is being processed.
Business Rule	None
Dependencies	FR-15.3
Priority	Medium

3.3.16 Functional Requirements for UC-16: Apply for and Track Leave Requests

Users apply for leave and monitor the status of their leave requests through the system.

3.3.16.1 FR-16.1: Access Leave Request Section

Attribute	Details
Identifier	FR-16.1
Title	Access Leave Request Section
Requirement	The system shall provide users with access to the leave request section within the HR management system after they log in.
Source	User
Rationale	To allow users to enter and manage their leave requests efficiently from a designated section within the system.
Business Rule	None
Dependencies	None
Priority	High

3.3.16.2 FR-16.2: Input Leave Details

Attribute	Details
Identifier	FR-16.2
Title	Input Leave Details
Requirement	The system shall allow users to select the type of leave they are requesting and input the start and end dates of the leave.
Source	User
Rationale	To capture all necessary details for processing leave requests according to organizational policies.
Business Rule	BR-1: Leave requests must be submitted at least 5 days in advance.
Dependencies	FR-16.1
Priority	High

3.3.16.3 FR-16.3: Validate Leave Request

Attribute	Details
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Identifier	FR-16.3
Title	Validate Leave Request
Requirement	The system shall validate the leave request to ensure compliance with leave policies, including checking leave balances and the advance notice period.
Source	System
Rationale	To ensure that all leave requests adhere to company policies and that users have sufficient leave balance.
Business Rule	None
Dependencies	FR-16.2
Priority	High

3.3.16.4 FR-16.4: Submit and Notify

Attribute	Details
Identifier	FR-16.4
Title	Submit and Notify
Requirement	Upon successful validation, the system shall submit the leave request, update the leave status, and notify the concerned person for approval.
Source	System
Rationale	To process the leave request officially and inform relevant personnel for further action, ensuring timely responses and record-keeping.
Business Rule	BR-1: Notifications must be sent to the appropriate personnel upon leave request submission.
Dependencies	FR-16.3
Priority	High

3.3.16.5 FR-16.5: Track Leave Status

Attribute	Details
Identifier	FR-16.5
Title	Track Leave Status
Requirement	The system shall allow users to track the status of their leave request, including pending approval, approved, or denied statuses.
Source	User
Rationale	To provide transparency and allow users to monitor the progress of their leave requests through the system.

Business Rule	None
Dependencies	FR-16.4
Priority	Medium

3.3.17 Functional Requirements for UC-17: Manage Timesheets

Users create and update timesheets on a daily, weekly, or monthly basis, recording regular and overtime hours worked

3.3.17.1 FR-17.1: Access Timesheet Section

Attribute	Details
Identifier	FR-17.1
Title	Access Timesheet Section
Requirement	The system shall provide users with access to the timesheet section where they can enter or update their hours worked.
Source	User
Rationale	To allow users to log into and navigate to the appropriate section for timesheet management seamlessly.
Business Rule	None
Dependencies	None
Priority	High

3.3.17.2 FR-17.2: Enter and Update Work Hours

Attribute	Details
Identifier	FR-17.2
Title	Enter and Update Work Hours
Requirement	The system shall allow users to enter or update regular and overtime hours worked for a selected period (daily, weekly, or monthly).
Source	User
Rationale	To enable users to record and adjust their work hours accurately within the system.
Business Rule	None
Dependencies	FR-17.1

Priority	High
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3.3.17.3 FR-17.3: Validate and Save Timesheet

Attribute	Details
Identifier	FR-17.3
Title	Validate and Save Timesheet
Requirement	The system shall validate the entered timesheet details for completeness and correctness, and save the updated timesheet if validation is successful.
Source	System
Rationale	To ensure that timesheet entries meet required standards and are stored correctly within the system for payroll and reporting purposes.
Business Rule	BR-1: Timesheets must be completed by the end of each pay period.
Dependencies	FR-17.2
Priority	High

3.3.17.4 FR-17.4: Display Confirmation Message

Attribute	Details
Identifier	FR-17.4
Title	Display Confirmation Message
Requirement	Upon successful saving of the timesheet, the system shall display a confirmation message to the user.
Source	System
Rationale	To confirm to users that their timesheet has been successfully updated and stored in the system.
Business Rule	None
Dependencies	FR-17.3
Priority	Medium

3.3.17.5 FR-17.5: Calculate Working Hours

Attribute	Details
Identifier	FR-17.5
Title	Calculate Working Hours

Requirement	The system shall automatically calculate the total working hours, including regular and overtime, based on the entered timesheet data.
Source	System
Rationale	To provide accurate calculation of working hours for payroll processing and compliance with labor regulations.
Business Rule	None
Dependencies	FR-17.3
Priority	High

3.3.18 Functional Requirements for UC-18: Review and Approve Requests

Admin can review, approve, or reject submitted expense reports and leave requests.

3.3.18.1 FR-18.1: Review Pending Requests

Attribute	Details
Identifier	FR-18.1
Title	Review Pending Requests
Requirement	The system shall provide the Admin with a list of all pending expense reports and leave requests that require review.
Source	Admin
Rationale	To enable Admins to efficiently review each request and make informed decisions based on the details provided.
Business Rule	BR-1: Only authorized personnel can approve or reject requests.
Dependencies	None
Priority	High

3.3.18.2 FR-18.2: Approve or Reject Requests

Attribute	Details
Identifier	FR-18.2
Title	Approve or Reject Requests
Requirement	The system shall allow the Admin to approve or reject each pending request and optionally provide a reason for the decision, especially in cases of rejection.
Source	Admin

Rationale	To process requests promptly and ensure that each decision is well-documented for transparency and record-keeping.
Business Rule	None
Dependencies	FR-18.1
Priority	High

3.3.18.3 FR-18.3: Notify Users

Attribute	Details
Identifier	FR-18.3
Title	Notify Users
Requirement	Upon approval or rejection of requests, the system shall update the request status and notify the corresponding User of the decision along with any relevant comments or reasons.
Source	System
Rationale	To ensure that Users are promptly informed about the status of their requests, maintaining transparency in the approval process.
Business Rule	None
Dependencies	FR-18.2
Priority	High

3.3.19 Functional Requirements for UC-19: Generate HR Reports

Admin can generate detailed reports and analytics on User attendance, expenses, and overall workforce productivity.

3.3.19.1 FR-19.1: Specify Report Parameters

Attribute	Details
Identifier	FR-19.1
Title	Specify Report Parameters
Requirement	The system shall provide an interface for the HR Analyst to specify the type of report and set parameters such as date range, user group, and other relevant filters.
Source	HR Analyst

Rationale	To enable precise customization of reports based on specific needs and criteria, ensuring relevance and accuracy of the generated data.
Business Rule	None
Dependencies	None
Priority	High

3.3.19.2 FR-19.2: Generate Report

Attribute	Details
Identifier	FR-19.2
Title	Generate Report
Requirement	The system shall generate the report based on the specified parameters, processing data to reflect attendance, expenses, or productivity metrics.
Source	System
Rationale	To transform raw data into structured reports that provide meaningful insights into workforce management and operational efficiency.
Business Rule	BR-1: Reports must adhere to data privacy and protection guidelines.
Dependencies	FR-19.1
Priority	High

3.3.19.3 FR-19.3: Validate Report Parameters

Attribute	Details
Identifier	FR-19.3
Title	Validate Report Parameters
Requirement	Before generating the report, the system shall validate the specified parameters to ensure they are correct and sufficient for report generation.
Source	System
Rationale	To prevent errors and ensure that the generated reports are accurate and based on valid inputs.
Business Rule	None
Dependencies	FR-19.1
Priority	Medium

3.3.19.4 FR-19.4: Save and Display Report

Attribute	Details
Identifier	FR-19.4
Title	Save and Display Report
Requirement	Once generated, the system shall save the report within the system for future access and immediately display the report to the HR Analyst.
Source	System
Rationale	To ensure that generated reports are readily available for review and analysis, and securely stored within the system.
Business Rule	None
Dependencies	FR-19.2
Priority	High

3.3.20 Functional Requirements for UC-20: Generate Custom Reports

Users generate real-time reports based on specific criteria such as financial performance, sales trends, and User productivity.

3.3.20.1 FR-20.1: Specify Report Criteria

Attribute	Details
Identifier	FR-20.1
Title	Specify Report Criteria
Requirement	The system shall provide an interface for users to select or define the criteria for the report, such as financial performance metrics, sales trends, or user productivity.
Source	User
Rationale	To enable users to tailor reports to specific needs or investigative purposes, ensuring the reports provide relevant and targeted insights.
Business Rule	None
Dependencies	None
Priority	High

3.3.20.2 FR-20.3: Display Generated Report

Attribute	Details
Identifier	FR-20.2
Title	Display Generated Report
Requirement	Upon successful generation, the system shall display the report to the user, offering insights into the analyzed data.
Source	System
Rationale	To provide immediate access to generated reports, allowing users to review and use the information for business operations or strategic planning.
Business Rule	None
Dependencies	None
Priority	High

3.3.21 Functional Requirements for UC-21: Export Reports

Users export reports in various formats such as PDF, Excel, or CSV for further analysis or sharing.

3.3.21.1 FR-22.1: Choose Export Format

Attribute	Details
Identifier	FR-22.1
Title	Choose Export Format
Requirement	The system shall provide users with the option to select the desired export format for the report, including PDF, Excel, or CSV.
Source	User
Rationale	To allow users to export reports in a format that suits their needs for further analysis or sharing outside the system.
Business Rule	BR-1: All exports must comply with data export and security standards.
Dependencies	None
Priority	High

3.3.21.2 FR-22.2: Process and Export Report

Attribute	Details
Identifier	FR-22.2
Title	Process and Export Report
Requirement	Once a format is selected, the system shall process and convert the report into the chosen format, preparing it for export.
Source	System
Rationale	To generate a downloadable file or a file that can be emailed, ensuring that the report is in the correct format as requested by the user.
Business Rule	None
Dependencies	FR-22.1
Priority	High

3.3.22 Functional Requirements for UC-22: Customize Dashboards by Role

Admins create and assign different dashboards for different user roles, ensuring each user sees tailored data.

3.3.22.1 FR-22.1: Customize Dashboard Elements

Attribute	Details
Identifier	FR-22.1
Title	Customize Dashboard Elements
Requirement	The system shall provide an interface for Admins to customize dashboard elements and data views, ensuring these elements are suitable for specific user roles.
Source	Admin
Rationale	To enable Admins to tailor the dashboard's layout and information presentation according to the needs and privileges of different user roles.
Business Rule	BR-1: Customizations must not violate data access policies.
Dependencies	None
Priority	High

3.3.22.2 FR-22.2: Select User Role for Dashboard Assignment

Attribute	Details
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Identifier	FR-22.2
Title	Select User Role for Dashboard Assignment
Requirement	The system shall allow Admins to select a user role to which the customized dashboard settings will be applied.
Source	Admin
Rationale	To ensure that the customized dashboards are correctly assigned to the appropriate user roles, enhancing user experience and data security.
Business Rule	None
Dependencies	FR-22.1
Priority	High

3.3.22.3 FR-22.3: Apply Customized Settings to Role

Attribute	Details
Identifier	FR-22.3
Title	Apply Customized Settings to Role
Requirement	Upon selection of a user role, the system shall apply the customized dashboard settings to the specified role.
Source	System
Rationale	To automatically update the dashboard configuration for selected user roles, ensuring that each role has access to tailored data and functionalities.
Business Rule	None
Dependencies	FR-22.2
Priority	High

3.3.22.4 FR-22.4: Confirm and Log Customization

Attribute	Details
Identifier	FR-22.4
Title	Confirm and Log Customization
Requirement	The system shall confirm the successful application of dashboard customizations to the selected roles and log these changes for auditing purposes.
Source	System

Rationale	To provide administrative accountability and traceability for changes made to dashboard configurations, ensuring compliance with governance and policy standards.
Business Rule	None
Dependencies	FR-22.3
Priority	Medium

3.3.23 Functional Requirements for UC-23: Interactive Data Visualizations

Dashboards support interactive visualizations such as charts, graphs, tables that allow users to drill down into specific data points.

3.3.23.1 FR-23.1: Interact with Visualization Elements

Attribute	Details
Identifier	FR-23.1
Title	Interact with Visualization Elements
Requirement	The system shall provide interactive visualization elements such as charts, graphs, and tables that users can manipulate to view and analyze specific data points.
Source	Admin
Rationale	To enable Admins to dynamically interact with data visualizations, enhancing the analytical capabilities of the dashboard by allowing detailed exploration of data.
Business Rule	BR-1: Interactive elements must be accessible and provide accurate data.
Dependencies	None
Priority	High

3.3.23.2 FR-23.2: Display Detailed Data

Attribute	Details
Identifier	FR-23.2
Title	Display Detailed Data
Requirement	When a user interacts with a visualization element, the system shall display detailed data or further breakdowns related to the selected data point.

Source	System
Rationale	To provide deeper insights into the visualized data, allowing users to make informed decisions based on comprehensive data analysis.
Business Rule	None
Dependencies	FR-23.1
Priority	High

3.3.23.3 FR-23.3: Customize Visualization Placement

Attribute	Details
Identifier	FR-23.3
Title	Customize Visualization Placement
Requirement	The system shall allow the Admin to select and customize the position of the visualization elements on the dashboard.
Source	Admin
Rationale	To facilitate personalized dashboard layouts that cater to the user's preferences or organizational requirements, enhancing the usability of the dashboard.
Business Rule	None
Dependencies	FR-23.1
Priority	Medium

3.3.24 Functional Requirements for UC-24: Customize and Update Dashboards

Admin can customize the layout and content of their dashboards according to their needs, and visualizations are kept updated with real-time data.

3.3.24.1 FR-24.1: Select Dashboard Elements

Attribute	Details
Identifier	FR-24.1
Title	Select Dashboard Elements
Requirement	The system shall provide an interface for admins to add, remove, or modify elements on their dashboard according to their needs.
Source	Admin

Rationale	To allow admins to tailor the dashboard by choosing elements that are most relevant to their tasks or the information they need to display.
Business Rule	None
Dependencies	None
Priority	High

3.3.24.2 FR-24.2: Arrange Dashboard Layout

Attribute	Details
Identifier	FR-24.2
Title	Arrange Dashboard Layout
Requirement	The system shall enable admins to rearrange the layout of the dashboard elements to fit their viewing preferences.
Source	Admin
Rationale	To provide flexibility in how information is displayed on the dashboard, enhancing user experience and accessibility.
Business Rule	None
Dependencies	FR-24.1
Priority	High

3.3.24.3 FR-24.3: Save Customizations and Refresh Dashboard

Attribute	Details
Identifier	FR-24.3
Title	Save Customizations and Refresh Dashboard
Requirement	After customization, the system shall save the new settings and refresh the dashboard to display up-to-date data with the new configuration.
Source	System
Rationale	To ensure that all changes are preserved and immediately reflected on the dashboard, providing admins with the latest data in their preferred layout.
Business Rule	BR-1: Customizations must not interfere with the overall performance of the dashboard.
Dependencies	FR-24.2
Priority	High

3.3.25 Functional Requirements for UC-25: Create Invoices

Admin can create invoices for their customer products or services, ensuring accurate billing records.

3.3.25.1 FR-25.1: Enter Customer and Product Details

Attribute	Details
Identifier	FR-25.1
Title	Enter Customer and Product Details
Requirement	The system shall provide an interface for admins to enter or select existing customer details and the products or services to be billed.
Source	Admin
Rationale	To ensure all necessary information is included for accurate invoicing, including customer contact information and specific products or services rendered.
Business Rule	None
Dependencies	None
Priority	High

3.3.25.2 FR-25.2: Calculate Total Cost

Attribute	Details
Identifier	FR-25.2
Title	Calculate Total Cost
Requirement	The system shall automatically calculate the total cost of the products or services, including any applicable taxes or discounts, based on the details entered.
Source	System
Rationale	To provide an accurate calculation of charges for the services or products offered, ensuring that the invoice reflects the correct amount due.
Business Rule	BR-1: Invoices must comply with local financial regulations.
Dependencies	FR-25.1
Priority	High

3.3.25.3 FR-25.3: Review and Finalize Invoice

Attribute	Details
Identifier	FR-25.3
Title	Review and Finalize Invoice
Requirement	After cost calculation, the system shall allow the admin to review all invoice details for accuracy before final issuance.
Source	Admin
Rationale	To ensure that all information on the invoice is correct and complete before it is finalized and sent to the customer.
Business Rule	None
Dependencies	FR-25.2
Priority	High

3.3.26 Functional Requirements for UC-26: Manage Receipts and Payments

Users either download receipts or make payments through a payment gateway, tracking the status of each transaction.

3.3.26.1 FR-26.1: Select Transaction Option

Attribute	Details
Identifier	FR-26.1
Title	Select Transaction Option
Requirement	The system shall provide an interface for users to choose whether to manage payments or download receipts for their transactions.
Source	User
Rationale	To allow users to specify their transaction needs, whether processing payments or accessing financial documents.
Business Rule	None
Dependencies	None
Priority	High

3.3.26.2 FR-26.2: Enter Payment Details

Attribute	Details
Identifier	FR-26.2
Title	Enter Payment Details
Requirement	For payment processing, the system shall provide a secure form for users to enter their payment details, such as credit card information.
Source	User
Rationale	To capture necessary payment information securely, enabling the processing of transactions.
Business Rule	BR-1: Payments must be processed securely using approved gateways.
Dependencies	FR-26.1
Priority	High

3.3.26.3 FR-26.3: Process Payment

Attribute	Details
Identifier	FR-26.3
Title	Process Payment
Requirement	Upon entering payment details, the system shall process the payment through an approved payment gateway and update the transaction status.
Source	System
Rationale	To ensure that all payments are processed efficiently and securely, reflecting the transaction status immediately in the user's account.
Business Rule	None
Dependencies	FR-26.2
Priority	High

3.3.26.4 FR-26.4: Generate and Download Receipt

Attribute	Details
Identifier	FR-26.4
Title	Generate and Download Receipt
Requirement	The system shall generate a receipt for completed transactions and provide an option for users to download it in a suitable format.
Source	System
Rationale	To provide legal documentation of transactions for users, allowing them to download and save or print receipts as needed.

Business Rule	None
Dependencies	FR-26.1
Priority	High

3.3.27 Functional Requirements for UC-27: Monitor Invoice Status

Users monitor the status of open invoices, receiving notifications and alerts for overdue payments.

3.3.27.1 FR-27.1: Display Invoice Status

Attribute	Details
Identifier	FR-27.1
Title	Display Invoice Status
Requirement	The system shall display the current status of each invoice, including whether they are paid, pending, or overdue.
Source	System
Rationale	To provide a clear and updated overview of invoice statuses, allowing users to track payment progress and identify any issues promptly.
Business Rule	None
Dependencies	None
Priority	High

3.3.27.2 FR-27.3: Alert for Overdue Invoices

Attribute	Details
Identifier	FR-27.2
Title	Alert for Overdue Invoices
Requirement	The system shall alert users to any invoices that are overdue, specifying the amount and the duration of the overdue status.
Source	System
Rationale	To ensure that users are immediately aware of overdue invoices, facilitating timely follow-up actions to manage potential financial risks.
Business Rule	BR-1: System must alert users to overdue invoices within one day of the due date.

Dependencies	FR-27.1
Priority	High

3.3.28 Functional Requirements for UC-28: Create Income and Expense Category

User can create a new income and expense category in the accounting module.

3.3.28.1 FR-28.1: Access Income and Expense Category Section

Attribute	Details
Identifier	FR-28.1
Title	Access Income and Expense Category Section
Requirement	The system shall provide users with access to the income and expense category section for managing financial categorizations.
Source	User
Rationale	To enable users to navigate to the correct section within the ERP system to manage and create new financial categories.
Business Rule	None
Dependencies	None
Priority	High

3.3.28.2 FR-28.2: Add New Category

Attribute	Details
Identifier	FR-28.2
Title	Add New Category
Requirement	The system shall allow users to create new income and expense categories by specifying a category name, description, and associated vendor.
Source	User
Rationale	To facilitate the addition of new financial categories to the system, allowing for better organization and tracking of income and expenses.
Business Rule	BR-1: Income and expense categories must be unique and properly defined.
Dependencies	FR-28.1
Priority	High

3.3.28.3 FR-28.3: Validate Category Details

Attribute	Details
Identifier	FR-28.3
Title	Validate Category Details
Requirement	The system shall validate the details entered for the new category to ensure uniqueness and completeness before submission.
Source	System
Rationale	To prevent the creation of duplicate or incomplete categories, ensuring data integrity and usability within the financial tracking system.
Business Rule	None
Dependencies	FR-28.2
Priority	High

3.3.28.4 FR-28.4: Submit and Confirm Category Creation

Attribute	Details
Identifier	FR-28.4
Title	Submit and Confirm Category Creation
Requirement	Upon successful validation, the system shall save the new category and confirm its addition to the user.
Source	System
Rationale	To confirm the successful creation of the new financial category and ensure it is available for immediate use in the system.
Business Rule	None
Dependencies	FR-28.3
Priority	High

3.3.29 Functional Requirements for UC-29: Add Account Incomes and Expense

User can record a new income entry in the system.

3.3.29.1 FR-29.1: Access Account Income and Expense Section

Attribute	Details
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Identifier	FR-29.1
Title	Access Account Income and Expense Section
Requirement	The system shall provide users with access to the account incomes and expenses section where they can manage financial entries.
Source	User
Rationale	To enable users to navigate to the correct module within the ERP system for managing financial transactions.
Business Rule	None
Dependencies	None
Priority	High

3.3.29.2 FR-29.2: Input Financial Details

Attribute	Details
Identifier	FR-29.2
Title	Input Financial Details
Requirement	The system shall allow users to input details related to an income or expense entry, including vendor name, category, expense name, date, amounts, and payment mode.
Source	User
Rationale	To capture all necessary information for accurate financial tracking and categorization within the system.
Business Rule	None
Dependencies	FR-29.1
Priority	High

3.3.29.3 FR-29.3: Validate Financial Entry

Attribute	Details
Identifier	FR-29.3
Title	Validate Financial Entry
Requirement	Before saving the entry, the system shall validate the accuracy and completeness of all financial data entered by the user.
Source	System
Rationale	To ensure that all financial records are accurate, complete, and compliant with regulatory standards before being saved.

Business Rule	BR-1: The system must validate the accuracy of all income data before saving.
Dependencies	FR-29.2
Priority	High

3.3.29.4 FR-29.4: Submit and Update Financial Records

Attribute	Details
Identifier	FR-29.4
Title	Submit and Update Financial Records
Requirement	After validation, the system shall submit the financial entry and update the overall financial records accordingly.
Source	System
Rationale	To ensure that the financial database is promptly updated with new entries, maintaining an accurate and current financial status within the ERP system.
Business Rule	None
Dependencies	FR-29.3
Priority	High

3.3.30 UC-30: Manage To-Do List

Admin can create a list of task and assign it to user

3.3.30.1 FR-30.1: Access To-Do List Section

Attribute	Details
Identifier	FR-30.1
Title	Access To-Do List Section
Requirement	The system shall provide admins with access to the to-do list page where they can manage tasks.
Source	Admin
Rationale	To enable admins to easily navigate to the to-do list module for creating and managing tasks.
Business Rule	None
Dependencies	None
Priority	High

3.3.30.2 FR-30.2: Add and Specify Task Details

Attribute	Details
Identifier	FR-30.2
Title	Add and Specify Task Details
Requirement	The system shall allow admins to add new tasks with specific details such as title, priority, and deadline.
Source	Admin
Rationale	To capture all necessary information for task management, facilitating organization and prioritization of tasks.
Business Rule	BR-1: Tasks must have a title and can have optional deadlines and priority levels.
Dependencies	FR-30.1
Priority	High

3.3.30.3 FR-30.3: Edit Task Details

Attribute	Details
Identifier	FR-30.3
Title	Edit Task Details
Requirement	The system shall provide an option for admins to edit existing tasks, allowing modification of details like title, priority, and deadline.
Source	Admin
Rationale	To enable easy updates to task details, ensuring that task information remains relevant and accurate.
Business Rule	None
Dependencies	FR-30.2
Priority	Medium

3.3.30.4 FR-30.4: Delete Task

Attribute	Details
Identifier	FR-30.4
Title	Delete Task
Requirement	The system shall allow admins to delete tasks from the to-do list, including a confirmation step to prevent accidental deletions.
Source	Admin

Rationale	To maintain a clean and updated to-do list by removing tasks that are no longer relevant or have been completed.
Business Rule	None
Dependencies	FR-30.3
Priority	Medium

3.3.31 Functional Requirements for UC-31: Schedule Events in Calendar

User can add an event like meeting and any other event in the calendar for remainder purpose.

3.3.31.1 FR-31.1: Add New Event

Attribute	Details
Identifier	FR-31.1
Title	Add New Event
Requirement	The system shall allow users to add new events to the calendar by specifying details such as date, time, title, description, and recurrence.
Source	User
Rationale	To facilitate the creation of detailed and recurring events within the calendar, enhancing organizational efficiency.
Business Rule	BR-1: Events must have a title and can optionally include a description.
Dependencies	None
Priority	High

3.3.31.2 FR-31.2: Modify and Delete Events

Attribute	Details
Identifier	FR-31.2
Title	Modify and Delete Events
Requirement	The system shall provide functionality for users to modify or delete existing events from the calendar.
Source	User
Rationale	To allow users to maintain accurate and current calendars by updating or removing events as needed.
Business Rule	None

Dependencies	FR-31.1
Priority	Medium

3.3.32 Functional Requirements for UC-32: Set Reminders and Receive Notifications

User get the notification about the event added in the calendar.

3.3.32.1 FR-32.1: Access Reminder Settings

Attribute	Details
Identifier	FR-32.1
Title	Access Reminder Settings
Requirement	The system shall provide users with access to the reminder settings interface.
Source	User
Rationale	To enable users to easily set reminders for upcoming tasks or events directly from the ERP system.
Business Rule	None
Dependencies	None
Priority	High

3.3.32.2 FR-32.2: Specify Reminder Details

Attribute	Details
Identifier	FR-32.2
Title	Specify Reminder Details
Requirement	The system shall allow users to specify reminder details, including the event, time, and method of notification.
Source	User
Rationale	To ensure that users receive timely notifications about important events, supporting better time management and task prioritization.
Business Rule	BR-1: Reminders can be set for any task or event that has a defined date and time.
Dependencies	FR-32.1
Priority	High

3.3.33 Functional Requirements for UC-33: Chatbot Assistance for Schema Creation

The chatbot provides instant answers to common questions regarding schema creation for new modules, enhancing user understanding and efficiency.

3.3.33.1 FR-33.1: Access Chatbot Interface

Attribute	Details
Identifier	FR-33.1
Title	Access Chatbot Interface
Requirement	The system shall provide a user-friendly interface for admins to interact with the chatbot regarding schema creation queries.
Source	Admin
Rationale	To enable easy access to automated guidance for schema creation, enhancing user understanding and efficiency in module development.
Business Rule	None
Dependencies	None
Priority	High

3.3.33.2 FR-33.2: Process Schema Queries

Attribute	Details
Identifier	FR-33.2
Title	Process Schema Queries
Requirement	The chatbot shall process queries regarding schema creation and respond with guidance based on the latest development guidelines.
Source	System
Rationale	To provide immediate, accurate assistance to admins, supporting effective schema design for new modules.
Business Rule	None
Dependencies	FR-33.1
Priority	High

3.3.34 Functional Requirements for UC-34: Format Guidance for Excel Schema

The chatbot advises users on the appropriate format and structure required for schemas within an Excel sheet, ensuring data consistency and correctness.

3.3.34.1 FR-34.1: Provide Format Guidance

Attribute	Details
Identifier	FR-34.1
Title	Provide Format Guidance
Requirement	The chatbot shall offer guidance on the appropriate Excel format and structure for schemas, ensuring data consistency and correctness.
Source	Admin
Rationale	To assist users in correctly formatting Excel schemas for the creation of multiple database tables, reducing errors and ensuring data integrity.
Business Rule	None
Dependencies	None
Priority	High

3.3.35 Functional Requirements for UC-35: Collect Feedback for Performance Improvement

The chatbot collects user feedback and ratings to assess its performance and improve user experience.

3.3.35.1 FR-35.1: Collect Feedback

Attribute	Details
Identifier	FR-35.1
Title	Collect Feedback
Requirement	After interactions, the chatbot shall prompt users to provide feedback on their experience, collecting insights on its performance and user satisfaction.
Source	System

Rationale	To gather user feedback for ongoing assessment and improvement of chatbot functionalities, ensuring alignment with user needs and expectations.
Business Rule	None
Dependencies	None
Priority	Medium

3.3.35.2 FR-35.2: Analyze Feedback

Attribute	Details
Identifier	FR-35.2
Title	Analyze Feedback
Requirement	The system shall analyze the collected feedback to identify trends, issues, and areas for improvement, informing future updates to the chatbot.
Source	System
Rationale	To use user feedback effectively for enhancing chatbot performance and accuracy, facilitating better user experiences over time.
Business Rule	None
Dependencies	FR-35.1
Priority	High

3.4 Non-Functional Requirements

3.4.1 Reliability

- **REL-1:** The ModuFlex ERP application shall maintain availability and accessibility to users 99.9% of the time.
- **REL-2:** In the event of application crashes, ModuFlex ERP shall prompt users to send anonymized diagnostic reports, including failure details and device information, for swift issue resolution and continuous improvement.

3.4.2 Usability

- **USE-1:** ModuFlex ERP shall provide an interactive onboarding process for new users, requiring no more than 10 minutes to introduce key functionalities, aimed at reducing initial learning curve and enhancing early user engagement.

- **USE-2:** The system shall ensure that all primary functions are accessible within three clicks from the home screen, promoting ease of use and efficient navigation.
- **USE-3:** ModuFlex ERP shall support customizable user interfaces, allowing users to adjust layouts, themes, and data presentations according to their preferences and accessibility needs.

3.4.3 Performance

- **PER-1:** ModuFlex ERP will ensure efficient performance, initiating its launch within 10 seconds on first use and 5 seconds for subsequent accesses, thereby providing rapid access to its services and significantly reducing waiting times for users.
- **PER-2:** The system will consistently maintain high responsiveness, reacting to user inputs within 2 seconds, to offer a fluid and delay-free user experience.
- **PER-3:** Should there be a network disruption, ModuFlex ERP will alert users to restore their internet connection and will continue attempting to reconnect for up to 10 seconds before timing out, keeping users well-informed of connectivity issues and reducing potential frustration.
- **PER-4:** Financial transactions conducted through ModuFlex ERP, using designated payment gateways, will be processed within 10 seconds to ensure quick and secure financial operations. If a transaction cannot be completed within this timeframe, the system will automatically return to the previous page to avoid user inconvenience and maintain transaction reliability.

3.4.4 Security

- **SEC-1:** ModuFlex ERP shall employ verification to confirm the user's email for authentication, enhancing account security and preventing unauthorized access.
- **SEC-2:** To change login credentials, users shall undergo OTP verification, adding an extra layer of security to account management processes and safeguarding user accounts against unauthorized modifications

3.5 External Interface Requirements

3.5.1 User Interfaces Requirements

User Interface Requirements for the ModuFlex ERP web application, the UI will be crafted to ensure ease of navigation and operational efficiency.

- Adhere to the latest Web Content Accessibility Guidelines (WCAG) and HTML5/CSS3 standards to ensure a consistent experience across various web browsers.
- Include a consistent placement of 'Home' and 'Back' buttons across all views. Utilize a standard navigation drawer for additional settings and features.
- Error messages, alerts, and confirmation prompts will be clear and brief, utilizing toast notifications for transient messages and modal dialogs for important interactions.
- Initially provide support for English, with infrastructure in place to include additional languages based on user demographics and demand.
- Ensure that the UI design is responsive, adapting seamlessly to various screen sizes and resolutions commonly used in web environments

3.5.2 Software interfaces

SI-1: ModuFlex ERP will interact with the following software components:

- Payment Gateway API (stripe)

3.5.3 Hardware interfaces

The ModuFlex ERP system is accessible via devices capable of running modern web browsers such as Chrome 90+, Firefox 88+, and Safari 14+. This ensures that the system is usable on PCs, laptops, tablets, and smartphones.

3.5.4 Communications interfaces

CI-1: ModuFlex ERP shall send emails for account verification, password resets, and event notification.

4 Design and Architecture

4.1 Architectural Design

The ModuFlex ERP System will adopt a Hybrid Architecture that combines 3-tier architecture with the MVC pattern, emphasizing modularity, simplicity, and efficiency. The system is structured around independent modules, each handling specific functionalities like user management, financial tracking, HR management, and task scheduling. The Presentation Layer features web and mobile-friendly interfaces for user interaction, dashboards, and notifications. The Application Layer manages business logic, including user authentication, role assignments, and module integration, while the Data Layer securely stores and retrieves information such as user profiles, financial records, and HR data. This architecture ensures scalability, cost-effectiveness, and reliability, providing a user-friendly and adaptable ERP solution for diverse business needs.

Figure 4.1 shows the Model View Controller (MVC) of use ModuFlex ERP System.

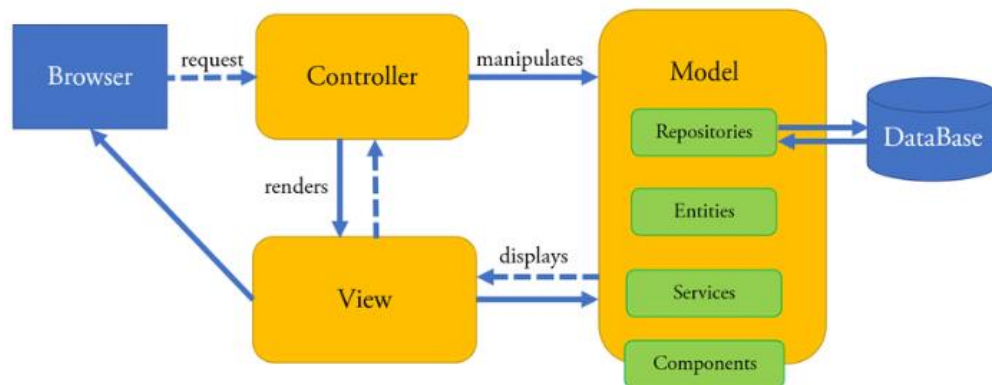


Figure 4-1 MVC Diagram of ModuFlex ERP System

4.2 Design Models

4.2.1 Activity Diagram

The activity diagrams of modules are as follows:

4.2.1.1 Login

Figure 4.2 shows the activity diagram of “Login” use case which Illustrates secure user login, error handling, and password recovery via email-based reset, ensuring smooth access management.

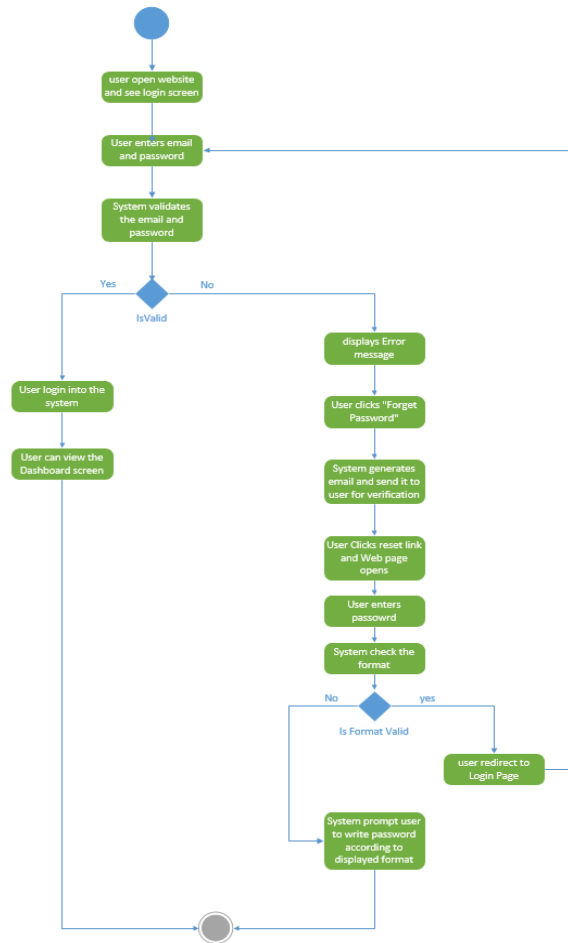


Figure 4-2 Login Activity Diagram

4.2.1.2 Edit Profile

Figure 4.3 shows the activity diagram of “Edit profile” use case which Demonstrates user actions to securely update profile details and passwords, highlighting data validation and clear user feedback.

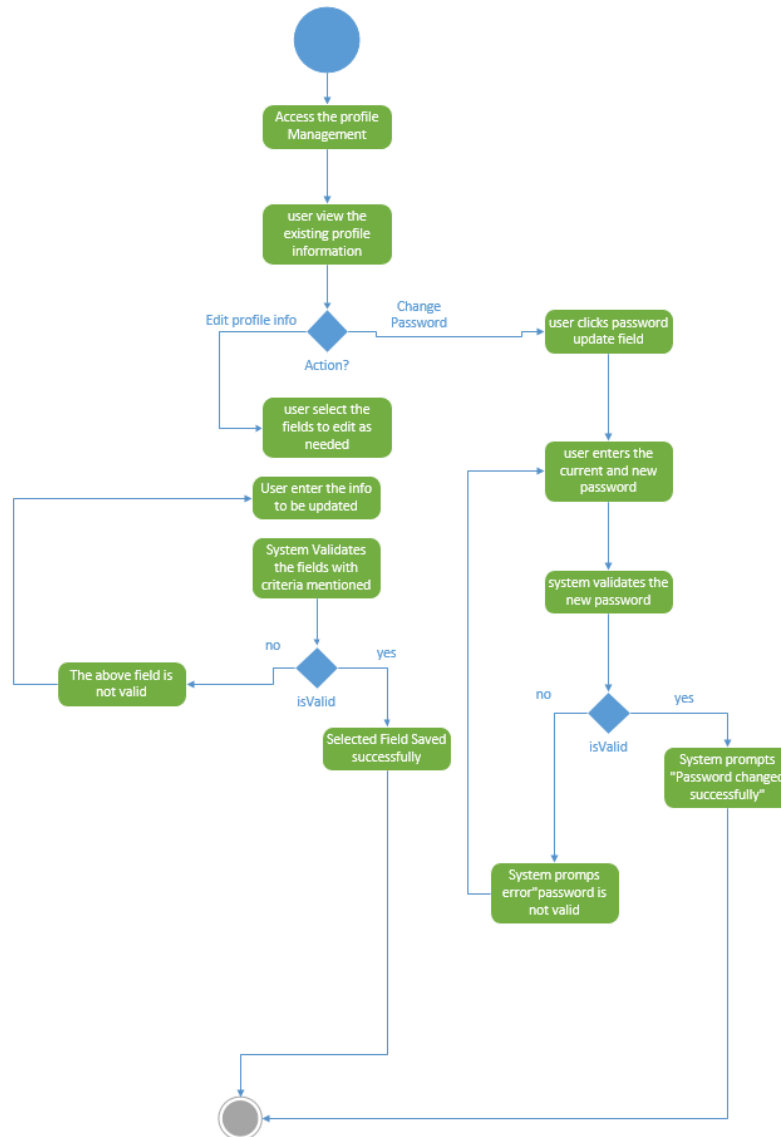


Figure 4-3 Edit profile Activity diagram

4.2.1.3 Delete User Account

Figure 4.4 shows the activity diagram of “Delete User Account and logout” use case which Shows secure session termination and role-based account deletion, emphasizing proper authorization and user communication.

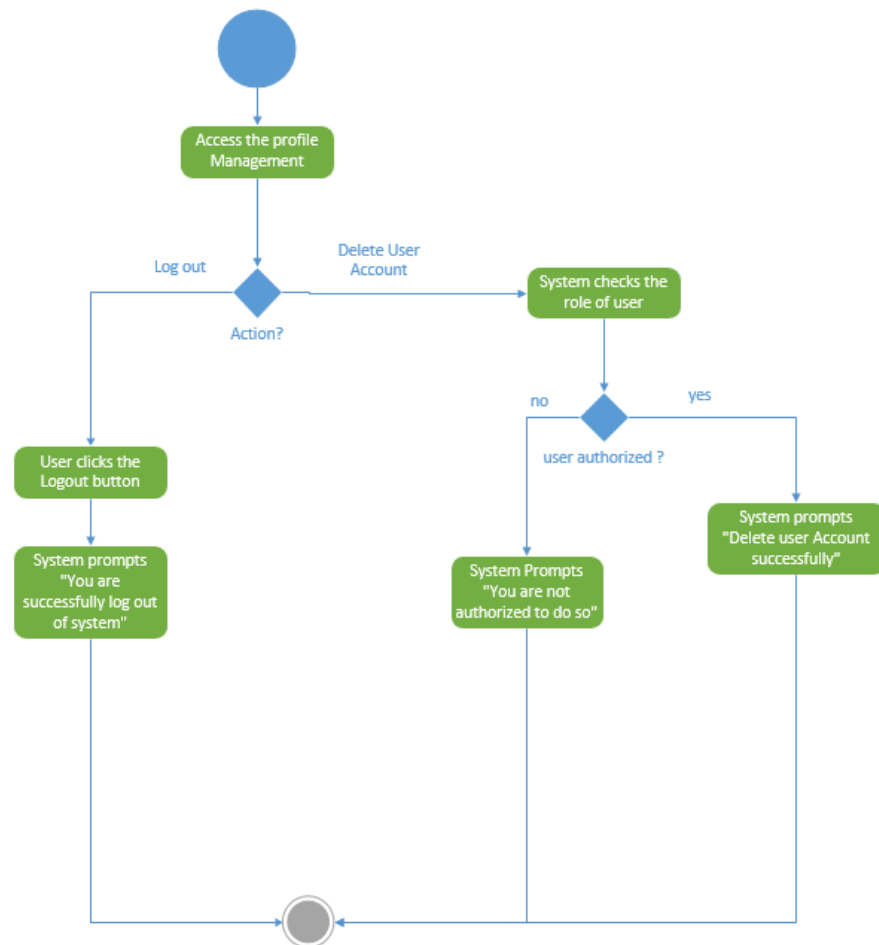


Figure 4-4 Delete User Account Activity Diagram

4.2.1.4 Role Creation and Assignment

Figure 4.5 shows the activity diagram of “role and permission” use case which Demonstrates role creation, assignment, and custom dashboards, ensuring roles have appropriate access and visualization tailored to user responsibilities.

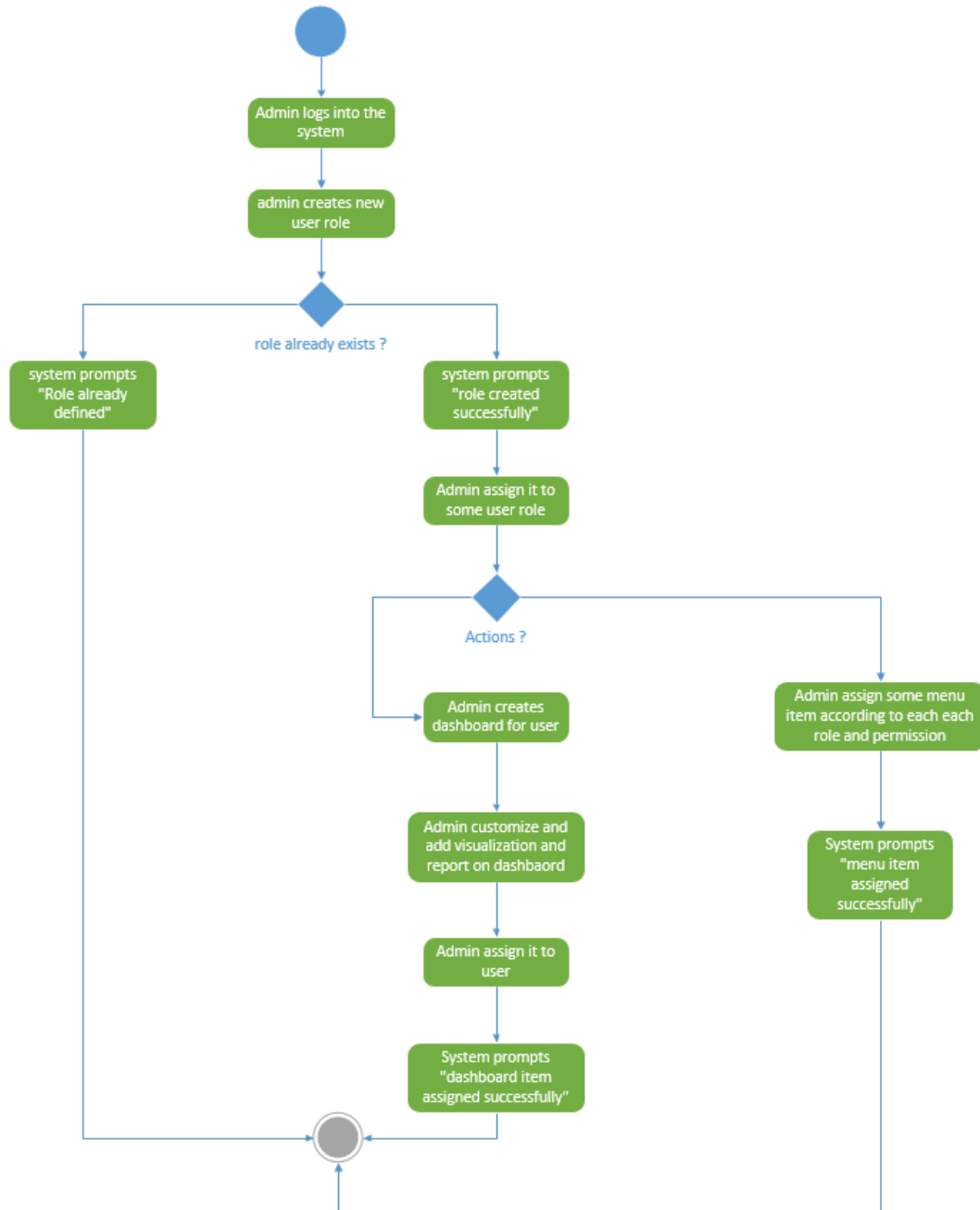


Figure 4-5 Role and Permission Activity Diagram

4.2.1.5 CRUD Generator

Figure 4.6 shows the activity diagram of “Integration and development” use case which Illustrates manual and Excel-based database schema creation, table relationships, and automated code generation, ensuring consistency and efficient integration into the system.

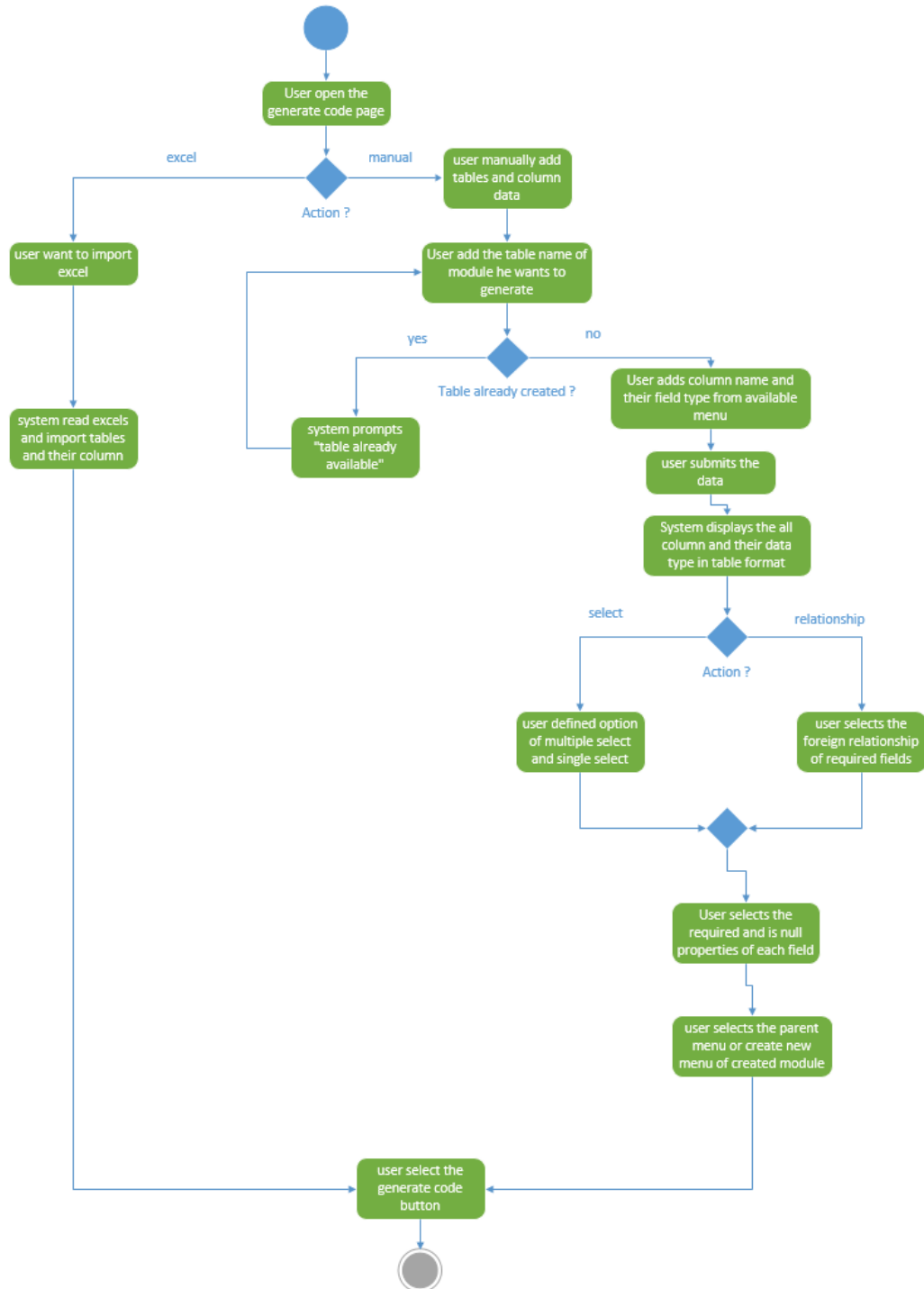


Figure 4-6 Integration and development Activity Diagram

4.2.1.6 Create New User Profile

Figure 4.7 shows the activity diagram of “create new user profile” use case which Depicts the process of creating and validating new user roles within the HR management module, ensuring complete and accurate user data entry.

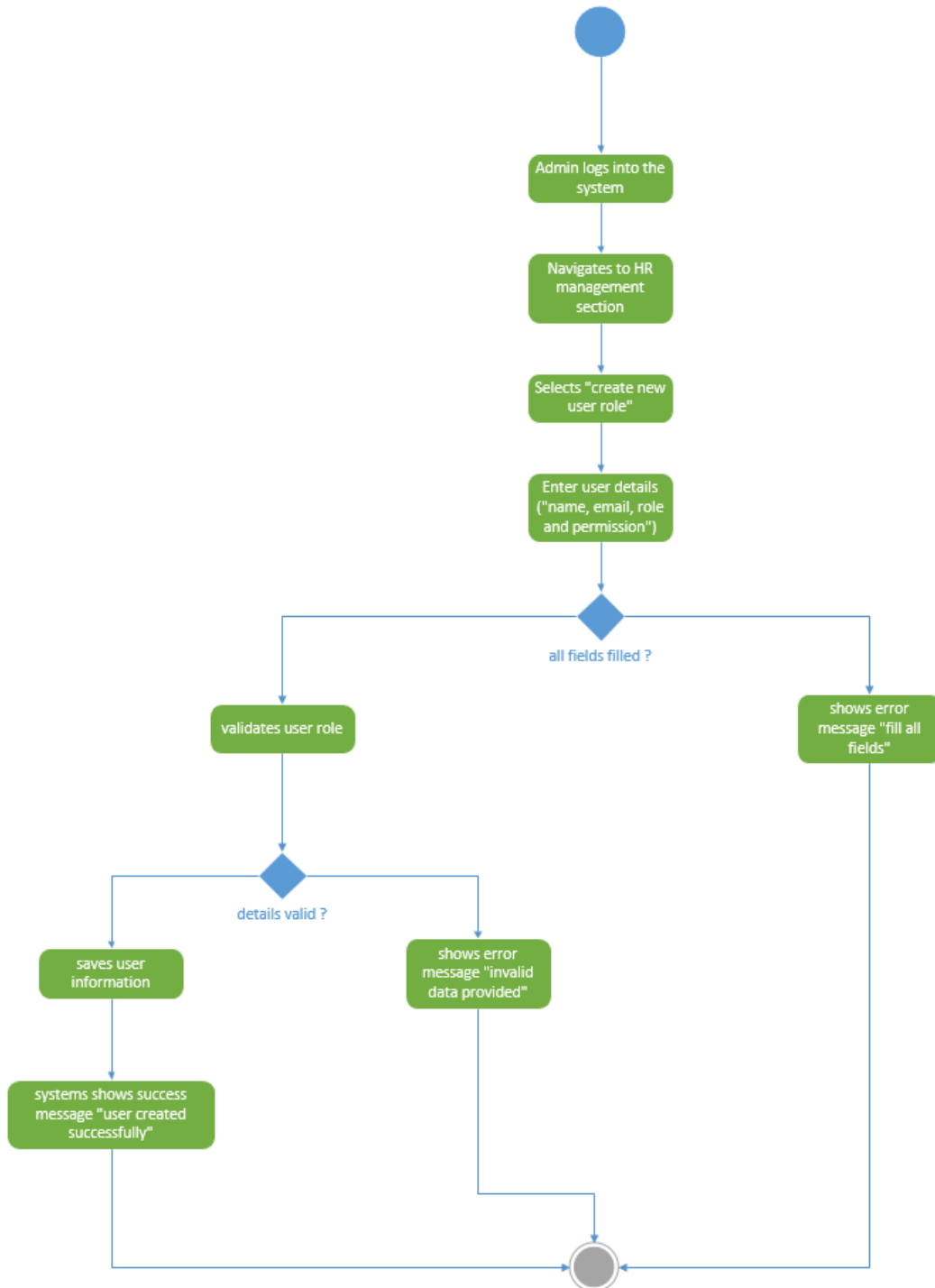


Figure 4-7 Create new user profile Activity Diagram

4.2.1.7 Submit and Track Expense Report

Figure 4.8 shows the activity diagram of “submit and track expense report” use case which Shows the creation, validation, tracking, and administrative approval/rejection process of expense reports, enhancing accuracy and accountability in financial management.

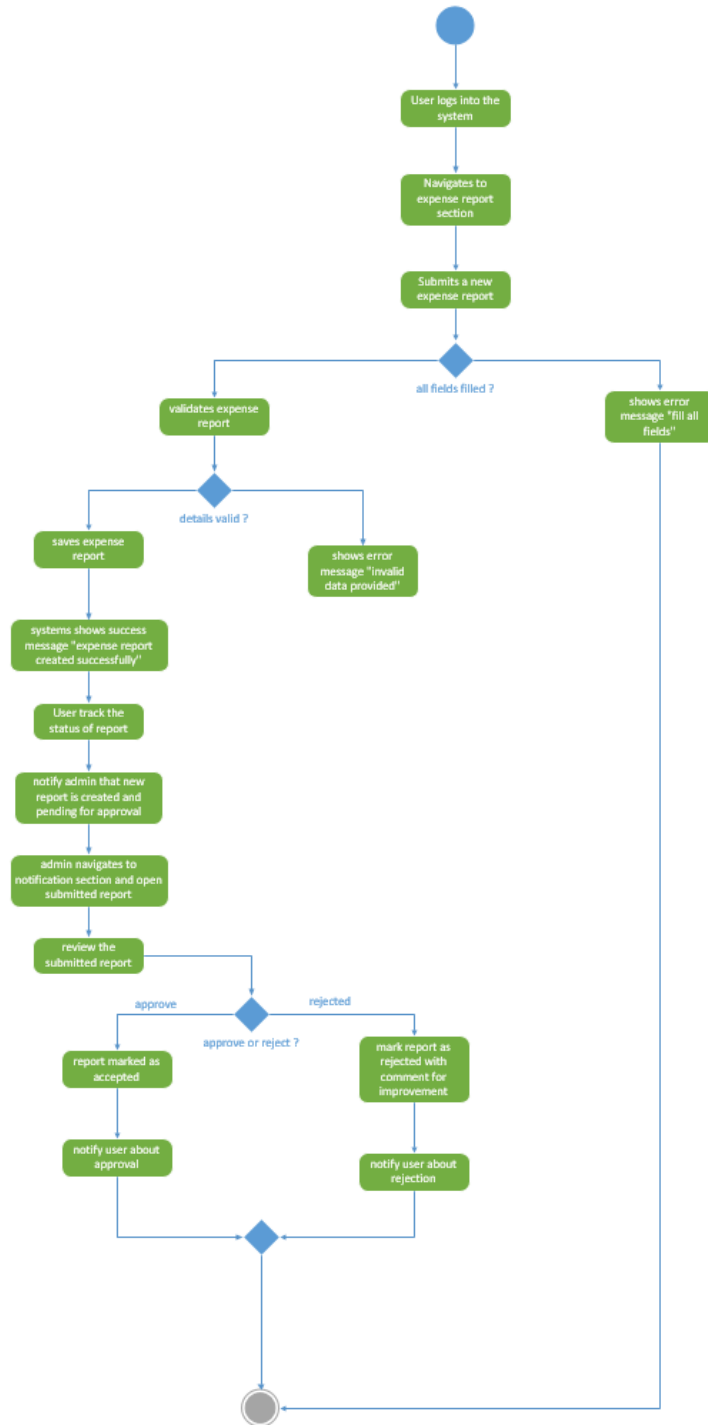


Figure 4-8 Submit and track expense report Activity Diagram

4.2.1.8 Leave Request

Figure 4.9 shows the activity diagram of “Leave request” use case which Outlines leave request creation, validation, submission, and administrative approval workflows, providing clear status tracking and streamlined HR processes

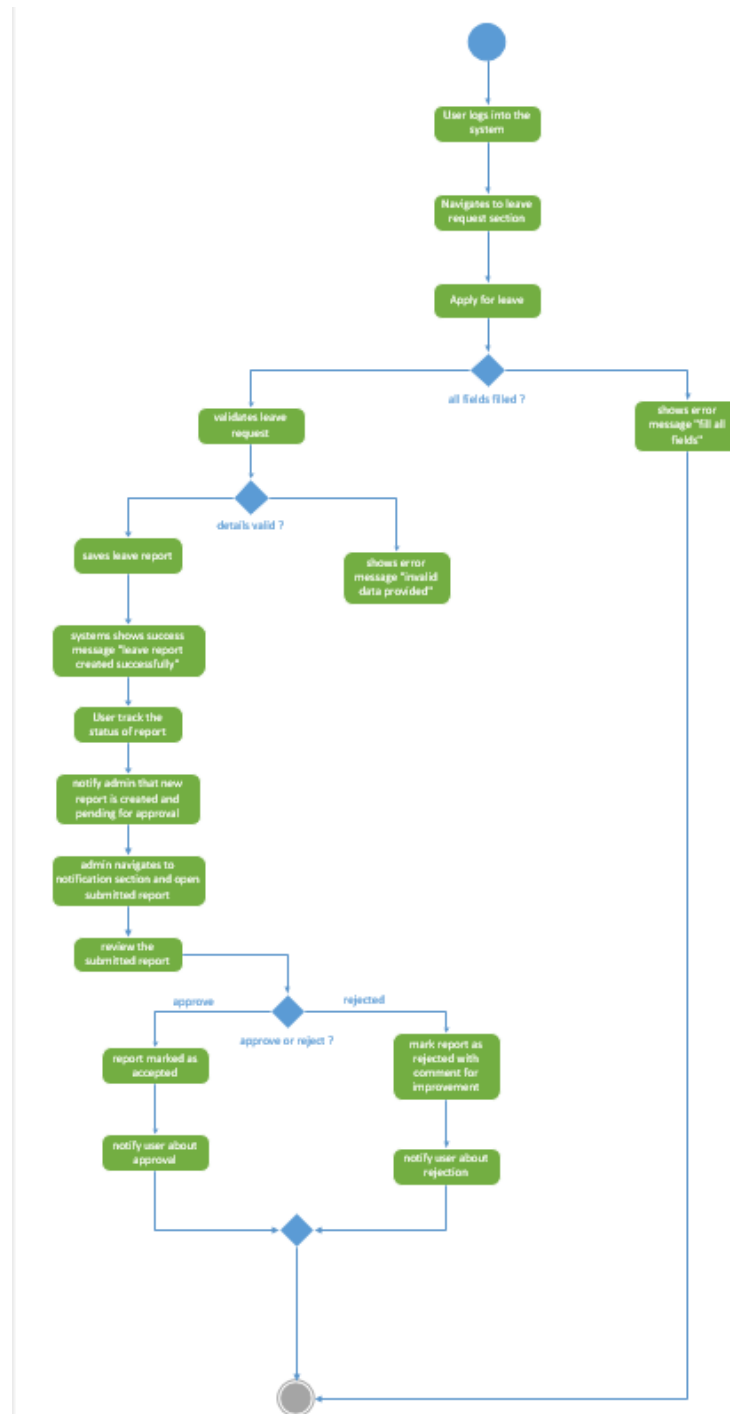


Figure 4-9 Leave Request Activity Diagram

4.2.1.9 Timesheets

Figure 4.10 shows the activity diagram of “Timesheets” use case which Displays timesheet creation, validation, approval or modification processes, and ensures accurate employee time tracking for payroll and productivity reporting.

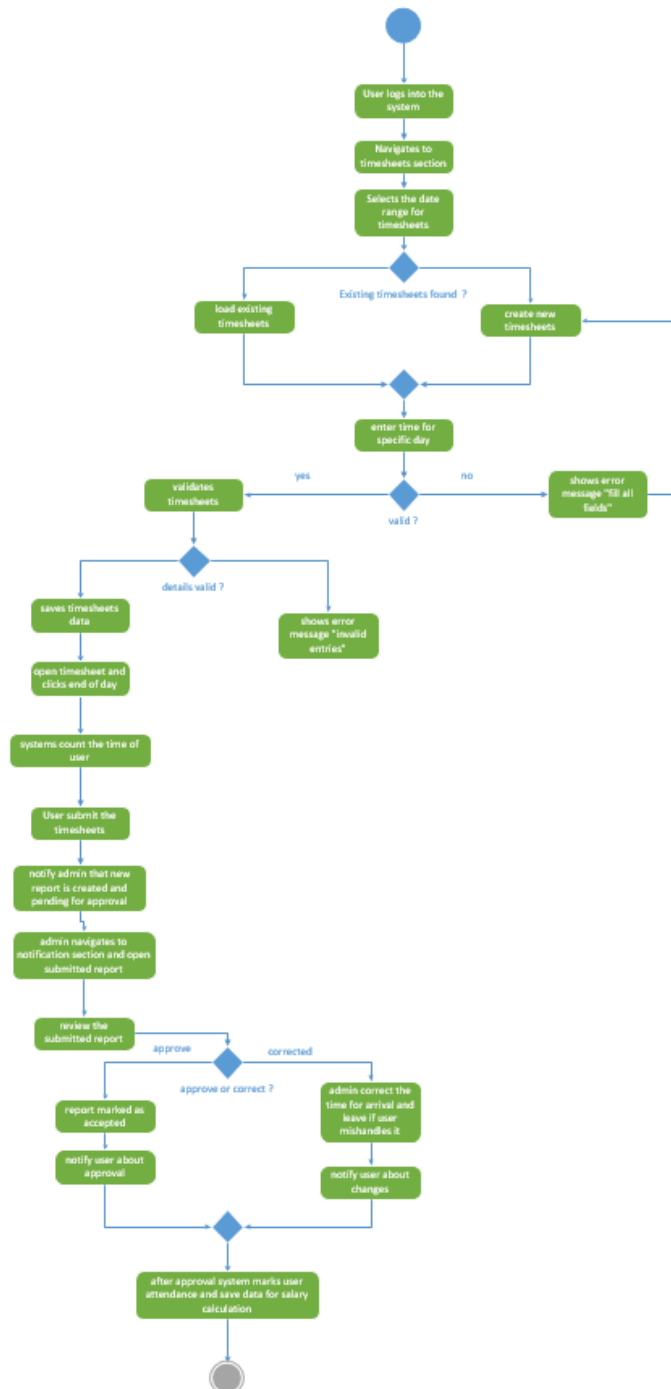


Figure 4-10 Timesheets Activity Diagram

4.2.1.10 Generate Report

Figure 4.11 shows the activity diagram of “Generate Report” use case which Demonstrates flexible HR and custom report generation, viewing, and exporting options, supporting insightful decision-making through real-time data visualization.

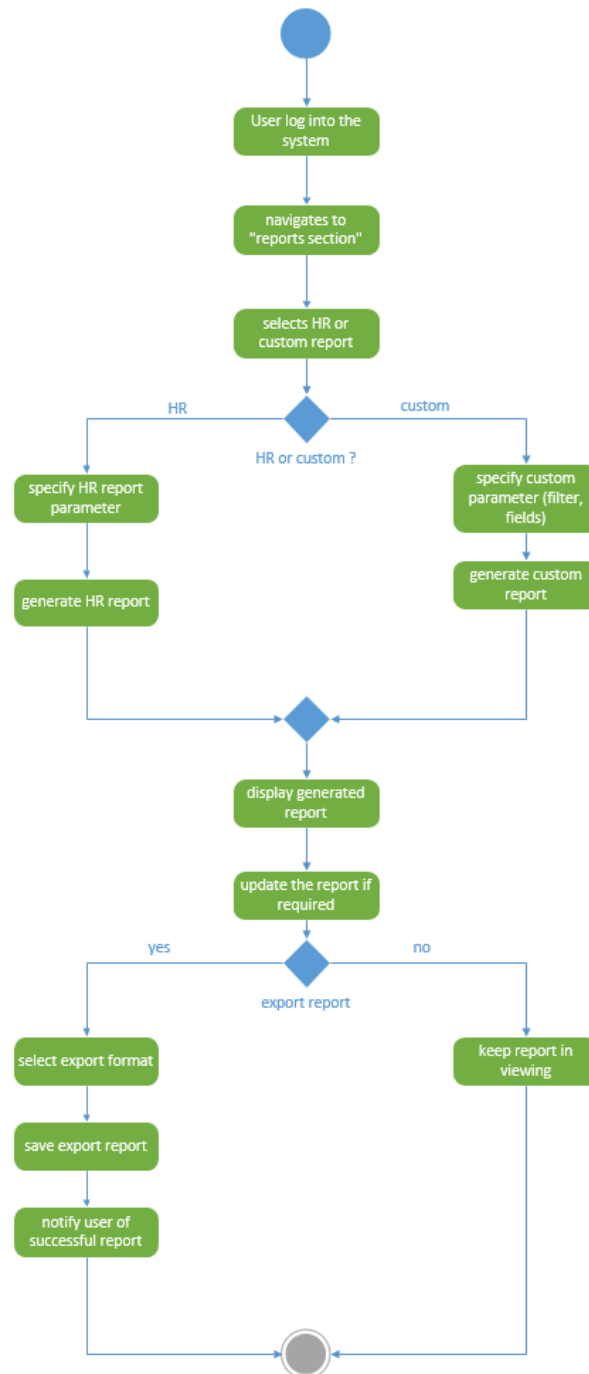


Figure 4-11 Generate Reports Activity Diagram

4.2.1.11 Create and Customize Dashboard

Figure 4.12 shows the activity diagram of “Create and Customize dashboard” use case which Illustrates the administrative workflow for creating, customizing, and assigning interactive dashboards tailored to user roles for insightful, real-time data monitoring.

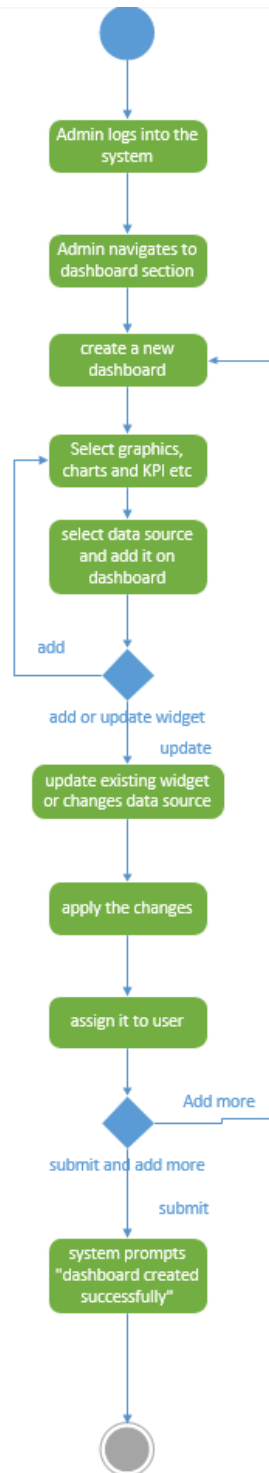


Figure 4-12 Dashboard creation Activity Diagram

4.2.1.12 Invoices Payment and Generate Receipts

Figure 4.13 shows the activity diagram of “Invoices payment and generate receipt” use case which Shows invoice creation and validation, payment method handling (card/cash), receipt generation, and invoice status tracking, ensuring streamlined financial operations.

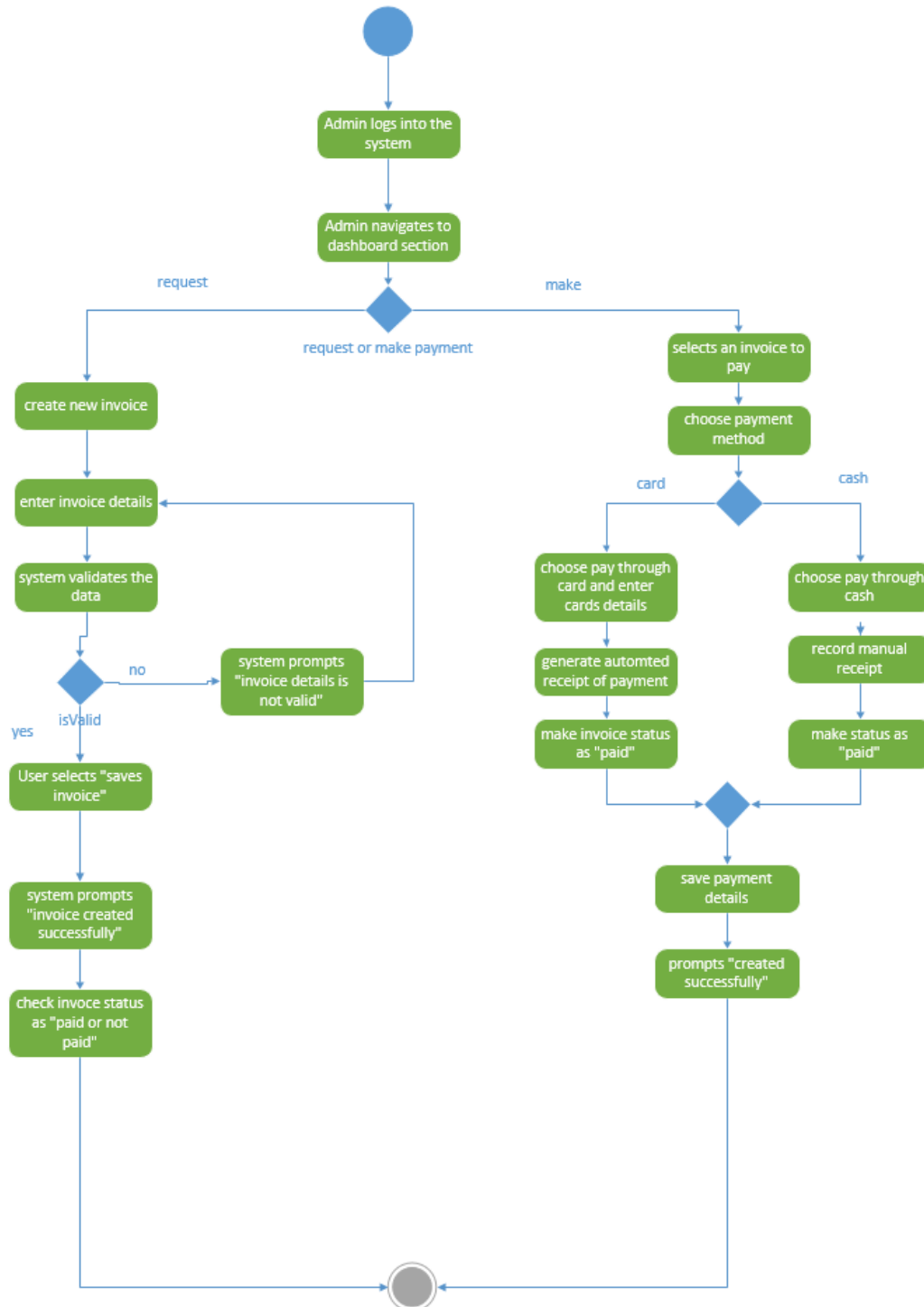


Figure 4-13 Invoices and payment Activity Diagram

4.2.1.13 Financial Inputs

Figure 4.14 shows the activity diagram of “Financial Inputs” use case which Presents the categorization, recording, and validation of income and expense transactions, enhancing accurate financial management and reporting within the system.

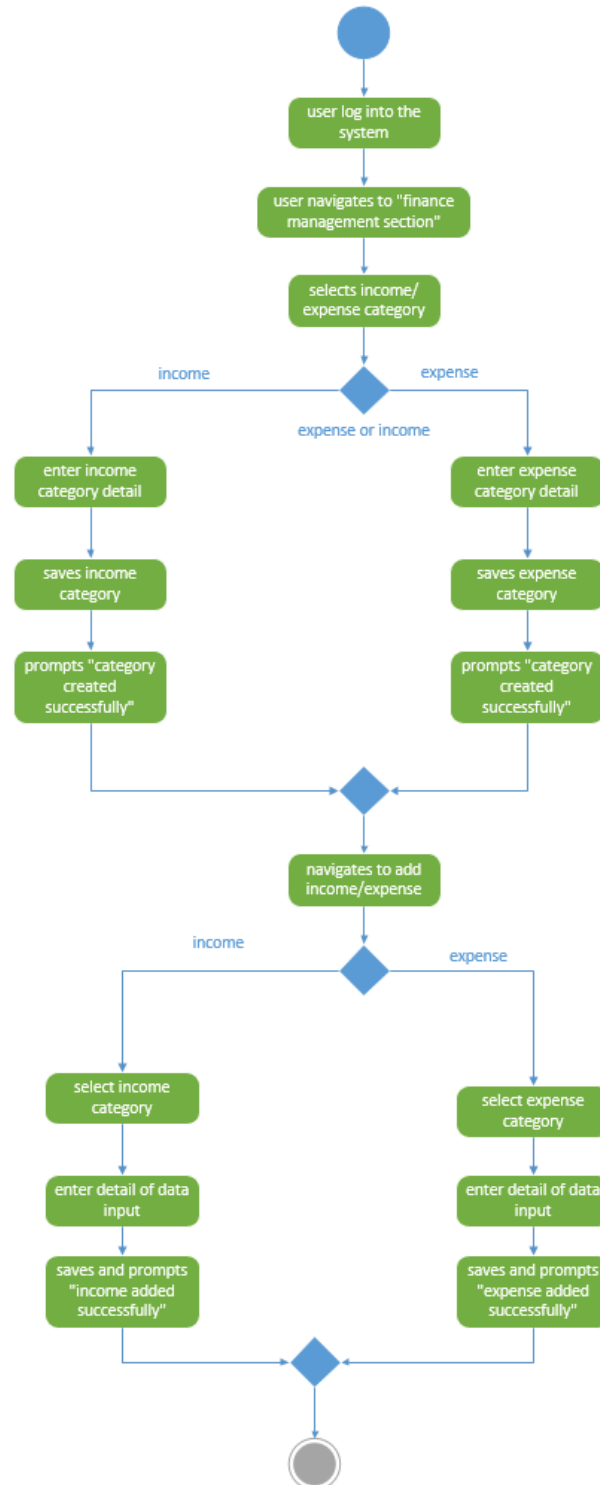


Figure 4-14 Financial Inputs Activity Diagram

4.2.1.14 Task and Progress Tracking

Figure 4.15 shows the activity diagram of “Task and progress tracking” use case which Shows task creation, assignment, notification, and progress tracking, ensuring clear communication and effective task management.

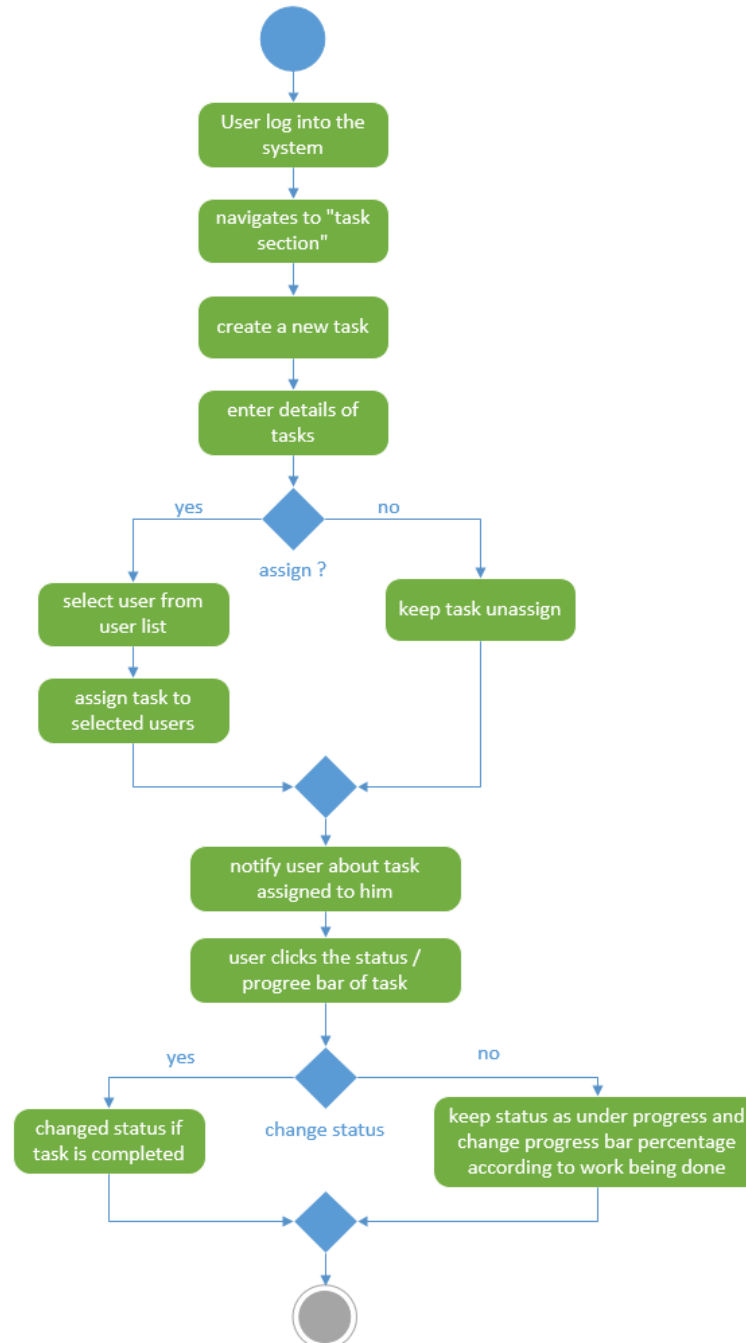


Figure 4-15 Task management Activity Diagram

4.2.1.15 Order and Inventory Management

Figure 4.16 shows the activity diagram of “order management” use case which illustrates order handling, discount calculations, stock verification, and inventory updates, ensuring accurate and efficient order processing.

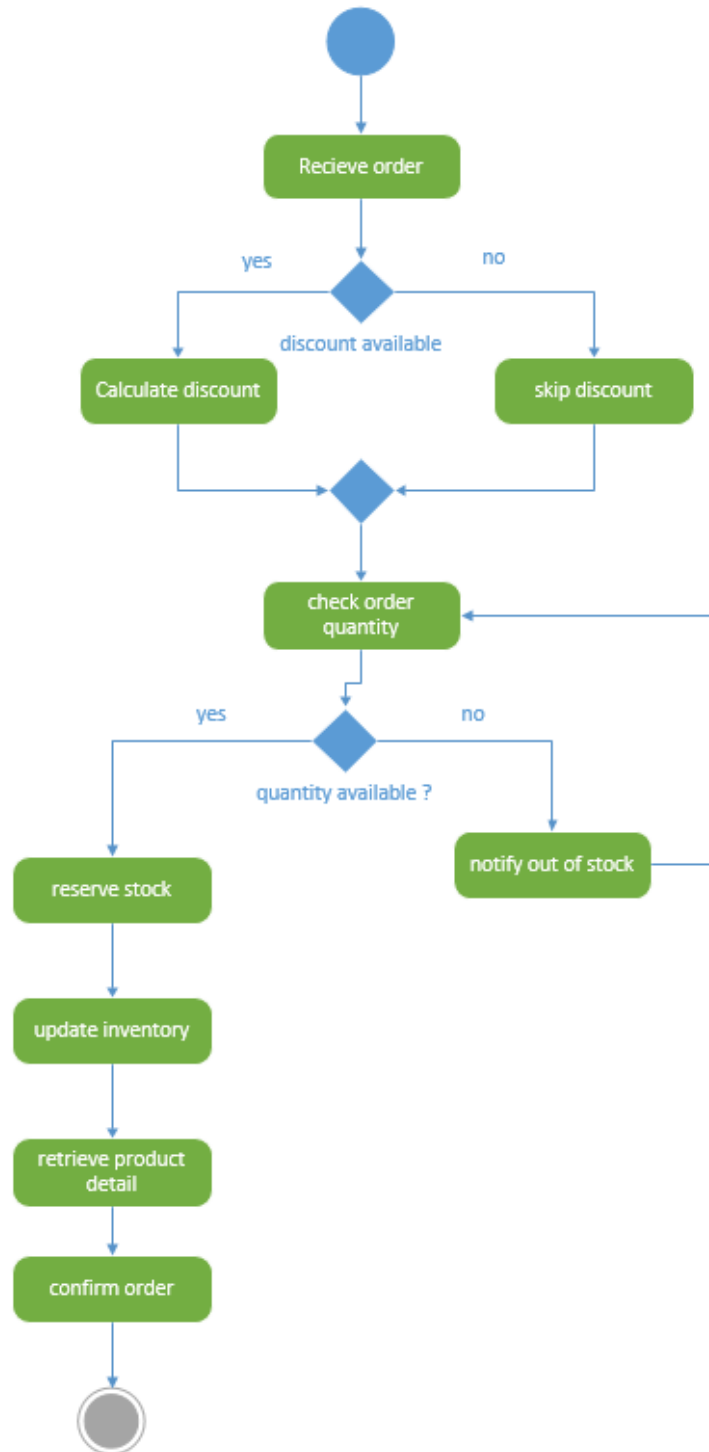


Figure 4-16 Order Management Activity Diagram

4.2.2 Data Flow Diagram

The Data Flow diagrams of modules are as follows:

4.2.2.1 Context Diagram

Figure 4.17 shows context diagram for ModuFlex ERP which Represents high-level interactions among Admin, User, Bank, and the ERP system, clearly depicting data flow and system-user interactions.

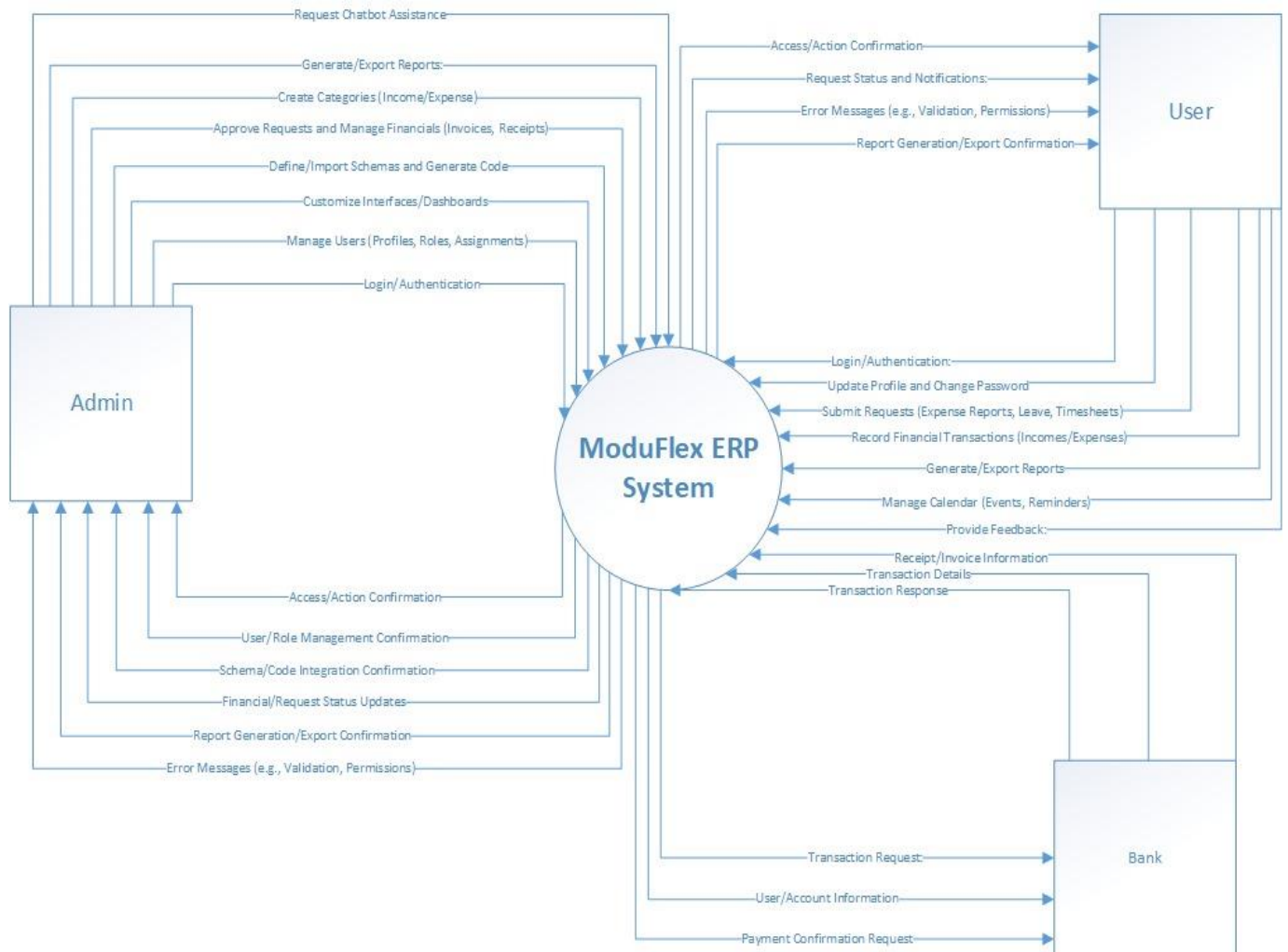


Figure 4-17 Context Diagram

4.2.2.2 Level 1 DFD

Figure 4.18 shows level 1 data flow diagram for user and role management of ModuFlex ERP which Depicts data interactions for user authentication, profile management, role assignment, and permissions, clarifying admin-user communication.

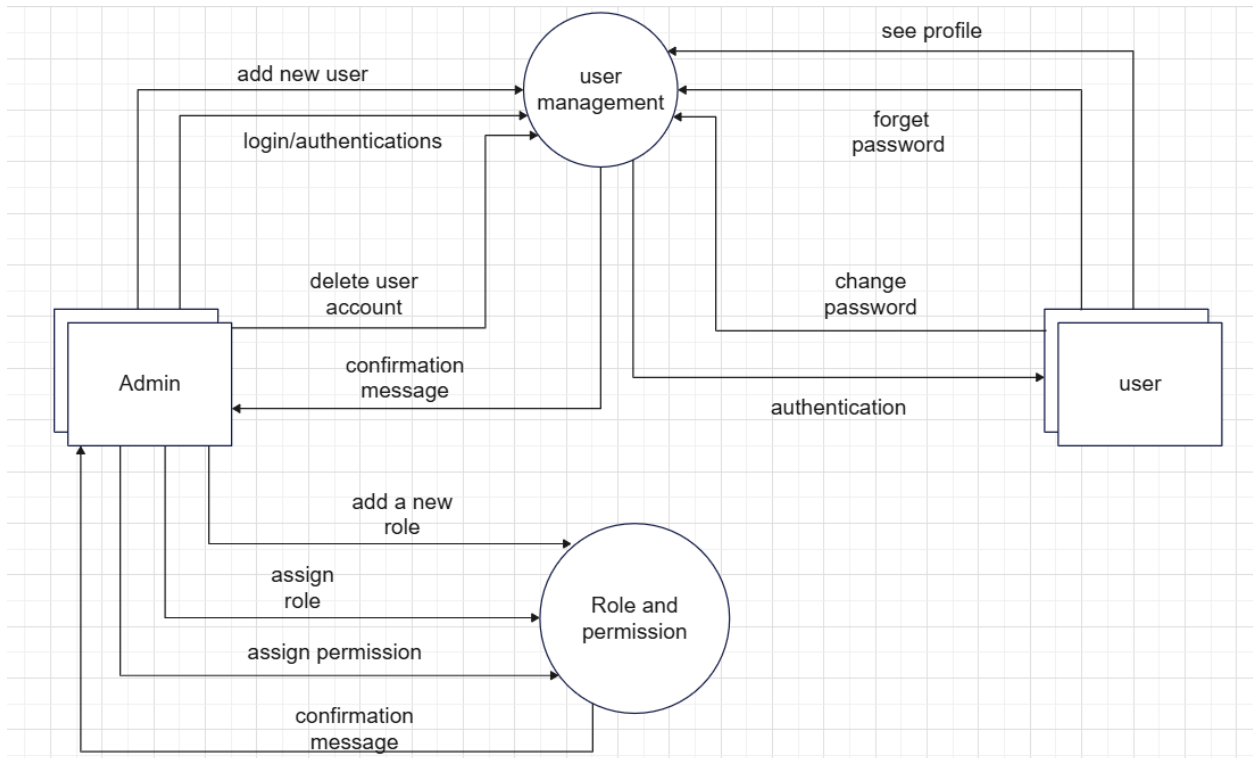


Figure 4-18 User and Role Management Level 1 DFD

Figure 4.19 shows level 1 data flow diagram for Human resource and Task management of ModuFlex ERP which Visualizes workflows for managing employee leave, timesheets, expenses, CRUD operations, and reporting integration, supporting efficient HR management.

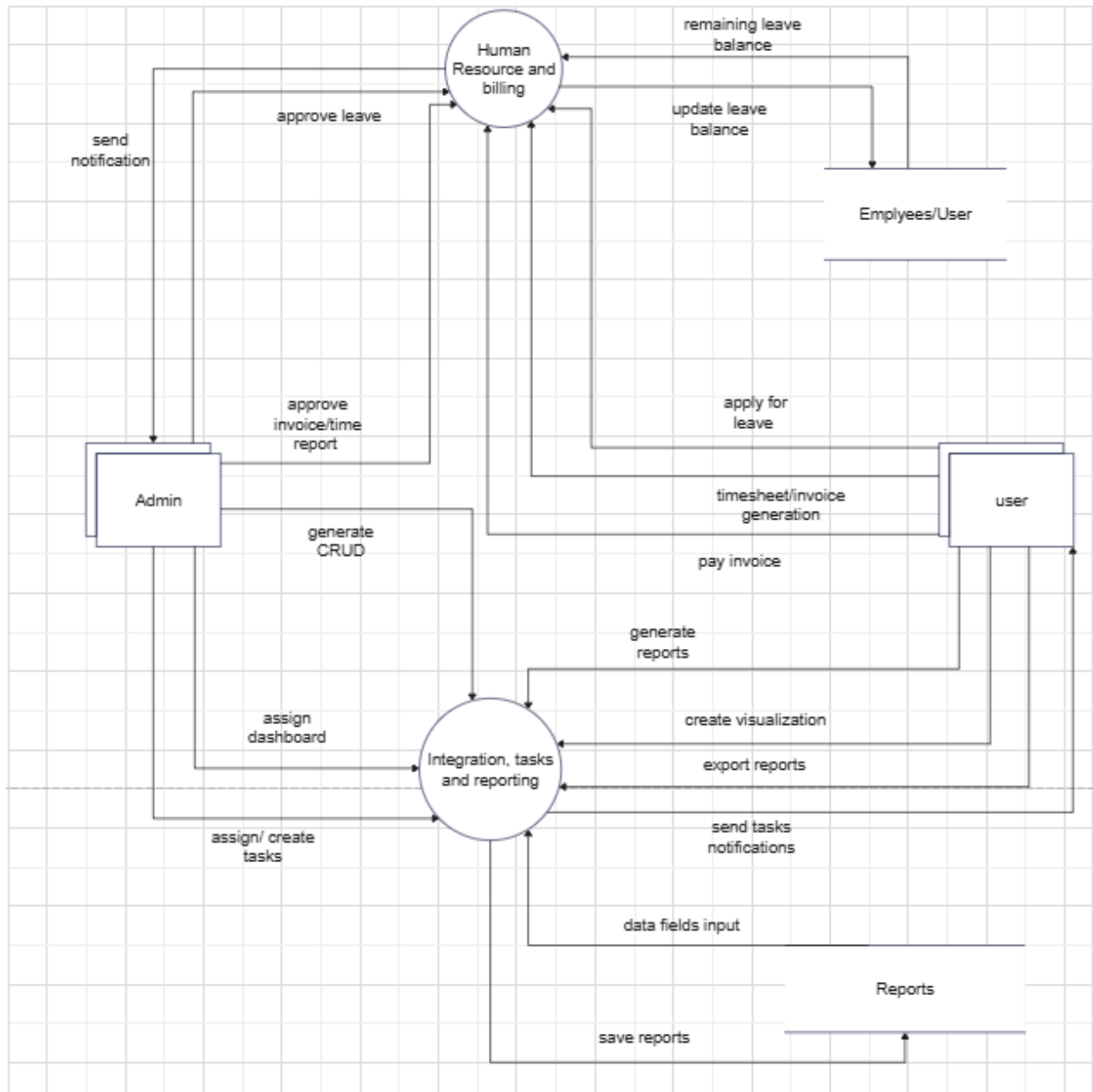


Figure 4-19 Development and HR management Level 1 DFD

4.2.2.3 Level 2 DFD

Figure 4.20 shows level 2 data flow diagram of login for ModuFlex ERP which Clearly represents user login validation, password handling, profile viewing, and admin-level user management, highlighting secure access controls.

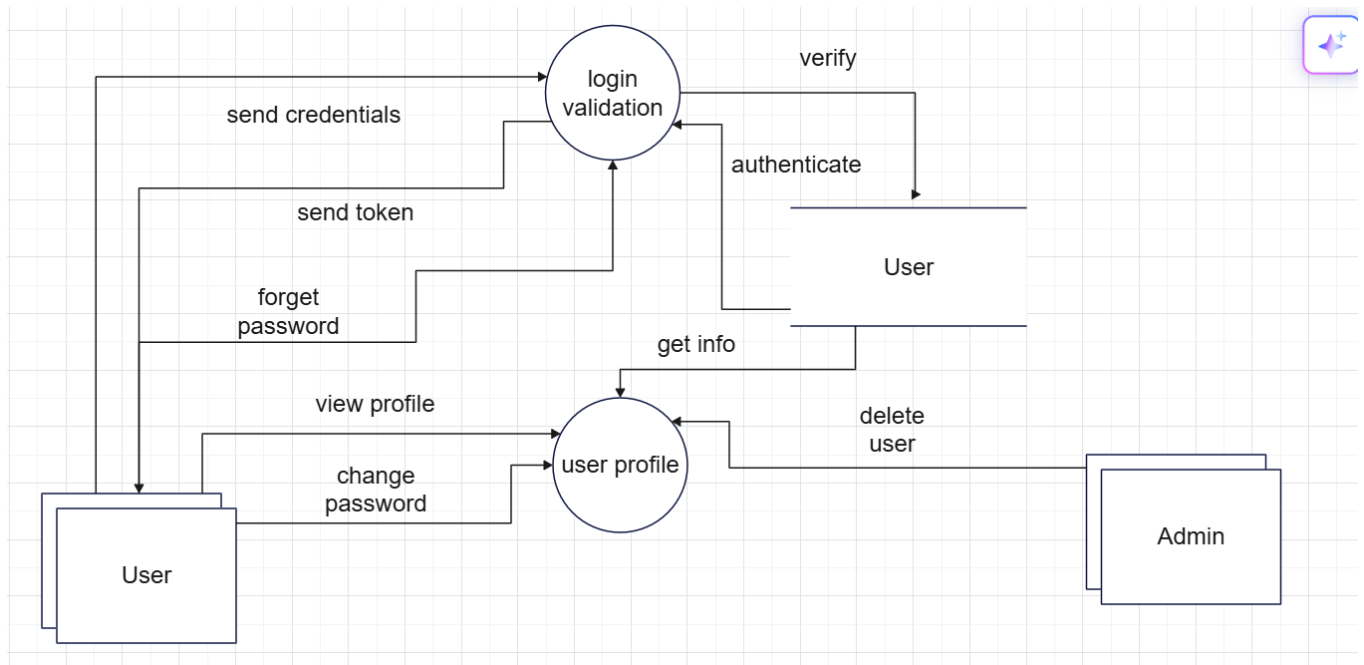


Figure 4-20 Login Level 2 DFD

Figure 4.21 shows level 2 data flow diagram of CRUD generator for ModuFlex ERP which Shows how schemas and Excel uploads create consistent database structures, emphasizing smooth integration and code generation within the system.

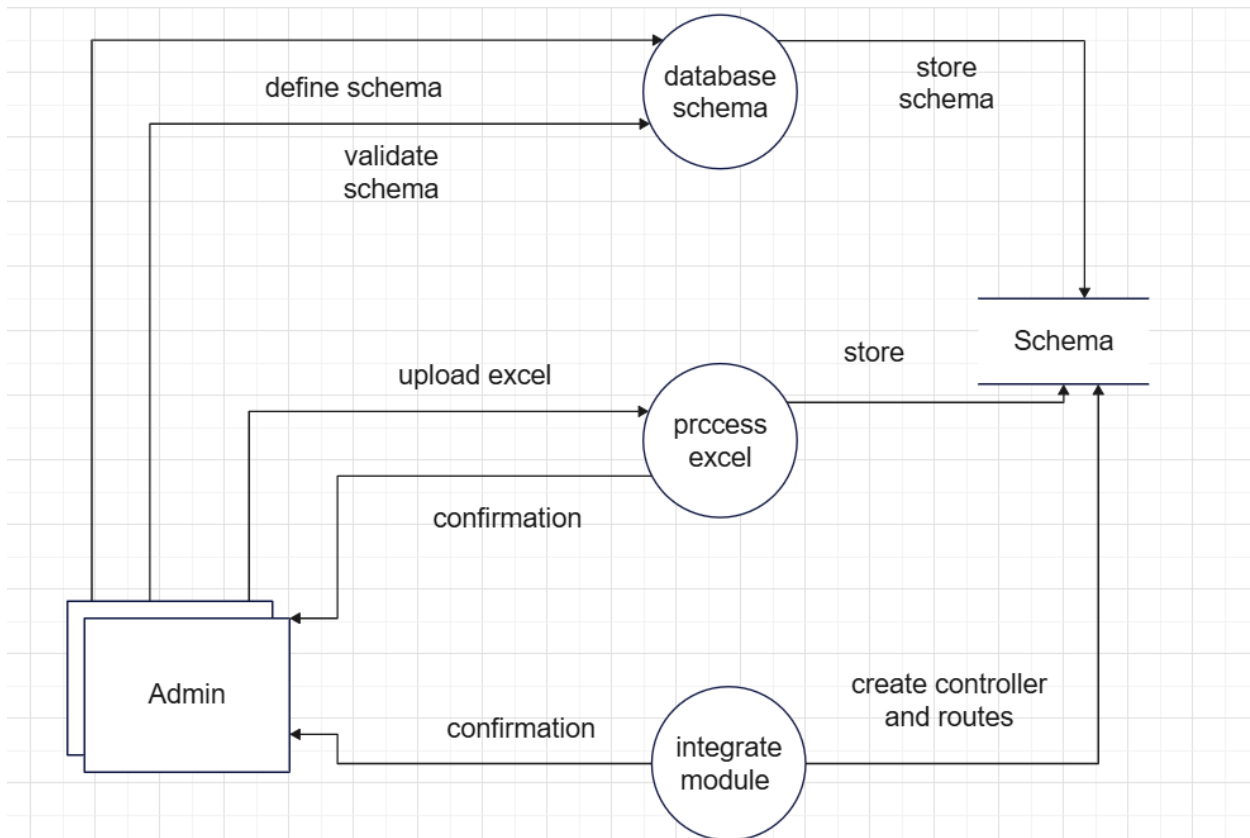


Figure 4-21 Integration and Development Level 2 DFD

Figure 4.22 shows level 2 data flow diagram of human resource management for ModuFlex ERP which Outlines processes involving employee profile management, leave requests, expense tracking, and timesheet approvals, streamlining administrative actions.

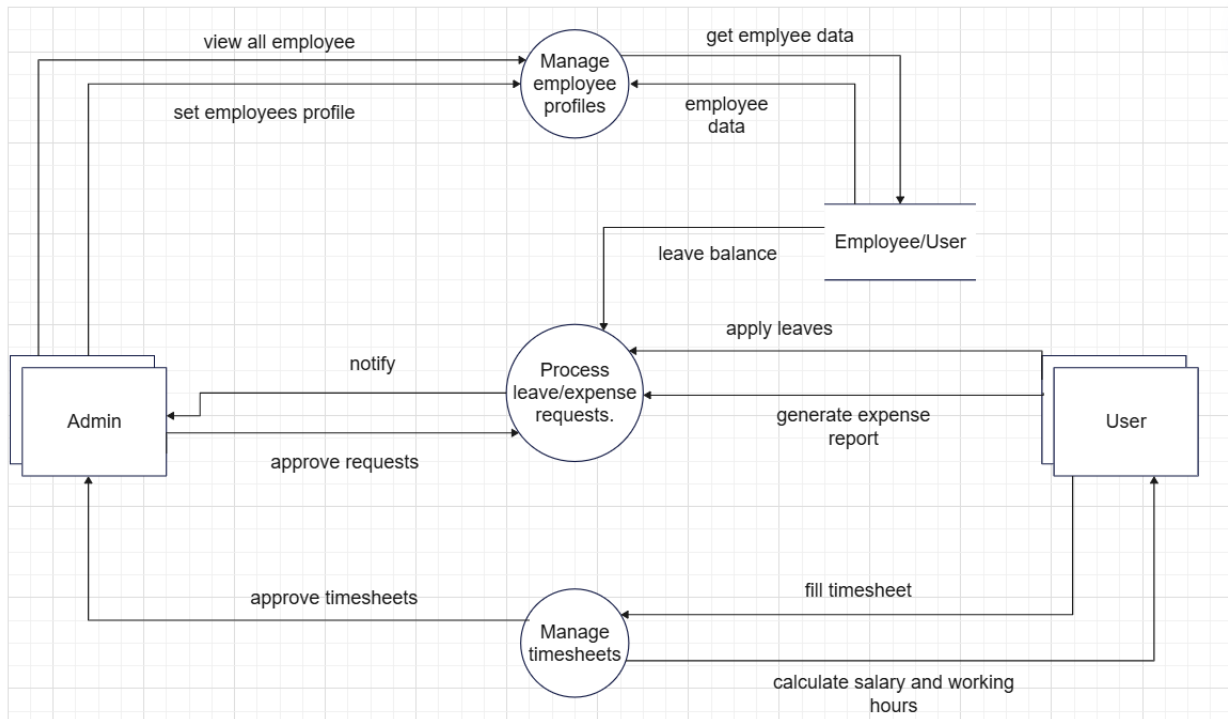


Figure 4-22 HR management Level 2 DFD

Figure 4.23 shows level 2 data flow diagram of dashboard and reports for ModuFlex ERP which Represents real-time report generation, dashboard configuration, and export functionalities, enabling informed decision-making and tailored data visualization.

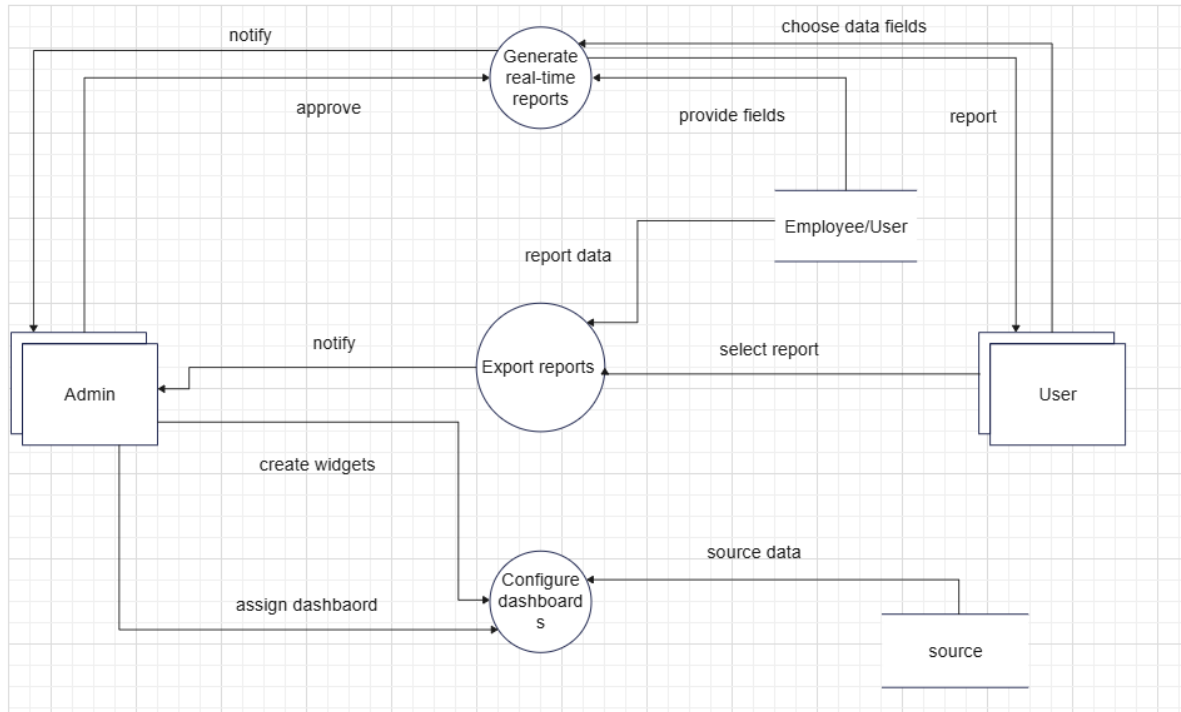


Figure 4-23 Dashboard Level 2 DFD

Figure 4.24 shows level 2 data flow diagram of tasks and payments for ModuFlex ERP which Shows clear financial tracking processes (invoices, income/expenses) along with task creation and event management, facilitating structured operational workflow.

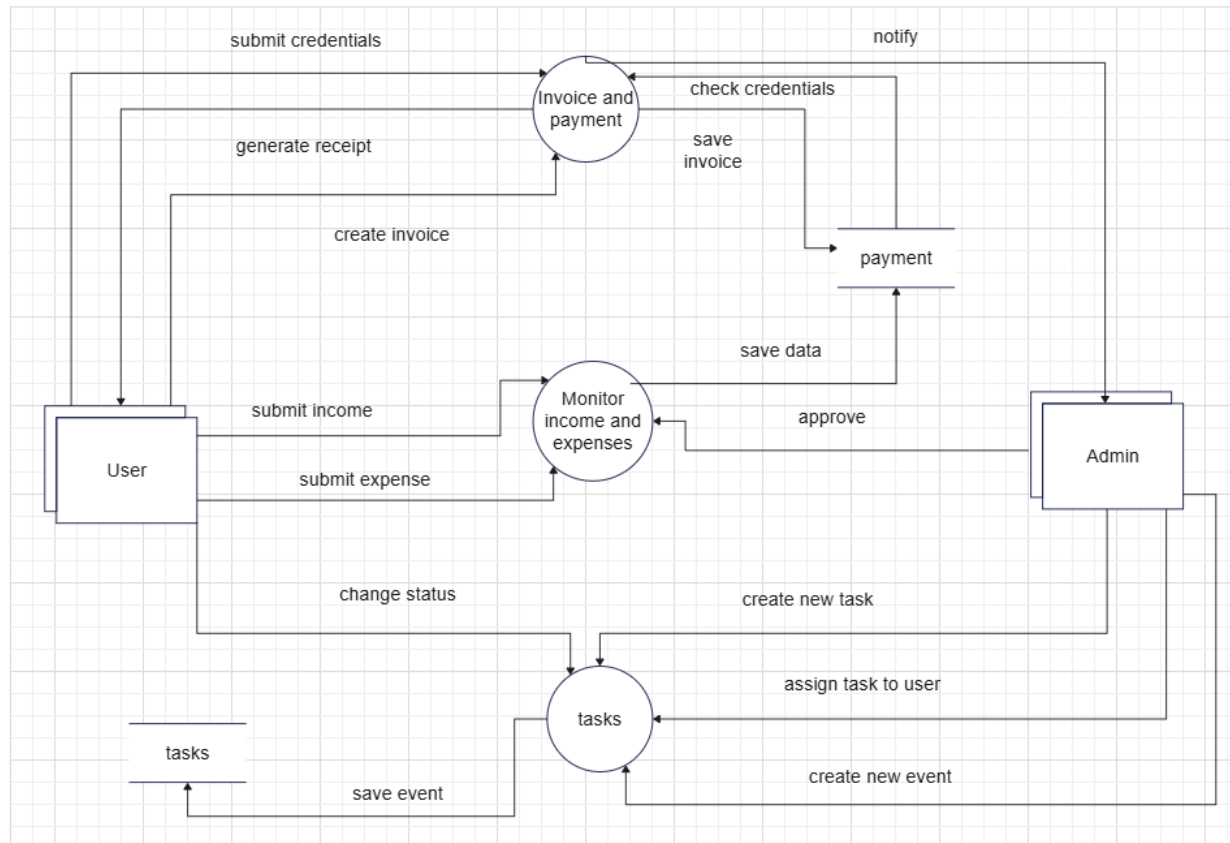


Figure 4-24 Task and billing Management Level 2 DFD

4.2.3 State Transition Diagram

4.2.3.1 User Authentication and Session Management

Figure 4.25 shows state transition diagram of user authentication and session management which illustrates user login, logout, and password recovery, providing clear authentication states and transitions to maintain secure system access.

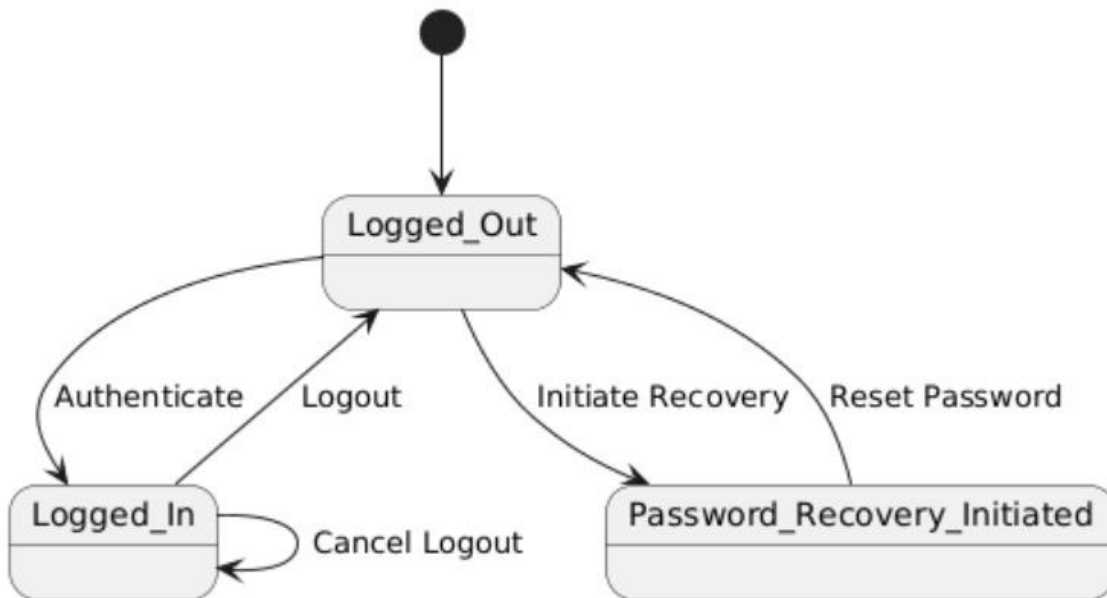


Figure 4-25 STATE TRANSITION DIAGRAM user authentication

4.2.3.2 Database Schema Generation

Figure 4.26 shows state transition diagram of database schema generation which Shows the states for defining, validating, correcting, and saving database schemas, ensuring accurate table creation and data consistency.

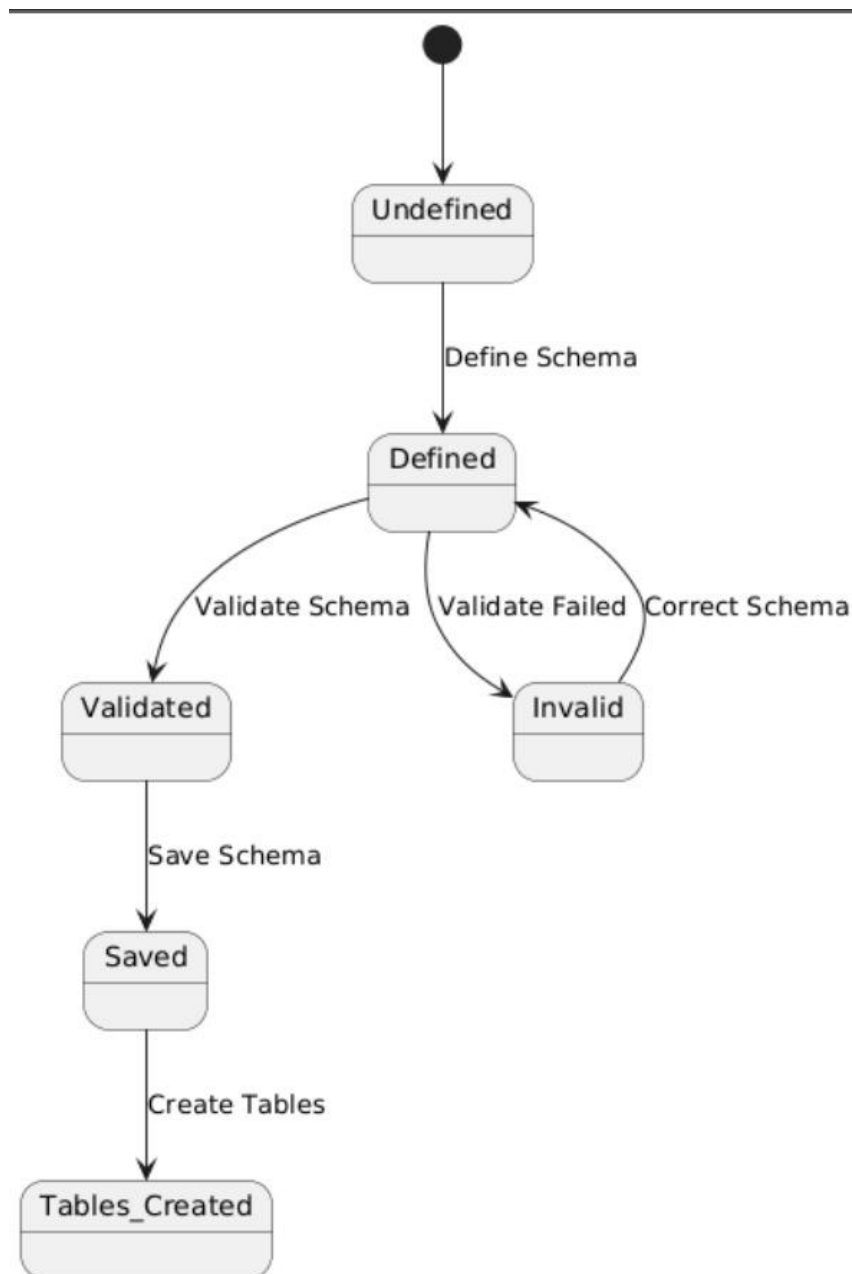


Figure 4-26 STATE TRANSITION DIAGRAM database schema generation

4.2.3.3 Submit and Track Expense Reports

Figure 4.27 shows state transition diagram of submit and track expenses which Clearly defines report creation, validation, submission, and approval/rejection, enabling effective tracking and management of submitted reports.

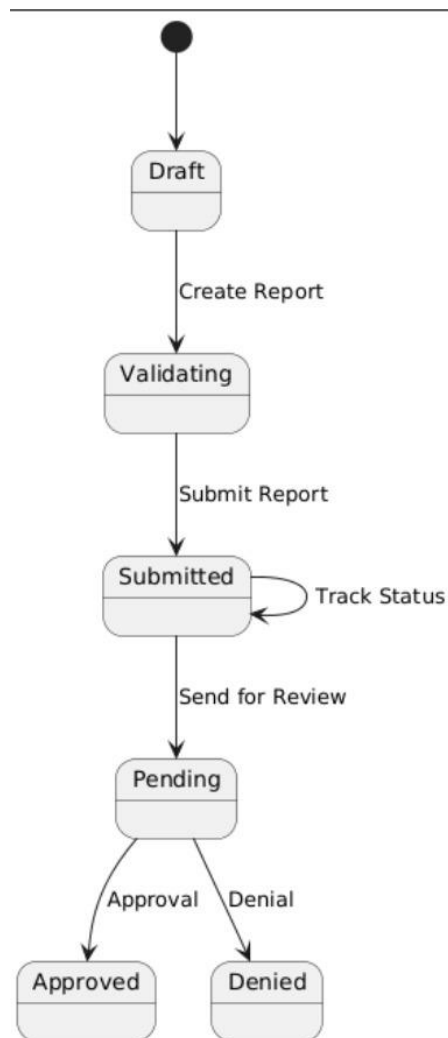


Figure 4-27 STATE TRANSITION DIAGRAM submit and track expense report

4.2.3.4 Managing Timesheets and Reports

Figure 4.28 shows state transition diagram of managing timesheets and reports which Depicts timesheet editing, submission, and the approval/rejection process, emphasizing structured tracking and accurate payroll preparation.

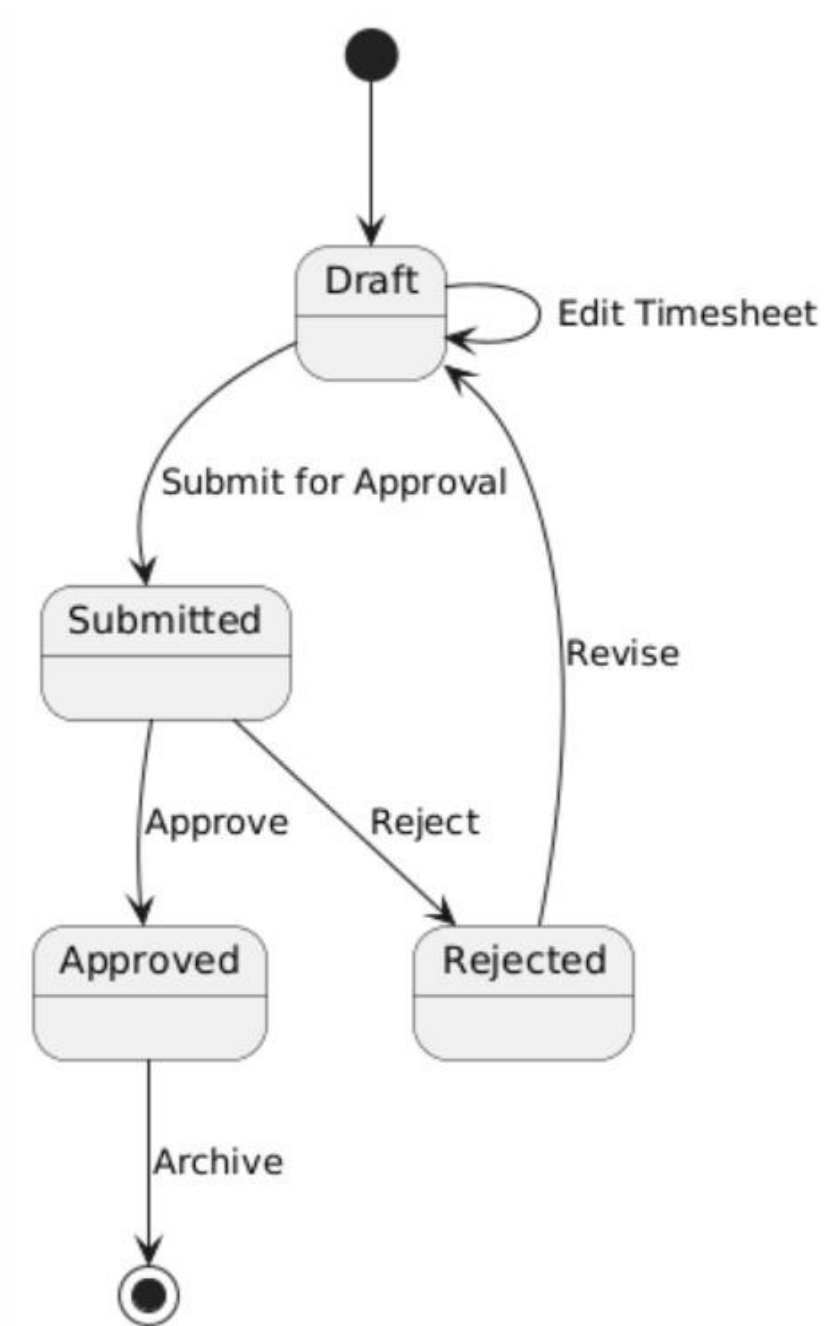


Figure 4-28 STATE TRANSITION DIAGRAM timesheets and reports

4.2.3.5 Manage Tasks

Figure 4.29 shows state transition diagram of manage tasks which Shows comprehensive task states from creation and editing to assignment, deletion, and status tracking, enhancing clarity in task management workflows.

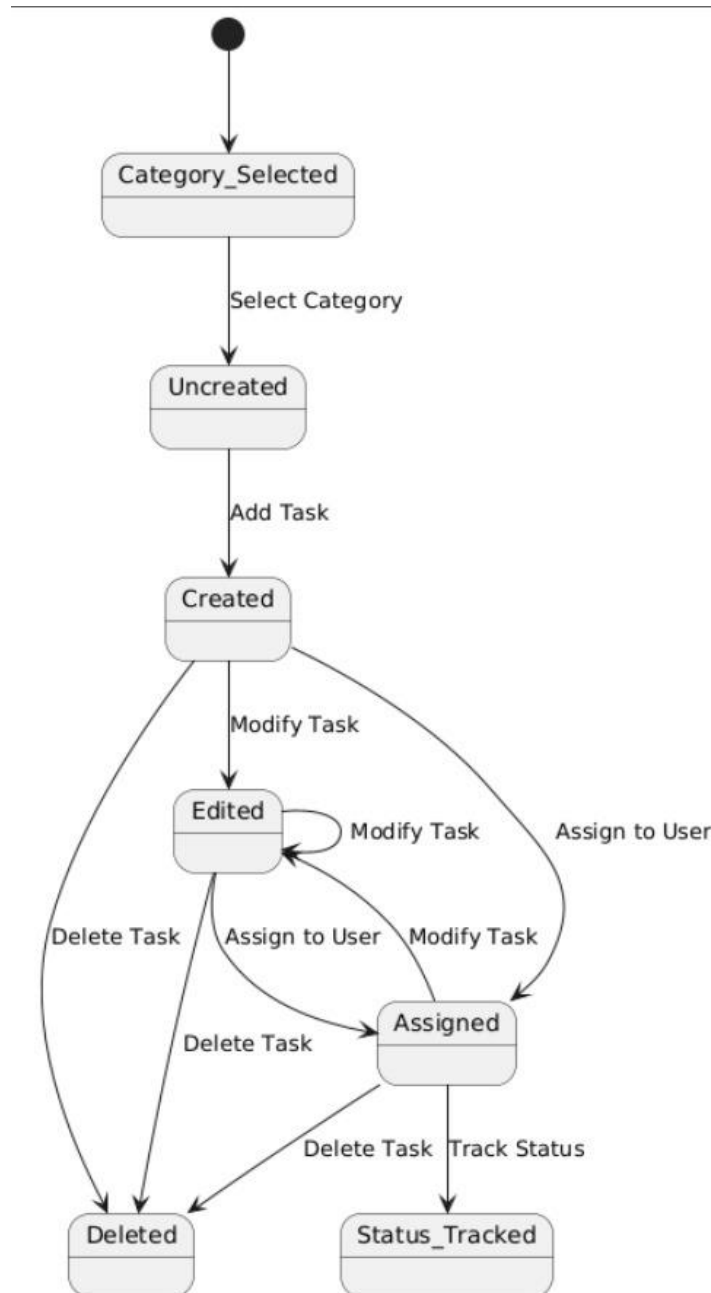


Figure 4-29 STATE TRANSITION DIAGRAM managing tasks

4.2.3.6 Events Scheduling

Figure 4.30 shows state transition diagram of event scheduling which Visualizes event lifecycle including scheduling, modification, notification, and deletion, clearly managing event updates and alerts.

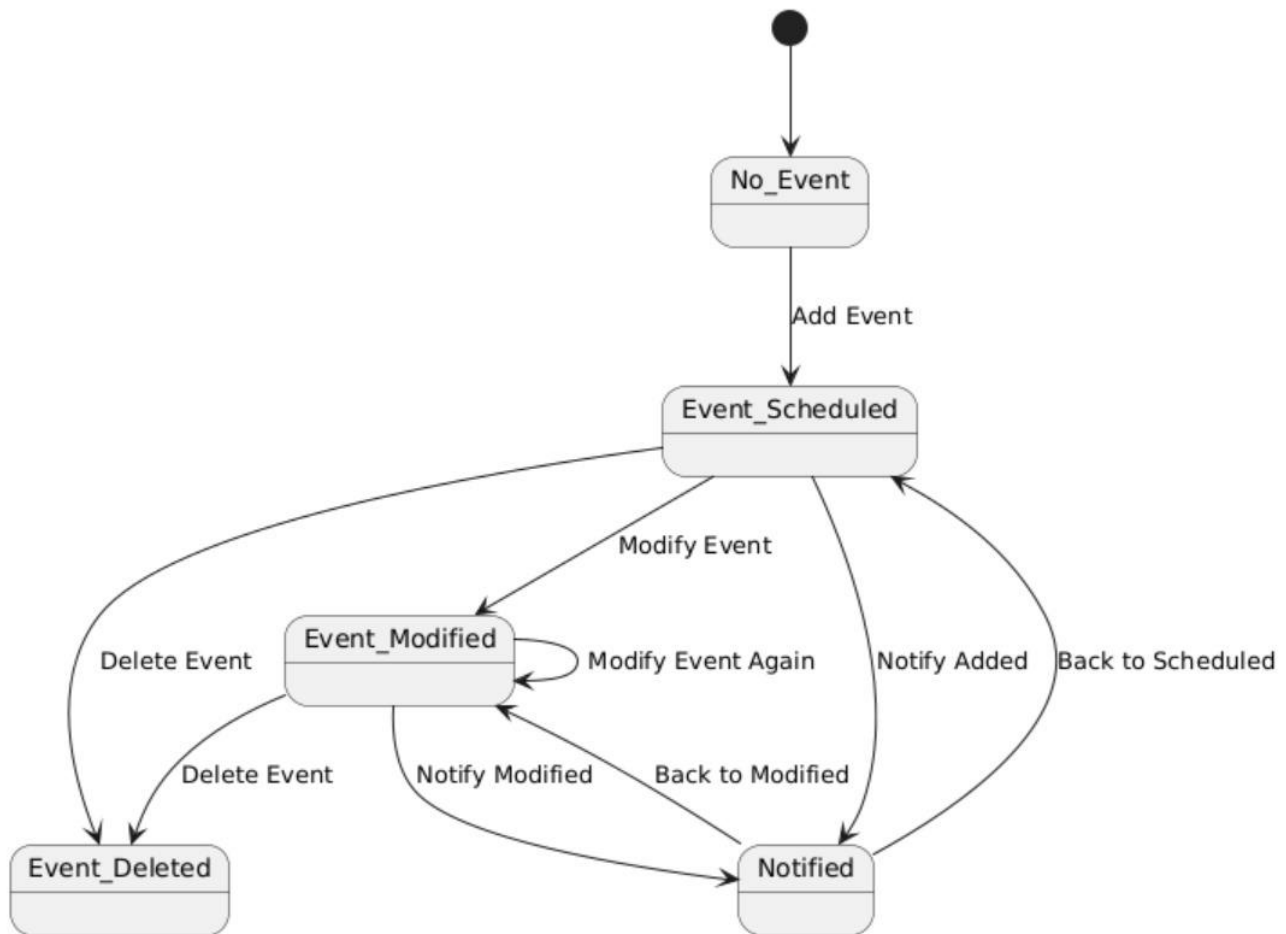


Figure 4-30 STATE TRANSITION DIAGRAM Event Scheduling

4.2.3.7 Predictive Analytics

Figure 4.31 shows state transition diagram of Predictive analytics which Demonstrates sequential stages from data collection and model training to prediction interpretation and actionable implementation, clearly defining the AI predictive analytics process.

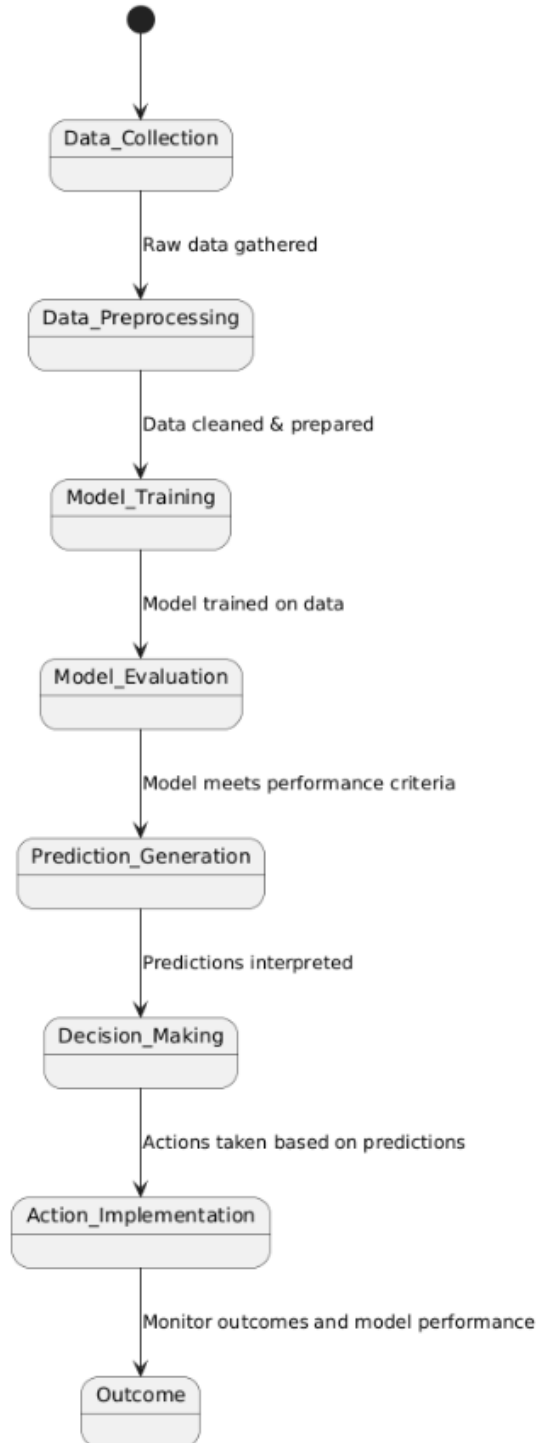


Figure 4-31 STATE TRANSITION DIAGRAM Predictive Analytics

4.2.3.8 Inventory Management

Figure 4.32 shows state transition diagram of Inventory management which Depicts inventory checking, low-stock alerts, reordering, and restocking, ensuring efficient inventory management and real-time stock control.

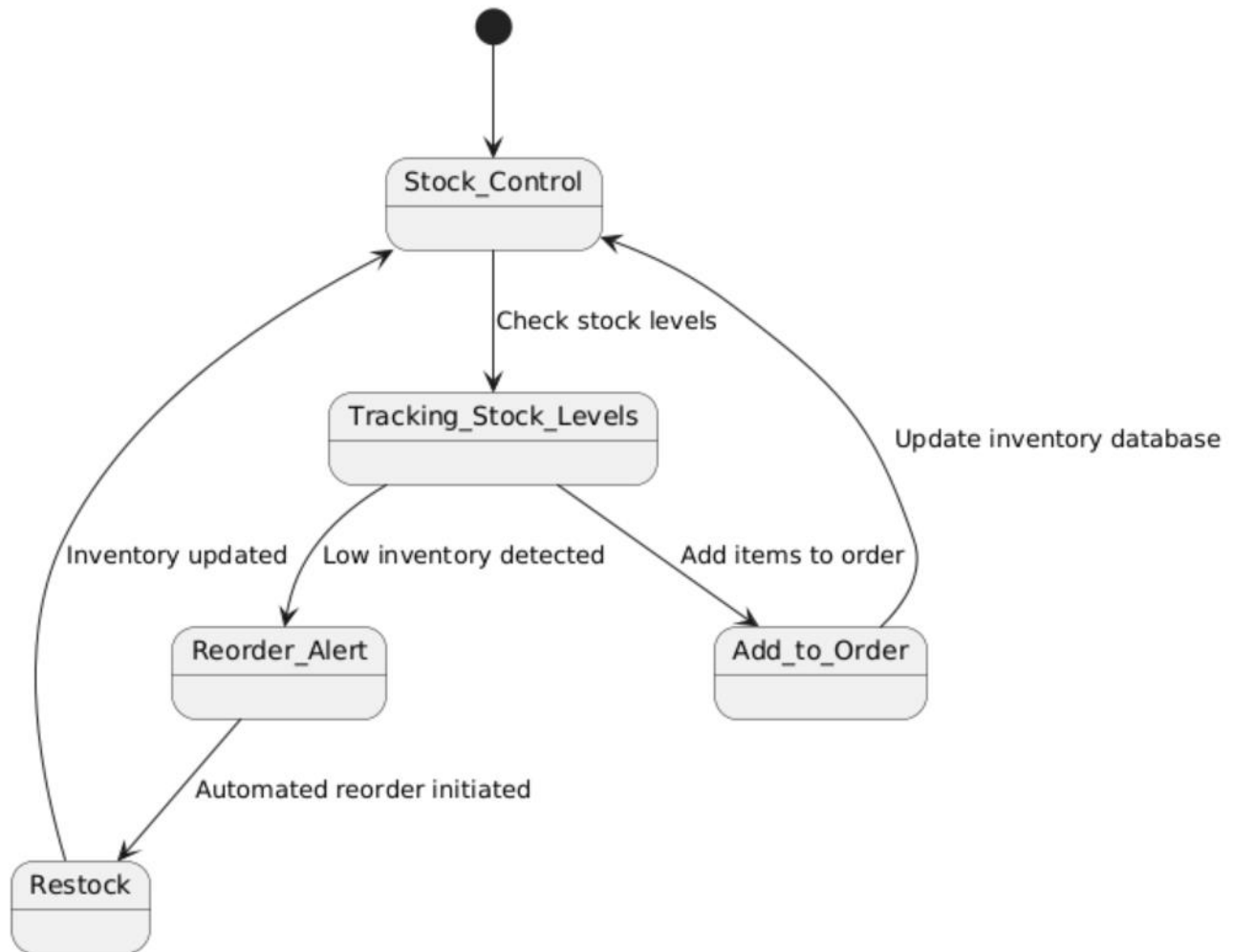


Figure 4-32 STATE TRANSITION DIAGRAM Inventory management

4.3 Data Design

4.3.1 User Management Schema

```
CREATE TABLE Users (
  UserID INT AUTO_INCREMENT PRIMARY KEY,
  Username VARCHAR(50) NOT NULL,
```

```
Email VARCHAR(100) UNIQUE NOT NULL,
PasswordHash VARCHAR(255) NOT NULL,
ProfilePicture VARCHAR(255),
ContactInfo VARCHAR(100),
RoleID INT NOT NULL,
IsActive BOOLEAN DEFAULT TRUE,
CreatedAt TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
UpdatedAt TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP
);
```

4.3.2 Role and Permission Schema

```
CREATE TABLE Roles (
    RoleID INT AUTO_INCREMENT PRIMARY KEY,
    RoleName VARCHAR(50) UNIQUE NOT NULL,
    Description TEXT
);

CREATE TABLE Permissions (
    PermissionID INT AUTO_INCREMENT PRIMARY KEY,
    PermissionName VARCHAR(50) UNIQUE NOT NULL,
    RoleID INT NOT NULL,
    Description TEXT
    FOREIGN KEY (RoleID) REFERENCES Roles(RoleID),
);
```

4.3.3 Human Resource Management Schema

```
CREATE TABLE Employees (
    EmployeeID INT AUTO_INCREMENT PRIMARY KEY,
    FirstName VARCHAR(50) NOT NULL,
    LastName VARCHAR(50) NOT NULL,
    Email VARCHAR(100) UNIQUE NOT NULL,
    JobTitle VARCHAR(100),
    Department VARCHAR(100),
    HireDate DATE,
    IsActive BOOLEAN DEFAULT TRUE
);

CREATE TABLE Timesheets (
    TimesheetID INT AUTO_INCREMENT PRIMARY KEY,
    EmployeeID INT NOT NULL,
```

```

    WorkDate DATE NOT NULL,
    HoursWorked DECIMAL(4, 2),
    OvertimeHours DECIMAL(4, 2),
    FOREIGN KEY (EmployeeID) REFERENCES Employees(EmployeeID)
);

CREATE TABLE LeaveRequests (
    LeaveID INT AUTO_INCREMENT PRIMARY KEY,
    EmployeeID INT NOT NULL,
    StartDate DATE NOT NULL,
    EndDate DATE NOT NULL,
    Status ENUM('Pending', 'Approved', 'Rejected') DEFAULT 'Pending',
    Reason TEXT,
    FOREIGN KEY (EmployeeID) REFERENCES Employees(EmployeeID)
);

```

4.3.4 Dashboard and Reporting Schema

```

CREATE TABLE Reports (
    ReportID INT AUTO_INCREMENT PRIMARY KEY,
    ReportName VARCHAR(100),
    GeneratedBy INT NOT NULL,
    Criteria JSON NOT NULL,
    GeneratedAt TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (GeneratedBy) REFERENCES Users(UserID)
);

```

4.3.5 Billing and Financing Schema

```

CREATE TABLE Invoices (
    InvoiceID INT AUTO_INCREMENT PRIMARY KEY,
    UserID INT NOT NULL,
    Amount DECIMAL(10, 2) NOT NULL,
    Status ENUM('Paid', 'Pending', 'Overdue') DEFAULT 'Pending',
    DueDate DATE NOT NULL,
    FOREIGN KEY (UserID) REFERENCES Users(UserID)
);

CREATE TABLE Transactions (
    TransactionID INT AUTO_INCREMENT PRIMARY KEY,
    InvoiceID INT NOT NULL,

```

```

    PaymentDate DATE,
    Amount DECIMAL(10, 2),
    PaymentMethod VARCHAR(50),
    FOREIGN KEY (InvoiceID) REFERENCES Invoices(InvoiceID)
);

```

4.3.6 Task and Notification Schema

```

CREATE TABLE Tasks (
    TaskID INT AUTO_INCREMENT PRIMARY KEY,
    UserID INT NOT NULL,
    TaskTitle VARCHAR(100) NOT NULL,
    Description TEXT,
    Priority ENUM('Low', 'Medium', 'High') DEFAULT 'Medium',
    Deadline DATE,
    Status ENUM('Pending', 'In Progress', 'Completed') DEFAULT 'Pending',
    CreatedAt TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (UserID) REFERENCES Users(UserID)
);

CREATE TABLE Notifications (
    NotificationID INT AUTO_INCREMENT PRIMARY KEY,
    UserID INT NOT NULL,
    Message TEXT NOT NULL,
    IsRead BOOLEAN DEFAULT FALSE,
    CreatedAt TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (UserID) REFERENCES Users(UserID)
);

```

4.3.7 Data Dictionary

Table 4.1 shows the data dictionary used in ModuFlex ERP

Table 4.1 Data dictionary

Collection	Property	Data Type	Description
Role	RoleID	INT	Unique identifier for a role.
Role	RoleName	STRING	Name of the role
User	UserID	INT	Unique identifier for a user
Collection	Property	Data Type	Description

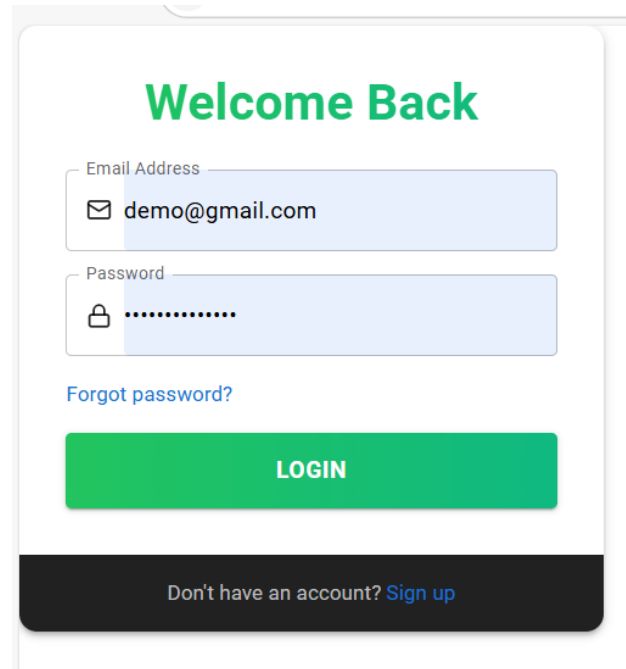
User	Username	STRING	Username chosen by the user.
User	PasswordHash	STRING	Hashed password for user authentication.
User	Email	STRING	Email address of the user.
User	ProfilePicture	STRING	URL or path to the user's profile picture.
User	RoleID	INT	Reference to the role assigned to the user.
User	IsActive	BOOLEAN	Status indicating active or deactivated user.
Task	TaskID	INT	Unique identifier for a task.
Task	UserID	INT	Reference to the user assigned to the task.
Task	Title	STRING	Title or name of the task.
Task	Description	TEXT	Detailed description of the task.
Task	Priority	ENUM	Priority level of the task
Task	Status	ENUM	Status of the task
Task	Deadline	DATE	The due date for the task completion.
Invoice	InvoiceID	INT	Unique identifier for an invoice.
Invoice	UserID	INT	Reference to the user who the invoice is generated for.
Invoice	Amount	DECIMAL	Total amount billed in the invoice.
Invoice	Status	ENUM	Status of the invoice
Invoice	DueDate	DATE	Due date for the payment of the invoice.
LeaveRequest	LeaveRequestID	INT	Unique identifier for a leave request.
LeaveRequest	EmployeeID	INT	Reference to the employee requesting leave.
LeaveRequest	LeaveStartDate	DATE	Start date of the leave.
LeaveRequest	LeaveEndDate	DATE	End date of the leave.
LeaveRequest	Status	ENUM	Status of the leave request
LeaveRequest	Reason	TEXT	Reason for the leave.
Notification	NotificationID	INT	Unique identifier for a notification.
Notification	UserID	INT	Reference to the user who will receive the notification.
Notification	Message	TEXT	The content of the notification message.

Notification	Timestamp	TIMESTAMP	The date and time when the notification was sent.
Notification	IsRead	BOOLEAN	Status indicating if the notification has been read.
Timesheet	TimesheetID	INT	Unique identifier for a timesheet.
Timesheet	UserID	INT	Reference to the user who filled out the timesheet.
Timesheet	WorkHours	DECIMAL	Total work hours logged by the user.
Timesheet	OvertimeHours	DECIMAL	Total overtime hours worked.
Timesheet	Date	DATE	The date for which the timesheet is logged.
Report	ReportID	INT	Unique identifier for a report.
Report	UserID	INT	Reference to the user who generated the report.
Report	Criteria	STRING	Criteria used for generating the report
Report	ReportData	TEXT	The generated report data, usually in a tabular or graphical format.
Report	Format	ENUM	Format of the report
Transaction	TransactionID	INT	Unique identifier for a transaction.
Transaction	InvoiceID	INT	Reference to the related invoice for which the payment was made.
Transaction	AmountPaid	DECIMAL	Amount paid in the transaction.
Transaction	PaymentMethod	STRING	Method used for payment
Transaction	Timestamp	TIMESTAMP	Date and time of the transaction.

4.4 Human Interface Design

4.4.1 Login Screen

Figure 4.33 shows Login screen of ModuFlex ERP

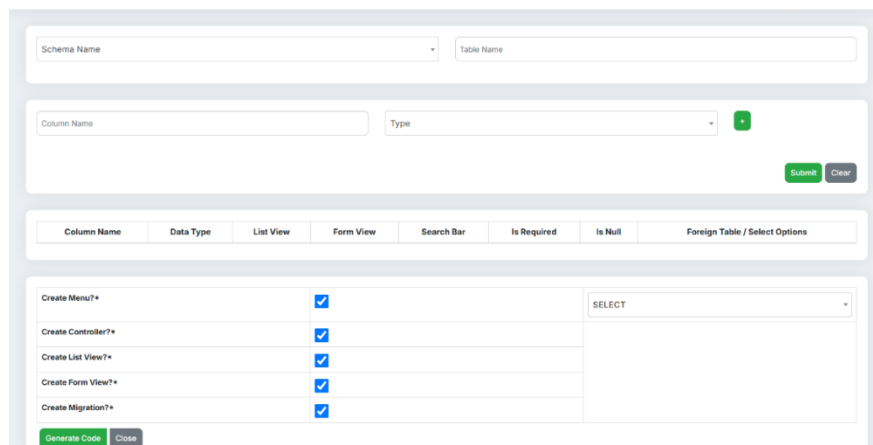


The login screen features a white card with a green header 'Welcome Back'. Below the header are two input fields: 'Email Address' with a mail icon and 'Password' with a lock icon. The email field contains 'demo@gmail.com' and the password field contains masked characters. A blue link 'Forgot password?' is positioned below the password field. A large green 'LOGIN' button is centered below the inputs. At the bottom, a dark grey bar contains the text 'Don't have an account? Sign up' with 'Sign up' as a blue link.

Figure 4-33 Login Mockup Screen

4.4.2 CRUD Generator Screen

Figure 4.34 shows CRUD generator screen of ModuFlex ERP



The CRUD Generator screen includes a form with 'Schema Name' and 'Table Name' dropdowns. Below is a section for 'Column Name' and 'Type' with a green '+' button. A 'Submit' button and a 'Clear' button are at the bottom right. A table with columns 'Column Name', 'Data Type', 'List View', 'Form View', 'Search Bar', 'Is Required', 'Is Null', and 'Foreign Table / Select Options' is present. Below the table, there are checkboxes for 'Create Menu?*', 'Create Controller?*', 'Create List View?*', 'Create Form View?*', and 'Create Migration?*', all of which are checked. A 'SELECT' dropdown is to the right of these checkboxes. At the bottom, there are 'Generate Code' and 'Close' buttons.

Figure 4-34 CRUD Generator Mockup Screen

4.4.3 Dashboard Screen

Figure 4.35 shows Dashboard screen of ModuFlex ERP

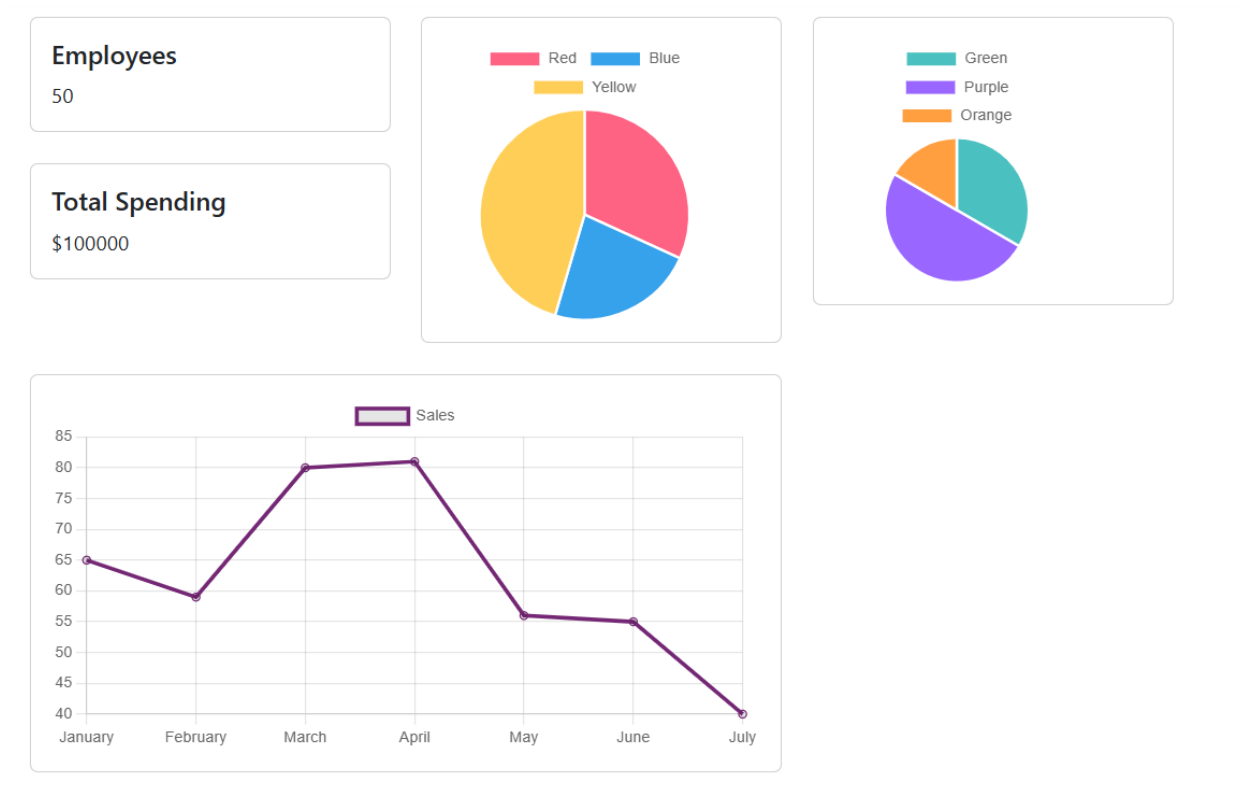


Figure 4-35 Dashboard Mockup Screen

4.4.4 Stripe Payment Screen

Figure 4.36 shows Stripe screen of ModuFlex ERP

The Stripe Payment Screen mockup displays the following form fields and buttons:

- Secure Payment** (Title)
- Amount:** Input field with the value 1000.
- Card Details:** Input field containing the card number 5200 0078 4000 0022 and the placeholder MM / YY CVC.
- Make Payment** (Button)

Figure 436 Stripe Mockup Screen

4.4.5 Role Screen

Figure 4.37 shows Role screen of ModuFlex ERP

Role Profile

Role Management

+ ADD NEW ROLE

SR#	NAME	DESCRIPTION	ACTIONS
1	admin	Administrator with full access	 
2	manager	Manager with project management permissions	 
3	employee	Regular employee with limited access	 

Figure 4-37 Role Mockup Screen

4.4.6 Expense Report Screen

Figure 4.38 shows Expense Report screen of ModuFlex ERP

New Expense Report

New expense report

Start date	mm/dd/yyyy 
End date	mm/dd/yyyy 
User	David Doe
User responsible for approval	Manager
Note (public)	
Note (private)	

CREATE EXPENSE REPORT

CANCEL

Figure 4-38 Expense Report Mockup Screen

5 Implementation.

5.1 Algorithm

Following are the Algorithms used in our core implemented modules.

5.1.1 Calculating WorkHours

Algorithm 4 Calculating WorkHours
Input: EmployeeID, WorkHours (daily hours), Date (for the specific day)
Output: TimesheetStatus, Calculated work hours
1: Employee $\leftarrow$ fetchEmployeeByID(EmployeeID) 2: if (Employee is not found) then 3: return "Employee not found" 4: end if 5: MonthlyWorkingHours $\leftarrow$ getMonthlyWorkingHours(EmployeeID, Date) 6: MonthlyLeaveDays $\leftarrow$ getMonthlyLeaveDays(EmployeeID, Date) 7: 8: // Calculate the working hours for the month including the current day 9: TotalWorkedHours $\leftarrow$ MonthlyWorkingHours + WorkHours 10: // Check if the current day is a leave day, reduce work hours accordingly 11: if (isLeaveDay(Date)) then 12: TotalWorkedHours $\leftarrow$ TotalWorkedHours - WorkHours 13: end if 14: 15: // Store the work hours for the day in the timesheet 16: Timesheet $\leftarrow$ createTimesheet(EmployeeID, WorkHours, Date) 17: saveTimesheetToDatabase(Timesheet) 18: 19: // Calculate Salary based on monthly hours worked and leave taken 20: DailyRate $\leftarrow$ Employee.BaseSalary / Employee.WorkingDaysPerMonth 21: TotalSalary $\leftarrow$ DailyRate * (TotalWorkedHours / Employee.WorkingHoursPerDay) 22: 23: // Apply any deductions for leave days if necessary (e.g., unpaid leave) 24: TotalSalary $\leftarrow$ TotalSalary - (Employee.LeaveDaysUnpaid * DailyRate) 25: 26: // Return status with calculated salary 27: return "Timesheet logged successfully", TotalSalary

5.1.2 Report Generation

Algorithm 5 GenerateReport
Input: UserID, ReportCriteria (specifies the fields and other criteria)
Output: GeneratedReport (generated report data in the specified format)
1: User $\leftarrow$ fetchUserByID(UserID) 2: if (User is not found) then 3: return "User not found" 4: end if 5: 6: // Fetch available fields based on the report type or category 7: AvailableFields $\leftarrow$ fetchAvailableFields(ReportCriteria.reportType) 8: 9: // Validate that selected fields exist in the available fields

```

10: SelectedFields ← validateSelectedFields(ReportCriteria.selectedFields, AvailableFields)
11: if (SelectedFields is empty) then
12:   return "No valid fields selected"
13: end if
14:
15: // Fetch data based on selected fields and report criteria
16: ReportData ← fetchReportData(SelectedFields, ReportCriteria.filters)
17:
18: // Format the data as per the report's structure
19: Report ← createReport(ReportData, SelectedFields)
20:
21: // Export report in the desired format (e.g., PDF, Excel, etc.)
22: exportReport(Report, ReportCriteria.format)
23:
24: return "Report generated successfully"

```

5.1.3 Leave Balance Calculation

Algorithm 6 RequestLeave

Input: EmployeeID, LeaveRequeState Transition Diagrametails

Output: LeaveRequestStatus (status of the leave request)

```

1: Employee ← fetchEmployeeByID(EmployeeID)
2: if (Employee is not found) then
3:   return "Employee not found"
4: end if
5:
6: // Fetch available paid and unpaid leaves for the employee
7: PaidLeaves ← fetchAvailablePaidLeaves(EmployeeID)
8: UnpaidLeaves ← fetchAvailableUnpaidLeaves(EmployeeID)
9:
10: // Calculate requested leave duration
11: RequestedLeaveDuration ← calculateLeaveDuration(LLeaveRequeState Transition Diagrametails)
12:
13: // Check if requested leave can be covered by paid leaves
14: if (RequestedLeaveDuration <= PaidLeaves) then
15:   // Deduct the paid leave balance
16:   PaidLeaves ← PaidLeaves - RequestedLeaveDuration
17:   updatePaidLeaveBalance(EmployeeID, PaidLeaves)
18:   LeaveRequestStatus ← "Leave granted with paid leaves"
19: else
20:   // If insufficient paid leaves, deduct from unpaid leaves and salary
21:   UnpaidLeaveDuration ← RequestedLeaveDuration - PaidLeaves
22:   UnpaidLeaves ← UnpaidLeaves - UnpaidLeaveDuration
23:   DeductSalary ← calculateSalaryDeduction(UnpaidLeaveDuration)
24:   updateUnpaidLeaveBalance(EmployeeID, UnpaidLeaves)
25:   deductSalary(EmployeeID, DeductSalary)
26:   LeaveRequestStatus ← "Leave granted with unpaid leave, salary deducted"
27: end if
28:
29: // Save the leave request details in the database
30: LeaveRequest ← createLeaveRequest(LLeaveRequeState Transition Diagrametails)
31: saveLeaveRequestToDatabase(LLeaveRequest)
32:
33: return LeaveRequestStatus

```

5.2 External APIs/SDKs

Table 5.1 shows the API's use in ModuFlex ERP

Table 5.1 Details of APIs used in the project

Name of API and version	Description of API	Purpose of usage	List down the API endpoint/function/class in which it is used
Stripe (version 2024-11-20)[2]	Credit Card Payment Integration	Used for processing payments, handling credit card transactions in the system.	stripe.paymentMethods.create
FastAPI	Web Framework for building APIs with Python	Used to build and expose an API for the regression model	fastapi.FastAPI, fastapi.APIRouter, model endpoint (/predict)

6 Testing and Evaluation

6.1 User management module

6.1.1 Unit Testing

Table 6.1 shows the unit test cases of user management module

Table 6.1 unit test cases

No.	Test Case/Test Script Description	Attribute and Value	Expected Result	Result
1	Validate email field in the login form for proper format.	Email: aqsam@gmail.com	Valid email, cursor moves to the password field.	Pass
2	Validate email field in the login form for invalid format.	Email: aqsam.gmail.com	Error: "Invalid email address format."	Pass
3	Validate new password for length criteria.	New Password: short	Error: "Password must be at least 8 characters long."	Pass
4	Validate existing password for incorrect input.	Existing Password: wrongPass	Error: "Current password is incorrect."	Pass
5	Validate profile field update with invalid data format.	Phone Number: 123abc	Error: "Phone number must contain only digits."	Pass

6.1.2 Functional Testing

Table 6.2 shows the Functional test cases of user management module

Table 6.2 Functional test cases

No.	Test Case/Test Script Description	Attribute and Value	Expected Result	Actual Result	Result
1	Login with valid credentials.	Email: sirdan@gmail.com, Password: 12345678	User is redirected to the dashboard page.	Successfully logged in and redirected to the dashboard.	Pass
2	Login with invalid credentials.	Email: sirdan@gmail.com,	Error message: "Invalid credentials."	Error displayed.	Pass

		Password: wrongpass			
3	Edit user profile with valid details.	Phone Number: 1234567890	Profile is successfully updated with the new phone number.	Profile updated successfully.	Pass
4	Change password with valid current and new password.	Existing Password: 12345678, New Password: Newpass123	Password is successfully updated in the system.	Successfully updated the password.	Pass
5	Logout from the system.	User clicks "Logout."	System redirects the user to the login page after confirmation.	Successfully logged out and redirected.	Pass

6.1.3 Business Rule Testing

Table 6.3 shows the Integration test cases of user management module

Table 6.3 Integration test cases

No.	Test Case/Test Script Description	Attribute and Value	Expected Result	Actual Result	Result
1	Edit profile with valid data	Name: John Doe, Email: aqsam@example.com	Profile is updated successfully.	Profile updated	Pass
2	Edit profile with invalid data	Email: invalid-email	Displays error message: "Invalid email format."	Error message displayed	Pass

6.1.4 Integration Testing

Table 6.4 shows the Integration test cases of user management module

Table 6.4 Integration test cases

No.	Test Case/Test Script Description	Attribute and Value	Expected Result	Actual Result	Result
1	Login and navigate to the profile page.	Email: aqsam@gmail.com, Password: 12345678	User is successfully logged in and navigates to the profile page.	Logged in and profile page loaded.	Pass

2	Edit profile and update phone number.	Phone Number: 1234567890	System validates input and updates the phone number in the profile.	Successfully updated and shown in the profile details.	Pass
3	Change password and test login with the new password.	Current Password: 12345678, New Password: Newpass123	Password is updated, and the user can log in with the new password.	Password updated, and login successful with new password.	Pass
4	Delete a user account and verify login.	Admin deletes the account: sirdan@gmail.com.	Account is removed, and the user cannot log in.	Login attempt fails with error: "Account not found."	Pass
5	Logout and verify session termination.	Click "Logout."	User session is terminated, and the system redirects to login.	Successfully logged out, and session invalidated.	Pass

6.2 Role and Permission Management

6.2.1 Unit Testing

Table 6.5 shows the unit test cases of Role and permission management module

Table 6.5 Unit test cases

No.	Test Case/Test Script Description	Attribute and Value	Expected Result	Result
1	Validate admin access to role creation feature.	Actor: Admin	Admin can access the "Create New Role" section.	Pass
2	Validate duplicate role name creation.	Role Name: Manager	Error: "Role name already exists."	Pass
4	Validate role assignment to a registered user.	User: sirdan@gmail.com , Role: Admin	Role is successfully assigned to the user.	Pass
5	Validate menu assignment to a specific role.	Menu Item: Reports	Menu item is successfully linked to the user or role.	Pass

6.2.2 Functional Testing

Table 6.6 shows the Functional test cases of Role and permission management module

Table 6.6 Functional test cases

No.	Test Case/Test Script Description	Attribute and Value	Expected Result	Actual Result	Result
1	Create a new user role with a valid name.	Role Name: Supervisor,	Role is created successfully and available for assignment.	Role created and displayed in the role list.	Pass
2	Attempt to create a new user role with an existing name.	Role Name: Admin	Error message: "Role name already exists."	Error displayed as expected.	Pass
3	Assign an existing role to a registered user.	User: aqsam@gmail.com, Role: Supervisor	Role is successfully assigned, and user permissions are updated.	User assigned to the role with updated access.	Pass
5	Customize user menus for a role and save.	Role: Supervisor, Menu Items: Tasks, Settings	Menu is successfully customized and saved for the role.	Menus assigned successfully.	Pass

6.2.3 Business Rule Testing

Table 6.7 shows the Integration test cases of Role and permission management module

Table 6.7 Integration test cases

No.	Test Case/Test Script Description	Attribute and Value	Expected Result	Actual Result	Result
1	Admin creates a new role with a unique name	Role Name: Analyst	Role is created successfully.	Role created	Pass
2	Admin creates a role with an existing name	Role Name: Manager	Displays message: "Role is created successfully.."	Error message displayed	Fails

6.2.4 Integration Testing

Table 6.8 shows the Functional test cases of Role and permission management module

Table 6.8 Integration test cases

No.	Test Case/Test Script Description	Attribute and Value	Expected Result	Actual Result	Result
1	Create a role and assign it to a user.	Role Name: Supervisor, User: aqsam@gmail.com	Role is created and assigned to the user successfully.	Role assigned, and permissions updated.	Pass
3	Customize and save a menu for a role.	Role: Manager, Menus: Dashboard, Settings	Menus are saved and assigned to the role successfully.	Menus assigned, and functionality verified.	Pass
4	Assign a duplicate role to a user.	User: aqsam@gmail.com, Role: Manager	System prompts that the user already holds this role.	Admin notified, and operation canceled.	Pass

6.3 Human Resource Management

6.3.1 Unit Testing

Table 6.9 shows the unit test cases of Human Resource management module

Table 6.9 Unit test cases

No.	Test Case/Test Script Description	Attribute and Value	Expected Result	Result
1	Create employee profile with valid data	Name: faheem, Department: Sales	Profile is created successfully.	Pass
2	Create employee profile with missing required field	Name: ``, Department: Sales	Displays error message: "Name is required."	Pass
3	Create employee profile with duplicate email	Email: faheem@example.com (already exists)	Displays error message: "Email already in use."	Pass

6.3.2 Functional Testing

Table 6.10 shows the Functional test cases of Human Resource management module

Table 6.10 Functional test cases

No.	Test Case/Test Script Description	Attribute and Value	Expected Result	Actual Result	Result
1	Submit expense report with valid data	Category: Travel, Amount: 100, Receipt: travel_receipt.jpg	Expense report is submitted successfully.	Report submitted	Pass
2	Submit leave request with valid data	Type: Vacation, Start Date: 2023-10-01, End Date: 2023-10-05	Leave request is submitted successfully.	Request submitted	Pass
3	Submit leave request with past start date	Type: Sick, Start Date: 2023-09-01, End Date: 2023-09-05	Displays error message: "Start date cannot be in the past."	Error message displayed	Pass
4	Submit leave request with end date before start date	Type: Vacation, Start Date: 2023-10-05, End Date: 2023-10-01	Displays error message: "End date must be after start date."	Error message displayed	Pass

6.3.3 Business Rule Testing

Table 6.11 shows the Integration test cases of Human Resource management module

Table 6.11 Integration test cases

No.	Test Case/Test Script Description	Attribute and Value	Expected Result	Actual Result	Result
1	Submit timesheet with valid work hours	Date: 2023-10-01, Hours: 8, Overtime: 2	Timesheet is submitted successfully.	Timesheet submitted	Pass
2	Approve expense report with valid data	Report ID: 123, Action: Approve	Expense report is approved successfully.	Report approved	Pass

6.3.4 Integration Testing

Table 6.12 shows the Functional test cases of Human Resource management module

Table 6.12 Integration test cases

No.	Test Case/Test Script Description	Attribute and Value	Expected Result	Actual Result	Result
1	Employee submits leave request and receives notification	Leave Request: Vacation, Dates: 2023-10-01 to 2023-10-05	Notification is sent to the employee and approver.	Notification sent	Pass
2	Expense report is approved, and employee receives notification	Report ID: 123, Action: Approve	Notification is sent to the employee about approval.	Notification sent	Pass
3	Timesheet deadline reminder notification	Timesheet Deadline: 2023-10-31	Notification is sent to the employee before the deadline.	Notification sent	Pass
4	Notification system fails to send alert	Leave Request: Sick, Dates: 2023-10-01 to 2023-10-03	System logs error and retries notification.	Error logged, retry attempted	Fail

6.4 Task Management

6.4.1 Unit Testing

Table 6.13 shows the unit test cases of Task management module

Table 6.13 Unit test cases

No.	Test Case/Test Script Description	Attribute and Value	Expected Result	Result	
1	Create a task with valid data	Task: Complete report, Priority: High, Deadline: 2023-10-15	Task is created successfully.	Task created	Pass
2	Create a task without a deadline	Task: Prepare presentation, Priority: Medium, Deadline: None	Displays error message: "Deadline is required."	Error message displayed	Pass
3	Create a task with invalid priority	Task: Review documents, Priority: Invalid,	Displays error message: "Invalid priority."	Error message displayed	Pass

		Deadline: 2023-10-20			
4	Create a task with a past deadline	Task: Submit feedback, Priority: Low, Deadline: 2023-09-01	Displays error message: "Deadline cannot be in the past."	Error message displayed	Pass

6.4.2 Functional Testing

Table 6.14 shows the Functional test cases of Task management module

Table 6.14 Functional test cases

No.	Test Case/Test Script Description	Attribute and Value	Expected Result	Actual Result	Result
1	Schedule a single event with valid data	Event: Team Meeting, Date: 2023-10-10, Time: 10:00 AM	Event is scheduled successfully.	Event scheduled	Pass
2	Schedule an event with a past date	Event: Project Review, Date: 2023-09-01, Time: 3:00 PM	Displays error message: "Event created."	Error message displayed	Fail
3	Set a reminder for a task	Task: Complete report, Reminder: 1 day before deadline	Reminder is set successfully.	Reminder set	Pass
4	Set a reminder without specifying time	Task: Prepare presentation, Reminder: None	Displays error message: "Reminder time is required."	Error message displayed	Pass

6.4.3 Integration Testing

Table 6.15 shows the Functional test cases of Task management module

Table 6.15 Integration test cases

No.	Test Case/Test Script Description	Attribute and Value	Expected Result	Actual Result	Result
1	Task reminder triggers notification	Task: Complete report, Reminder: 1 day before deadline	Notification is sent via email and in-app alert.	Notification sent	Pass

2	Event reminder triggers notification	Event: Team Meeting, Reminder: 30 minutes before	Notification is sent via email and in-app alert.	Notification sent	Pass
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6.5 Dashboard and Reporting

6.5.1 Unit Testing

Table 6.16 shows the unit test cases of Dashboard and Reporting module

Table 6.16 Unit test cases

No.	Test Case/Test Script Description	Attribute and Value	Expected Result	Result
1	Validate report criteria input (valid)	Criteria: Sales, Timeframe: Last Quarter	Input is accepted.	Pass
2	Validate report criteria input (invalid)	Criteria: Invalid, Timeframe: Last Month	Displays error: "Invalid criteria selected."	Pass
3	Validate dashboard widget selection (valid)	Widget: Sales Chart	Widget is added to the dashboard.	Pass

6.5.2 Functional Testing

Table 6.17 shows the Functional test cases of Dashboard and Reporting module

Table 6.17 Functional test cases

No.	Test Case/Test Script Description	Attribute and Value	Expected Result	Actual Result	Result
1	Generate a sales report	Criteria: Sales, Timeframe: Last Quarter	Report is generated successfully.	Pass	1
2	Export report as PDF	Format: PDF, Report: Sales Report	Report is exported as a PDF file.	Pass	2

3	Customize dashboard layout with valid widgets	Widgets: Sales Chart, Productivity Graph	Dashboard layout is updated successfully.	Pass	4
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6.5.3 Integration Testing

Table 6.18 shows the Functional test cases of Dashboard and Reporting module

Table 6.18 Integration test cases

No.	Test Case/Test Script Description	Attribute and Value	Expected Result	Result
1	Real-time update in sales dashboard	Data: New Sales Entry, Dashboard: Sales Dashboard	Dashboard reflects new data in real-time.	Pass
2	Drill down into sales chart for detailed data	Chart: Sales Chart, Data Point: Q3 Sales	Displays detailed sales data for Q3.	Pass
3	Real-time update fails due to server error	Data: New Sales Entry, Dashboard: Sales Dashboard	System logs error and retries synchronization.	Fail

6.6 Billing and Financing

6.6.1 Unit Testing

Table 6.19 shows the unit test cases of Billing and Financing module

Table 6.19 Unit test cases

No.	Test Case/Test Script Description	Attribute and Value	Expected Result	Result
3	Validate payment amount input (valid)	Amount: 100.00	Amount is accepted.	Pass
4	Validate payment amount input (invalid)	Amount: -100.00	Displays error: "Amount must be positive."	Pass

6.6.2 Functional Testing

Table 6.20 shows the Functional test cases of Billing and Financing module

Table 6.20 Functional test cases

No.	Test Case/Test Script Description	Attribute and Value	Expected Result	Result
1	Process payment via Stripe (successful)	Amount: 100.00, Card: 4242 4242 4242 4242	Payment is processed successfully, and receipt is generated.	Pass
2	Process payment via Stripe	Amount: 100.00, Card: 4000 0000 0000 0002	Displays error: "Payment failed. Please try again."	Pass

6.6.3 Integration Testing

Table 6.21 shows the Functional test cases of Billing and Financing module

Table 6.21 Integration test cases

No.	Test Case/Test Script Description	Attribute and Value	Expected Result	Result
1	Verify payment status sync with Stripe	Payment ID: 123456789	Payment status is synced with Stripe and displayed in the system.	Pass

6.7 Chatbot

6.7.1 Unit Testing

Table 6.19 shows the unit test cases of Chatbot module

Table 6.19 Unit test cases

No.	Test Case/Test Script Description	Attribute and Value	Expected Result	Result
1	Validate getOrderSummary(userId) returns list when orders exist	Mock DB: 3 orders for userId=5	• Order 101: 2 items, \$50.00; • Order 102: 1 item, \$30.00; • Order 103: 4 items, \$120.00	Pass
3	Validate getAttendanceSummary(start ,end) returns summary	Mock DB: present 80, absent 20 between dates	present: 80; absent: 20	Pass

5	Validate getProjectTaskSummary(proj ectId) returns breakdown	Mock DB: pending 5, in_progress 3, completed 12 for projectId=7	pending: 5; in_progress: 3; completed: 12	Pass
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6.7.2 Functional Testing

Table 6.20 shows the Functional test cases of Chatbot module

Table 6.20 Functional test cases

No.	Test Case/Test Script Description	Attribute and Value	Expected Result	Result
1	Process “Order summary for user 5” via API	POST /api/chatbot	HTTP 200, { answer: "• Order 101: 2 items, \$50.00; • Order 102: 1 item, \$30.00; • Order 103: 4 items, \$120.00" }	Pass
2	Process “Attendance summary between 2025-05-01 and 2025-05-10” via API	POST /api/chatbot { question: "Attendance summary between 2025-05-01 and 2025-05-10" }	HTTP 200, { answer: "present: 80; absent: 20" }	Pass
3	Process “Task summary by status for project 7” via API	POST /api/chatbot	HTTP 200, { answer: "pending: 5; in_progress: 3; completed: 12" }	Pass

6.7.3 Integration Testing

Table 6.21 shows the Functional test cases of chatbot module

Table 6.21 Integration test cases

No.	Test Case/Test Script Description	Attribute and Value	Expected Result	Result
1	End-to-end order-summary scenario	Seed real DB with orders for userId=5 → call /api/chatbot	UI shows: • Order 101: 2 items, \$50.00; • Order 102: 1 item, \$30.00; • Order 103: 4 items, \$120.00	Pass
2	End-to-end attendance-summary scenario	Seed real DB with attendance records → call /api/chatbot	UI shows: present: 80; absent: 20	Pass

3	End-to-end project-task-summary scenario	Seed real DB with tasks for projectId=7 → call /api/chatbot	UI shows: pending: 5; in_progress: 3; completed: 12	Pass
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6.8 Order and Inventory

6.8.1 Unit Testing

Table 6.19 shows the unit test cases of Order and Inventory module

Table 6.19 Unit test cases

No.	Test Case/Test Script Description	Attribute and Value	Expected Result	Result
1	Validate createOrder() with valid payload	items: [{id:1,qty:2,price:25.00}], userId:10	Order record saved with total \$50.00, status pending	Pass
2	Validate createOrder() with empty items	items: [], userId:10	Throws error: "Order must contain at least one item."	Pass
3	Validate isLowStock() returns true when below threshold	productId:2 stock=4, threshold=10	Returns true	Pass
4	Validate isOutOfStock() returns true when stock == 0	productId:3 stock=0	Returns true	Pass

6.8.2 Functional Testing

Table 6.20 shows the Functional test cases of Order and Inventory module

Table 6.20 Functional test cases

No.	Test Case/Test Script Description	Attribute and Value	Expected Result	Result
1	Update inventory level via API	PUT /api/inventory/1	Update inventory level via API	Pass
2	Place new order via REST API	POST /api/orders	HTTP 201; Location header /api/orders/{newId}; Body contains { id:newId, total:..., status:"pending" }	Pass

3	Retrieve order details via REST API	GET /api/orders/200	HTTP 200; Body contains { id:200, userId:10, items:[...], total:50.00, status:"pending" }	Pass
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6.8.3 Integration Testing

Table 6.21 shows the Functional test cases of Order and Inventory module

Table 6.21 Integration test cases

No.	Test Case/Test Script Description	Attribute and Value	Expected Result	Result
1	End-to-end order placement reduces inventory	Seed product ID 1 stock=100 → place order via /api/orders with qty 5 → fetch /api/inventory/1	Stock for product 1 reduced to 95	Pass
2	Restock via purchase order updates inventory and logs audit	POST /api/purchase-orders qty:20 for product 1 → check /api/inventory/1 and audit	Stock increased by 20; audit record { productId:1, delta:+20, ... }	Pass
3	Sync with external WMS updates stock correctly	Simulate external WMS callback { productId:4, stock:30 } → verify /api/inventory/4	Stock for product 4 updated to 30	Pass

7 Conclusion and Future Work

7.1 Conclusion

ModuFlex ERP is a comprehensive and customizable Enterprise Resource Planning (ERP) system designed to streamline operations for retailers and e-commerce businesses. By addressing the challenges of repetitive coding, inflexible systems, and high costs, ModuFlex ERP provides an efficient solution that simplifies complex business processes. The system incorporates key modules such as User Management, Role and Permission Management, Task and Notification Management, Human Resource Management, Billing and Financing, and Dashboard and Reporting, ensuring a seamless and integrated experience for users. With features like real-time data updates, customizable dashboards, and integration with payment gateways like Stripe, ModuFlex ERP enhances productivity, reduces operational costs, and improves decision-making. The successful implementation of this system demonstrates its potential to transform business operations, making it a valuable tool for businesses with both physical and online presence.

7.2 Future Work

The ModuFlex ERP platform is a forward-thinking project with significant potential for growth and enhancement.

Here are some areas for future work

1. **Expand Modules:** Add new modules such as **Supply Chain Management**, **Customer Relationship Management (CRM)**, and **Advanced Analytics** to cater to a wider range of business needs.
2. **Integrate AI Features:** Implement **predictive analytics** for inventory management and demand forecasting and Use **machine learning algorithms** for personalized product recommendations and sales trend analysis.
3. **Enhance Scalability:** Improve system architecture to support **larger enterprises** and **higher transaction volumes** and Optimize database and server performance for faster data processing and real-time updates.
4. **Advanced Reporting and Visualization:** Introduce **interactive and dynamic reporting tools** with advanced visualizations (e.g., heatmaps, geospatial analysis) for deeper business insights.
5. **Multi-Language and Multi-Currency Support:** Add support for **multiple languages** and **currencies** to make the system accessible to global businesses.
6. **Enhanced Security Features:** Implement **two-factor authentication (2FA)**, **data encryption**, and **role-based access control** to ensure robust security and compliance with data protection regulations.
7. **IoT Integration:** Integrate with **Internet of Things (IoT)** devices for real-time monitoring of inventory, equipment, and supply chain activities.

8. **User Experience Improvements:** Continuously refine the **user interface (UI)** and **user experience (UX)** based on feedback to ensure ease of use and accessibility.
9. **Automation of Routine Tasks:** Introduce **robotic process automation (RPA)** to automate repetitive tasks such as data entry, invoice generation, and report scheduling.
10. **Collaboration Tools:** Add **collaboration features** like team chat, task commenting, and file sharing to improve communication and teamwork within the system.

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9. Turnitin Report

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